

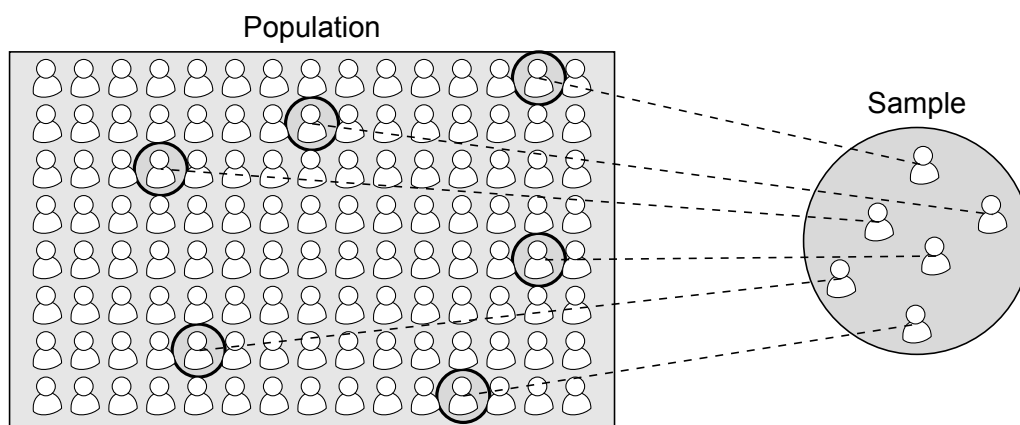
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Introduction

Statistics is the science of collecting, organising, presenting and analysing data. It is vital in helping us develop a better understanding of the world, and to allow decisions to be made in a rational manner based on the information that is available to us. Governments, organisations, businesses and individuals across a diverse range of fields rely on statistics and statisticians.



The *Advanced Higher Statistics* course aims to equip learners with a sound foundation of statistical knowledge. This textbook has been created to serve as a source of key facts, examples, and exercises to support progress through the course, covering all of the key stages involved in statistical research and investigations, including:

- Gathering data using appropriate *sampling techniques*.
- Initial exploration of data using *descriptive statistics*, including:
 - Visualising data with charts and graphs.
 - Calculating sample statistics.
- Using *statistical inference* to assess the evidence for claims about population parameters, including:
 - An introduction to *probability theory*, which underpins all statistical inference
 - Conducting hypothesis tests
 - Constructing confidence intervals

Techniques suitable for a range of contexts will be covered, and it is expected that by the end of the course students should have sufficient knowledge and confidence to collect their own data and conduct their own statistical investigations.

Suggested Use Of The Book

Other Resources

Acknowledgements

1

Exploratory Data Analysis

Collected data, without any form of processing having been applied, is referred to as *raw data*. Before conducting any statistical analysis, a statistician will typically want to get a quick sense of the data, checking for any potential issues and identifying any features of note. Common first steps are therefore often to: *organise* the data into suitable lists or tables; *visualise* the data using plots and graphs; and *measure* various properties of the data.

This process is called *exploratory data analysis*.

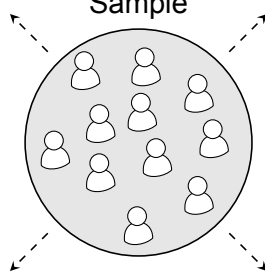
Patient data table (first five entries only)

age (yrs)	height (cm)	NHS health board
57	182.2	Fife
43	168.2	Tayside
48	179.4	Tayside
64	162.6	Forth Valley
52	171.5	Fife

Dot plot and box plot (age, NHS Fife patients)



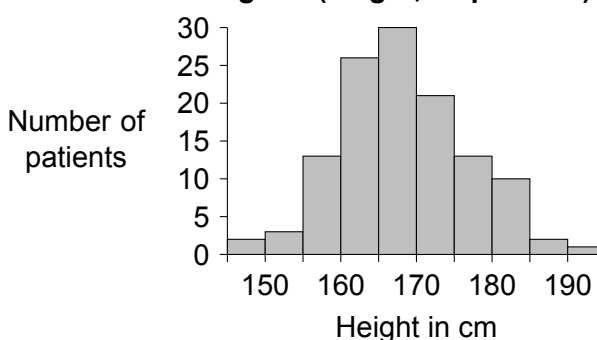
Sample



Summary (height in cm, all patients)

sample size	$n = 120$
mean	$\bar{x} = 168.5$
standard deviation	$s = 8.26$
median	$\text{med} = 168.9$
interquartile range	$\text{IQR} = 10.3$
minimum value	$\text{min} = 149.0$
maximum value	$\text{max} = 190.1$

Histogram (height, all patients)

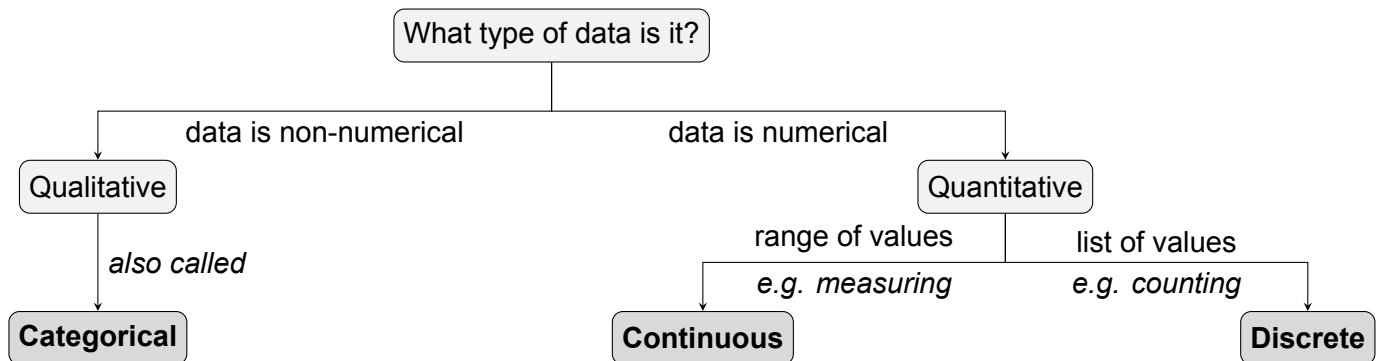


1.1 Types of Data

Knowing what *type* of data has been collected helps inform what kind of graphs and statistics may be appropriate.

Data that is numerical is called *quantitative*, of which there are two types. **Discrete** data consists of values that can be listed, such as someone's shoe size or the number of siblings they have. **Continuous** data can take *any* value within a range, though is often rounded to reflect the precision of measurements taken, such as the mass of a mouse.

Qualitative data (also called **categorical**) is *non-numerical*, such as whether a squirrel is red or grey.



For example, the table below shows data relating to some football players:

Position	Team	Height (m)	Caps	Goals
Midfielder	Aston Villa	1.78	66	18
Midfielder	Man United	1.93	49	8
Forward	Southampton	1.75	30	6
Forward	Bournemouth	1.78	49	6
Midfielder	Celtic	1.75	39	5
Categorical		Continuous	Discrete	

Exercise 1.1

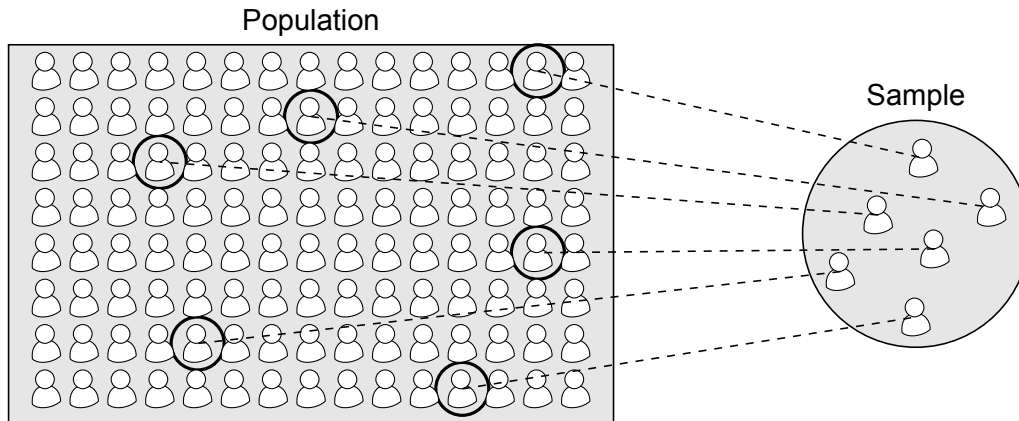
- A survey is carried out on the pets owned by a class. State whether the following data is qualitative or quantitative.
 - How many pets each pupil owns.
 - What kind of animal the pets are.
 - The pet's age.
 - The name of the pet.
 - The mass of the pet.
 - Whether the pet is a mammal.
- For each of the following, state whether the data collected would be discrete or continuous.
 - The number of people in a room.
 - The mass of a book.
 - The number of pages in a book.
 - The number of songs in an album.
 - The volume of water in a bottle.
 - Time taken to complete a puzzle.
- A random sample of members of a gym were surveyed and the first few results are shown in the table below.

Activity level	Smoker	Height (inches)	Mass (pounds)	Pulse (bpm)
Moderate	No	66	140	64
Moderate	No	72	145	58
A lot	Yes	73.5	160	62
Slight	Yes	73	190	66

State the type of data that was collected for each variable.

1.2 Population Parameters and Sample Statistics

In a statistical study, the term **population** refers to the entire set of objects of interest, whilst a **sample** is a selection of some objects chosen from the population. Recognising the difference is fundamental to statistics.



If data were available for an entire *population* then various **population parameters** could be calculated, such as the *population mean*. Since this is never really possible, instead *samples* are taken from populations. Measures calculated from a sample of data are called **sample statistics**, such as the *sample mean*.

Letters from the Greek alphabet are commonly used to represent population parameters.

Population Parameter

Population mean μ ("mu")

Population standard deviation σ ("sigma")

Sample Statistic

Sample mean $\bar{x}$ ("x bar")

Sample standard deviation s

It is good practice to refer *clearly* to whether a population parameter or sample statistic is being discussed.



Population parameters are seen as having fixed values, at the time of the study at least, whilst the values of sample statistics will depend on the sample chosen.

Sample statistics may be seen as *estimates* of the true value of their respective population parameters, and **inference** is the process by which claims may be made about population parameters (such as μ) based on observed sample statistics (such as $\bar{x}$).

Exercise 1.2

For each, state using the context of the question which population parameter or sample statistic has been obtained.

1. A year group at a school contains 120 pupils. Data is obtained for the number of merits each pupil in the year group obtained, and the mean of those values is calculated to be 3.4.
2. A bus company in Scotland checks the total mileage for some of its buses when they come in for a service. The mean mileage is calculated as 45,300 miles.
3. A medical trial for a new supplement to increase iron levels in the blood involves a number of patients being given the treatment over the course of a month, and then their blood iron levels being checked. The mean increase for the patients is found to be 27.3 micrograms per decilitre (mcg/dL).

1.3 Measures of Location and Spread

Summary statistics are a collection of measures taken of the data from a sample, such as the mean, or standard deviation. These offer a quick way to communicate some general sense of the data. For example, for the *raw data* of 0, 2, 2, 6, 12, 14, 16, 18, 18, 18, 20, the table below shows a selection of summary statistics.

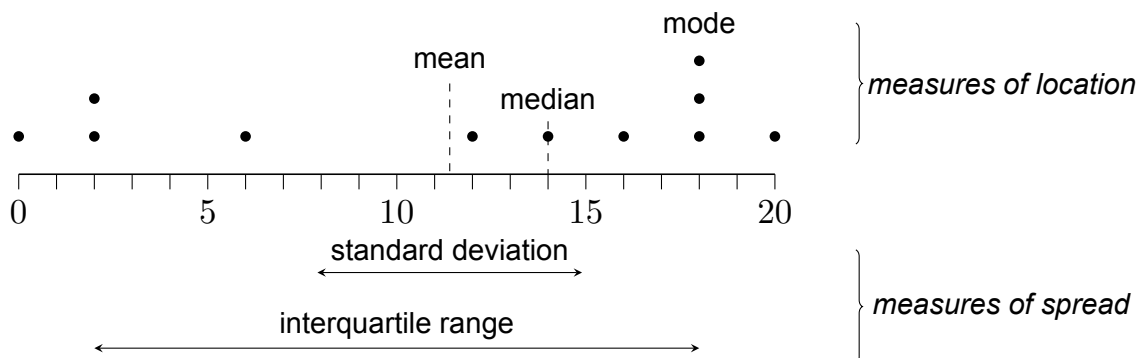
n	mean	sd	sum	sum of squares	min	LQ	median	UQ	max	range	IQR	mode
11	11.45	7.54	126	2012	0	4	14	18	20	20	14	18

There are two main types of summary statistic: those that measure the *central tendency* of data (or its **location**); and those that measure the *variability* of data (or its **spread**). It is common to wish to measure both the location *and* spread.

Measures of Location

The three main measures of location, or *averages*, should already be familiar. To avoid ambiguity, statisticians will typically avoid referring to “*the average*”, and will instead specify exactly *which* average they are referring to:

- **Mode** - the most commonly observed piece of data.
- **Median** - the middle value after the data has been put into numerical order.
- **Mean** - the sum of the values divided by the number of values.



Measures of Spread

The measure of spread which is usually encountered first in a maths classroom is the *range*, which is the difference between the highest and lowest values. In practice this measure is seldom used by statisticians since it only uses the two most extreme values from the data. There are three main measures of spread commonly used, two of which are likely to already be familiar:

- **Interquartile Range** - the difference between the upper quartile and the lower quartile.
- **Variance** - the *mean squared difference* between the values and the mean value.
- **Standard deviation** - the square root of the variance.

The Mean and the Standard Deviation

The *mean* and *standard deviation* are perhaps the most commonly used measures of location and spread, respectively, and both link strongly to the commonly-encountered *bell-shaped curve* (see page 13). The formulae for calculating the sample mean and sample standard deviation from sample data are given below using notation, where the Greek letter Σ (“*Sigma*”) means a *sum* should be calculated, x represents the all of the values in the sample (a shortened abbreviation of $x_1, x_2, \dots, x_n$ or x_i) and n represents the *number* of values in the sample.

The *sample mean*, $\bar{x}$, is calculated as: $\bar{x} = \frac{\Sigma x}{n}$

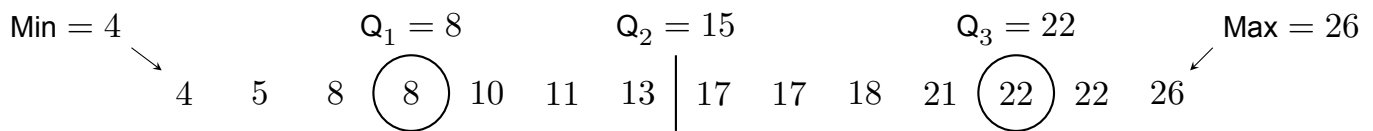
The *sample standard deviation*, s , is calculated as: $s = \sqrt{\frac{S_{xx}}{n-1}}$ where: $S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n}$

The *sample variance*, s^2 , is simply the *square of the standard deviation*.

The Median and the Interquartile Range

The median, Q_2 , and interquartile range, IQR, are often chosen to measure location and spread when the data is particularly *skewed* (see page 13), as they are less influenced by *extreme values* than the mean and standard deviation.

The *lower quartile* (LQ or Q_1) and the *upper quartile* (UQ or Q_3) split a set of values into quarters along with the median. From the lowest value to Q_1 contains the lowest 25% of the values, from Q_1 to Q_2 contains the next 25%, and so on.



The *interquartile range* is calculated as: $IQR = Q_3 - Q_1$

Exercise 1.3

1. The monthly cost of 10 drivers' car insurance (x , in £'s) are recorded. Calculate the mean and standard deviation.

$$\Sigma x = 359 \quad \Sigma x^2 = 15119 \quad n = 10$$

2. Daily fuel spend (f , in £'s) for 15 drivers is summarised below. Calculate the mean and standard deviation.

$$\Sigma f = 87 \quad \Sigma f^2 = 569 \quad n = 15$$

3. The marks, out of 40, of eight pupils sitting a chemistry test are below. Find the mean and standard deviation.

32 25 29 37 16 39 23 32

4. The ages (in years) of 18 customers in a shop are recorded. Calculate the median and interquartile range.

17 63 34 23 40 57 72 70 43
56 29 37 41 50 13 24 30 58

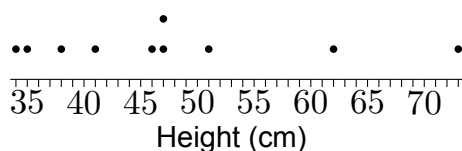
1.4 Displaying Data

Before any analysis of a data set is undertaken, it is common practice to *display* the data in a simple form so that an initial impression of what the data shows can be formed. Some key examples are shown below, using a sample taken by a gardener of the heights of 10 of their sunflowers 8 weeks after germination, measured to the nearest centimetre:

34 35 38 41 46 47 47 51 62 73

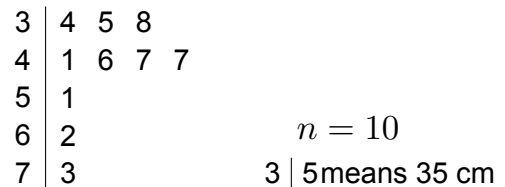
Dot Plots

Dot plots offer a quick way of presenting data in a manner similar to a bar chart, stacking repeated values.



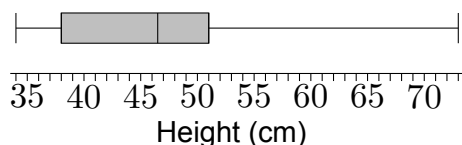
Stem-and-Leaf Diagrams

Stem-and-leaf diagrams display data in a manner similar to a grouped bar graph. Leaves should be in order, and a key (or *legend*) and the sample size should be included.



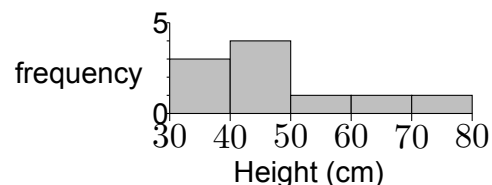
Box Plots

Box plots offer a visual representation of a five-figure summary, consisting of the minimum value, the lower quartile Q_1 , the median, the upper quartile Q_3 and the maximum value.



Histograms

Histograms put continuous numerical data into intervals (or *bins*) and display it in a manner similar to a bar graph.



Exercise 1.4

1. The ages (in years) of competitors in a junior golf tournament are below. Draw a dot plot to represent this data.

16 14 19 13 14 17 16 19 16 17 12 16 14 19

2. The price of an ice cream is recorded at 12 stalls, in £'s. Represent the data using a stem-and-leaf diagram.

1.50 2.20 2.50 2.10 1.80 1.40
2.00 1.90 3.60 0.90 1.70 1.50

3. The masses (in kg) of the luggage of 10 passengers on a plane to Barra were recorded. Construct a box plot.

15 18 14 22 19 14 10 21 17 16

1.5 Tables, Figures and Output

There are many ways in which data can be presented, extending beyond those covered on the previous page. In statistical reports they are often clearly labelled as **tables** or **figures**, and numbered so they can be more easily referred to.

Table 1: Contingency table of mode of transport.

	Year Group	Transport			
		walk	bus	car	bike
	3rd	12	15	3	4
	2nd	7	13	8	1
	1st	4	14	11	2

Figure 1: Bar chart of mode of transport.

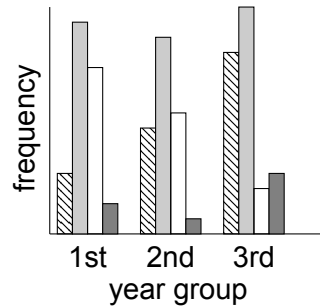
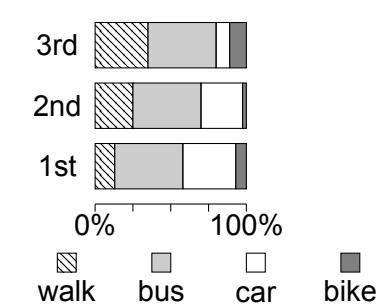


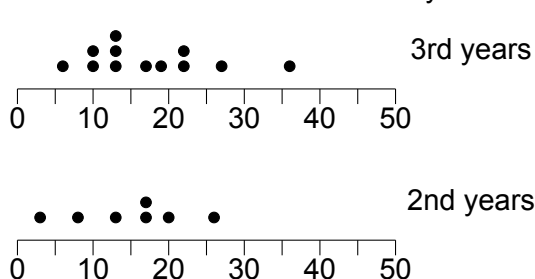
Figure 2: Chart of transport percentages.



Not all graphs encountered in real life will be up to scratch; some graphs may be unclear, accidentally misleading or even intentionally misleading. Being alert to potential flaws in tables and figures and identifying possible improvements is an important skill.

The widespread availability of computing power means that specialist software is often used by statisticians to avoid the need for time-consuming calculations where possible. Whilst this course requires no use of statistical software, examples of computer **output** will be displayed from which the required information must be extracted.

Figure 3: Dot plots of time (mins) to walk to school for 2nd and 3rd years.



Output 1: Summary data for walk times (mins) for 2nd and 3rd years.

```
data: walk time

3rd yr: n=12    mean=17.3    sd=8.42

2nd yr: n=**** mean=14.9    sd=7.65

alternative hypothesis: means are not equal

t=0.619, df=17, p-value=0.544
```

Whilst this textbook aims to enable familiarity with common figures, tables and computer output, the best way to practise obtaining information from them is to *read a wide range of relevant statistical articles and reports*.

Exercise 1.5

1. Use **Table 1** to calculate the proportion of pupils in the sample that walk to school.
2. Suggest two possible improvements to **Figure 1**.
3. Using **Figure 2**, compare the proportions of pupils in the sample who take the bus between each year group.
4. From **Figure 3**, state the approximate median time taken to walk for 3rd year pupils.
5. One piece of information from **Output 1** has been replaced by ****. State the missing value.

1.6 Outliers and Fences

Sometimes a set of data may contain values that seem *extremely high or low* in comparison to the rest of the values. It is important to be wary of such values and the potentially significant impact they could have on any statistical analysis performed. There are a range of objective approaches that can be used to determine whether one or more points are extreme enough to be classed as **outliers**. In this course, *fences* will be used.

Identifying Outliers

$$\text{Lower Fence} = Q_1 - 1.5 \times \text{IQR.}$$

$$\text{Upper Fence} = Q_3 + 1.5 \times \text{IQR.}$$

Any values *below the lower fence* or *above the upper fence* are **outliers**.

If an outlier is identified in a set of data, an investigation should be conducted to determine the cause. If it is concluded that the outlier is a rogue point, such as an error in measurement, it can be removed from the data set and summary statistics re-calculated. However, if an outlier is a valid point in the data set then it should not be removed.

Returning to the sunflower heights from page 6, the data can now be checked for possible outliers:

Raw data										Summary data					
Heights (cm)										n	min	Q1	median	Q3	max
34	35	38	41	46	47	47	51	62	73	10	34	38	46.5	51	73

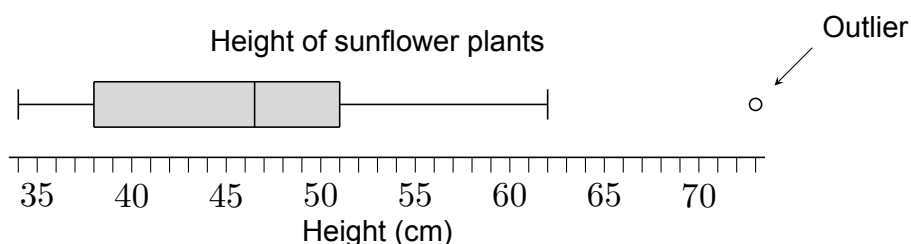
$$\text{IQR} = Q_3 - Q_1 = 51 - 38 = 13$$

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} = 38 - 1.5 \times 13 = 18.5$$

$$\text{Upper fence} = Q_3 + 1.5 \times \text{IQR} = 51 + 1.5 \times 13 = 70.5$$

Since $73 > 70.5$, it is an outlier.

On a box plot, outliers are indicated separately as shown below, with the “whiskers” extending only as far as the highest (and lowest) values which are *not* outliers. However the quartiles remain as those calculated using the *full* set of values, including any outliers. *Plotting outliers separately in this way is to highlight their presence, and is not a judgement on whether those values are rogue or valid.*



The outlier in this data set could be valid if, for instance, the sunflower plant is in a particular sunny spot in the garden. However, if the height was measured incorrectly then it can be removed from the data set.

Example

Problem: A vet conducts a survey, asking 10 randomly chosen cat owners registered with his practice to share the mass of their adult cat, in kilograms. The results are:

4.6 3.6 4.1 4.5 3.7 3.9 4.0 0.3 3.9 4.8

- (a) Show that the set of data contains an outlier.
 (b) Suggest a possible cause for this outlier which would mean it should be removed.

Solution:

- (a) $Q_1 = 3.75$, $Q_3 = 4.4$, $IQR = 4.4 - 3.75 = 0.65$.

$$\text{Lower fence} = 3.75 - 1.5 \times 0.65 = 2.775$$

$$\text{Upper fence} = 4.4 + 1.5 \times 0.65 = 5.375$$

Since $0.3 < 2.775$, the value of 0.3 is an outlier.

- (b) The owner could have mistakenly measured the mass of their cat in stones instead of kilograms.

Exercise 1.6

1. Strawberry tarts are sold every day in a bakery over the summer months. The shop manager monitors the sales and takes a random sample of the number of strawberry tarts sold on 10 randomly selected days over one month.

23 31 45 32 35 19 24 25 30 38

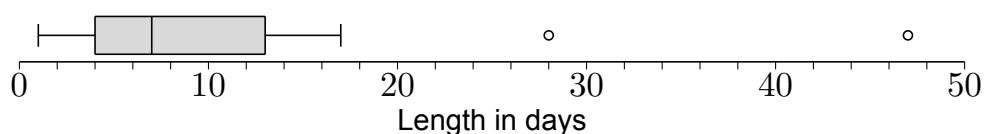
Calculate the upper and lower fences for this data set and identify any possible outliers.

2. The number of hours spent studying each week by a randomly selected group of 24 students is given below.

10 18 15 12 17 27 4 14 14 14 17 15
 15 12 17 17 13 15 15 16 16 9 11 15

- (a) Identify any possible outliers.
 (b) Draw a box plot to display this data.
3. A hospital trust compiled data relating to the length of stay of patients in a particular hospital. A random sample of 18 patients gave the following summary data, measured in days.

n	mean	sd	min	Q1	median	Q3	max
18	10.78	11.17	1	4	7	13	47

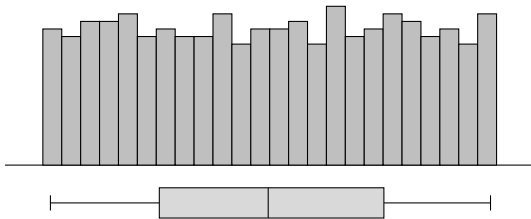


- (a) Show that the two values indicated by circles on the box plot are outliers.
 (b) Suggest a cause for these outliers that would make it valid to remove them from further analysis.

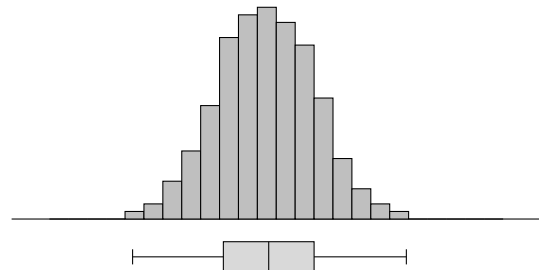
1.7 Describing Data Distributions

Summary statistics alone are not sufficient to describe the full complexity of the way data is distributed. It is also important to consider and be able to describe the **shape** of the distribution, accomplished most easily using a histogram. Whilst real-life rarely *perfectly* follows a precise mathematical model, a number of shapes are commonly observed and a number of descriptions are commonly considered. The following data was generated by *simulating sampling* from populations of data which followed the distributions described.

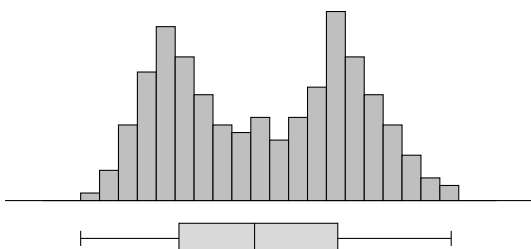
uniformly distributed data ($n = 103$)



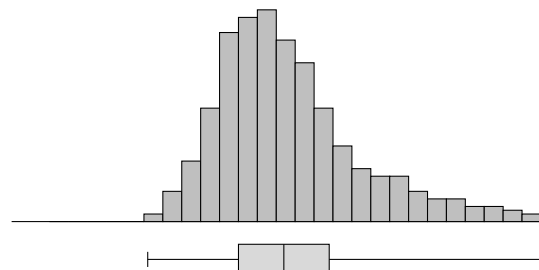
normally distributed data, also referred to as a bell-curve ($n = 215$)



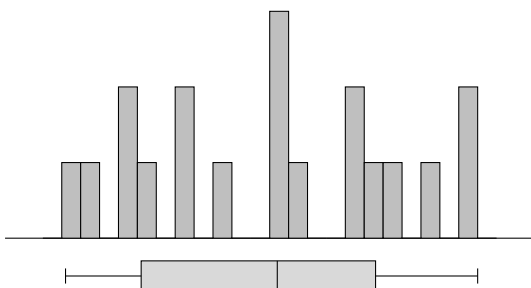
bimodal data ($n = 87$)



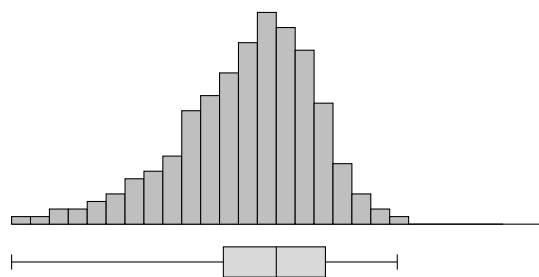
positively skewed data ($n = 134$)



very hard to infer distributions of data from small samples ($n = 19$)



negatively skewed data ($n = 97$)



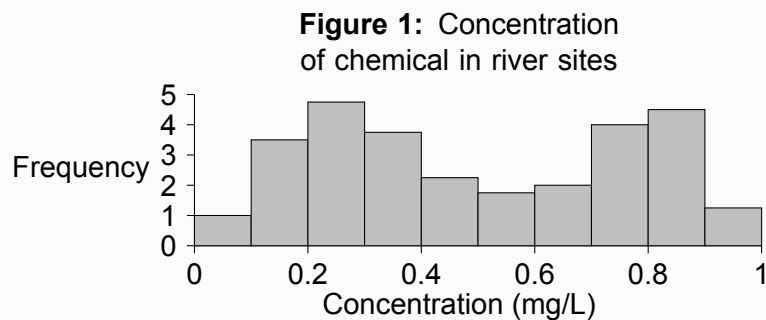
As well as considering whether the **underlying population distribution** of data appears to be *normal*, *skewed*, *uniform* or *bimodal*, the appropriateness of some statistical inference techniques will depend on whether the distribution is *symmetrical*, or whether two sets of data follow the *same distribution*, with matching shapes and spread.

Whilst the following example references techniques that will be introduced later in the course, it is intended to give an idea of the importance of studying distributions in selecting appropriate methods for statistical inference.

Example

Problem: A environmental scientist collected water samples from a number of rivers in a region to explore whether the concentration of a harmful chemical in the water has changed since farming practises used locally have changed. They are considering the following choices of hypothesis test:

- A *t*-test, the most appropriate choice if the data follows a normal distribution.
- A *Wilcoxon Signed Rank test*, appropriate if the data is distributed symmetrically.



- (a) Use **Figure 1** to make a comment on the distribution of chemical concentrations.
- (b) Comment on the appropriateness of each choice of hypothesis test suggested for this study.

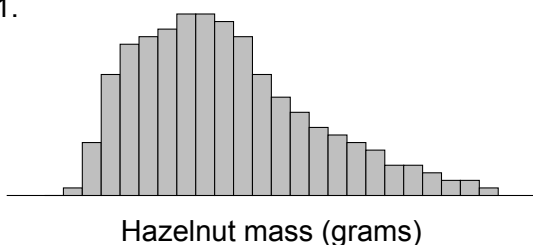
Solution:

- (a) Figure 1 suggests that the population distribution of concentrations of the chemical in river sites may be bimodal.
- (b) Since the histogram suggests the population distribution of chemical concentration may be symmetrical but not normal, a Wilcoxon Signed Rank test would be appropriate whilst a *t*-test would not.

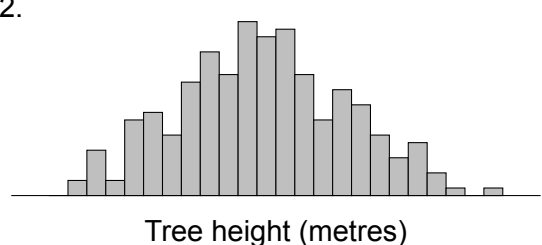
Exercise 1.7

Comment on each population distribution using the simplified histograms, which show data from samples.

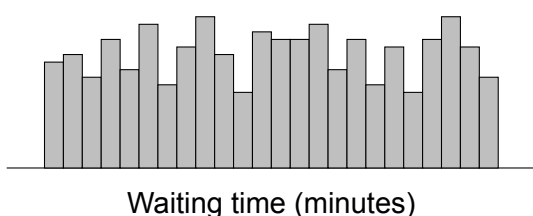
1.



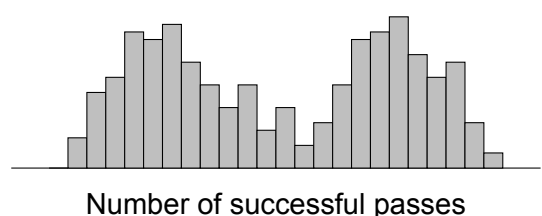
2.



3.



4.



1.8 Comparing Distributions

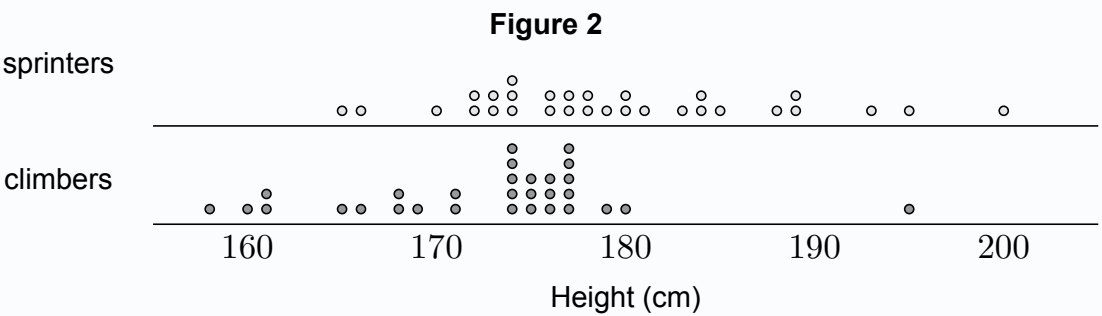
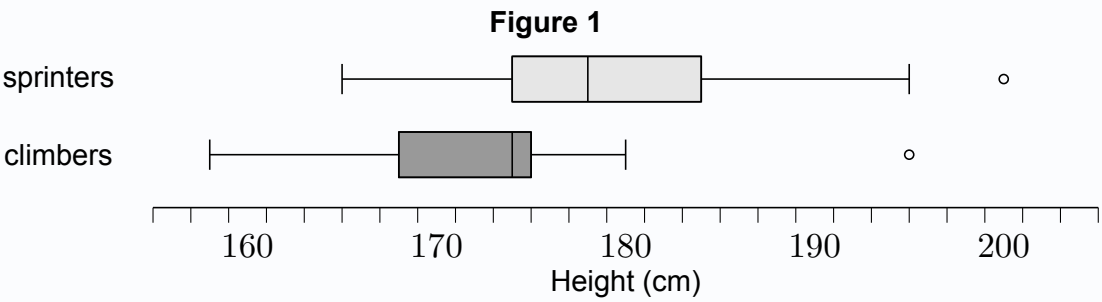
When comparing two or more samples of data, a good starting point is to create suitable plots and obtain summary statistics. Differences and similarities should be noted, considering the location, spread and shape of the distributions of each group.

Example

Problem: Within the sport of professional cycling, it is common for riders to specialise in certain types of cycling. *Sprinters* aim to win races that finish on flat road at high speed, whilst *climbers* aim to win races with an uphill finish. A researcher studying the characteristics of elite-level cyclists randomly selects samples of 30 *sprinters* and 30 *climbers*, recording the heights of each. The results are shown summarised in **Table 1**.

Table 1							
variable	n	Min	Q1	Median	Q3	Max	IQR
sprinters	30	165	174	179	184	200	10.5
climbers	30	158	168	174	177	195	9

Box plots for the two groups of cyclists are shown below in **Figure 1**, and dot plots in **Figure 2**.



- (a) Use **Figure 1** to make two comments comparing the heights of sprinters and climbers.
- (b) Use **Figure 2** to comment on the shape of the distribution of the heights of climbers.

Solution:

- (a) The sample median height of the sprinters was greater than that of the climbers.
The larger interquartile range suggests that the population heights of sprinters are more varied.
- (b) The dot plot suggests the underlying distribution of heights for climbers is negatively skewed.

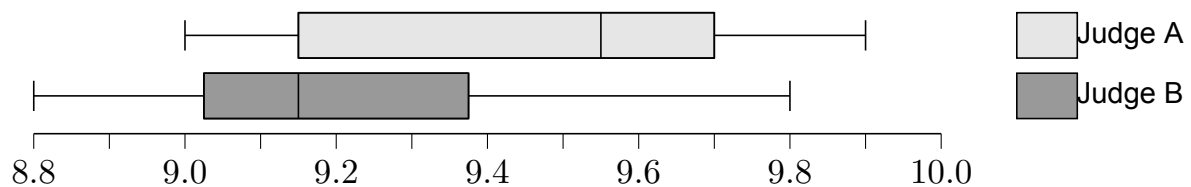
Note that this data is based on the heights of *samples* sprinters and climbers. Without the use of *statistical inference* techniques covered later in the course, it is only possible to *conjecture* rather than ever being able to make any definitive claims about the heights of the full *populations* of sprinters and climbers.

Exercise 1.8

1. Two models of coffee machines produced by the brand *DOLCAFÉ* are supposed produce a white coffee with the same volume of liquid each time, but natural variation means the actual amount dispensed is not quite the same every time. The volumes dispensed, in ml, are recorded for random samples from each model.

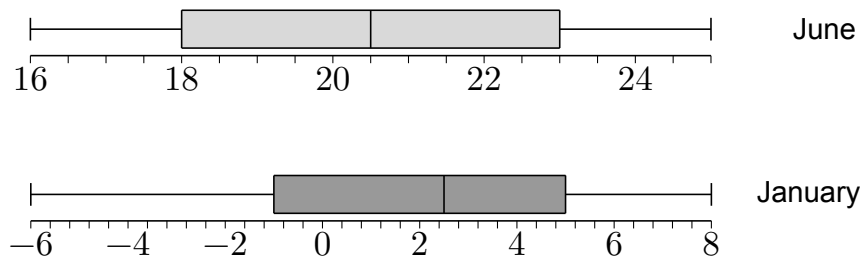
Model A	295	298	299	301	303	304	309
Model B	282	294	300	302	306	312	320

- (a) Determine the sample mean and standard deviation for each model.
 (b) Make two comments comparing the amount of coffee dispensed by the two models.
2. The scores awarded by two judges in a diving competition are recorded and displayed in box plots, shown below.



Using the box plots, make two comments comparing the scoring of the two judges.

3. The temperature at 2pm was noted for 14 randomly selected days in January and June at a local tourist spot and box plots drawn to display the results.



- (a) Using the box plots, make two comments comparing temperatures in January and June.
 (b) Explain why the box plots may be misleading.
4. The manager of a lemonade bottling plant is interested in comparing the performances of two production lines, one of which has only recently been installed. For each line she selects 10 one-hour periods at random and records the number of crates completed in each hour. The table below gives the results:

Production Line	Number of crates completed per hour									
New	78	87	79	82	87	81	85	80	82	83
Old	74	77	78	70	87	83	76	78	81	76

- (a) Draw dot plots to represent the data.
 (b) Calculate the mean and standard deviation for each sample.
 (c) Make two comments comparing the two lemonade production lines.

Review Exercise

1. A music fan wishes to investigate the characteristics of songs that are popular on a well-known music streaming app. She is able to obtain data on the thousand most-streamed songs of all time, a small selection of which are displayed in the table below.

Track	Artist	Released	Length (m:s)	No. of Streams
Creep	Radiohead	1992	3:59	1271293243
Blank Space	Taylor Swift	2014	3:52	1355959075
Something In The Way	Nirvana	1991	20:37	368646862
Nonsense	Sabrina Carpenter	2022	2:43	342897938
Drivers License	Olivia Rodrigo	2021	4:02	1858144199
Iris	The Goo Goo Dolls	1998	4:50	1284942608

- (a) State which of the five variables contain:
- Categorical data.
 - Discrete data.
 - Continuous data.
- (b) If it was suspected that one of the tracks is an outlier in terms of length, state what could be calculated in order to determine whether it is in fact an outlier.
2. A sports clothing store is seeking to better understand its sales of running shoes. For a random sample of twelve sales of pairs of running shoes, the sizes are recorded as follows:

4 7 9 11 6 9 8 4 10 9 6 5

- Construct a dot plot to display the data.
 - Find the median and interquartile range for the sizes of running shoes sold.
 - Calculate the mean and standard deviation for the sizes of running shoes sold.
 - State the type of data recorded.
3. The back-to-back stem-and-leaf diagram below shows the heights of trees in two areas of woodland.

Woodland A					Woodland B			
			5	2	1	3	3	8
		7	4	3	0	2	4	7
9	7	7	2	4	2	5		
	7	6	3	5				

- Suggest one improvement needed for the diagram.
 - Make one comment comparing the heights of trees in the two areas of woodland.
4. The petal lengths (in centimetres) for samples of flowers from two species of iris flowers is recorded: *iris setosa* and *iris versicolor*. The data is summarised in the table below using statistical software:

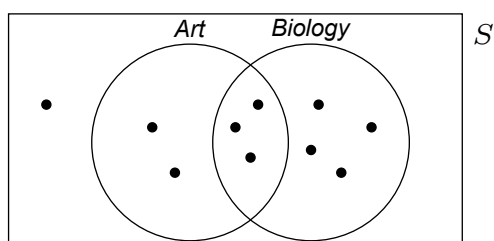
species	n	min	Q1	Q2	Q3	max
<i>iris setosa</i>	50	1.00	1.40	1.50	1.58	1.90
<i>iris versicolor</i>	50	3.00	4.00	4.35	4.60	5.10

- For the species *iris setosa*, determine whether the dataset contains any outliers.
- Make two comments comparing the petal lengths of the two species of iris.

2

Further Probability Theory

Suppose ten pupils in a class are asked whether they are taking Art, and whether they are taking Biology, and the results are displayed in both a Venn diagram and a table.



	Biology	$\overline{\text{Biology}}$
Art	3	2
$\overline{\text{Art}}$	4	1

From both the table and the Venn diagram, it can be seen that, of the ten pupils, five are taking Art. Therefore, the probability of randomly selecting a pupil from the class and that pupil turning out to be taking Art can be stated:

$$P(\text{Art}) = \frac{5}{10} = \frac{1}{2}$$

Now suppose a pupil is chosen at random and, before it is known whether or not they are taking Art, they reveal that they are taking Biology. The probability that the pupil is also taking Art is no longer $\frac{1}{2}$, since only three of the seven pupils taking Biology are taking Art. The probability now being sought is the probability that they are taking Art *given that they are taking Biology*. This is called a *conditional probability*, and can be notated as:

$$P(\text{Art} \mid \text{Biology}) = \frac{3}{7}$$

Conditional probability is fundamental to many of the statistical analysis techniques that will be introduced.

2.1 Conditional Probability from Tables and Diagrams

Example

Problem: A medical trial for drug treating a minor illness has patients given either the drug or a placebo, with a test after one month of treatment determining whether or not each patient has fully recovered.

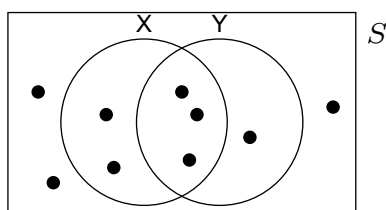
	drug	placebo
fully recovered	42	18
not fully recovered	8	12

A patient is selected at random to be interviewed. State $P(\text{fully recovered} \mid \text{placebo})$.

Solution: $P(\text{fully recovered} \mid \text{placebo}) = \frac{18}{30} = \frac{3}{5}$

Exercise 4.1

1. The Venn diagram below shows equally likely outcomes in a sample space and events X and Y . State:



- (a) $P(X)$ (e) $P(Y)$
 (b) $P(X|Y)$ (f) $P(Y|X)$
 (c) $P(X \cap Y)$ (g) $P(Y \cap X)$
 (d) $P(X \cup Y)$ (h) $P(Y \cup X)$

2. A basketball coach records the number of 2-pointer, 3-pointer and free-throw scoring attempts over the course of a season, broken down by whether those attempts scored or missed. An attempt is chosen at random to be analysed. Find:

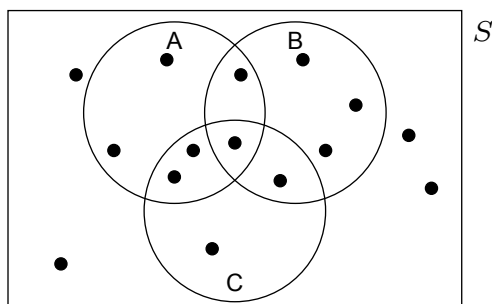
- (a) $P(\text{scored})$ (c) $P(3\text{-pointer})$
 (b) $P(\text{scored} \mid 3\text{-pointer})$ (d) $P(3\text{-pointer} \mid \text{scored})$

	2-pointer	3-pointer	free throw
scored	72	30	44
missed	60	52	12

3. On mornings when it is raining, the probability that a red squirrel will leave its drey is 0.4. On mornings when it is not raining, the probability that it will leave its drey is 0.9. It is rainy on 30% of mornings. State the value of:

- (a) $P(\text{rainy})$ (c) $P(\text{leaves} \mid \text{rainy})$ (e) $P(\text{leaves} \mid \text{not rainy})$
 (b) $P(\text{not rainy})$ (d) $P(\text{does not leave} \mid \text{rainy})$ (f) $P(\text{does not leave} \mid \text{not rainy})$

4. The Venn diagram below shows equally likely outcomes in a sample space and events A , B and C . Determine:



- (a) $P(A)$ (g) $P(A|C)$
 (b) $P(B)$ (h) $P(C|B)$
 (c) $P(C)$ (i) $P(\overline{B}|C)$
 (d) $P(A \cap B)$ (j) $P(C|\overline{A})$
 (e) $P(B \cup C)$ (k) $P((A \cap B)|C)$
 (f) $P(A \cap B \cap C)$

2.2 The Conditional Probability Formula

Where a conditional probability cannot be intuitively stated, the formula for conditional probability may be used:

Conditional Probability:
$P(A B) = \frac{P(A \cap B)}{P(B)}$

It may be helpful to note that $P(\bar{A}|B) = 1 - P(A|B)$.

Example

Problem: Given events V and W such that $P(V) = 0.7236$ and $P(V \cap W) = 0.1046$, find $P(W|V)$.

Solution: $P(W|V) = \frac{P(W \cap V)}{P(V)} = \frac{0.1046}{0.7236} = 0.1446$

Exercise 4.2

1. Find, by first writing out the conditional probability formula for these events:

- (a) $P(A|B)$ given $P(B) = 0.8$ and $P(A \cap B) = 0.16$.
- (b) $P(\text{red} | \text{blue})$ given $P(\text{red} \cap \text{blue}) = 0.1526$, $P(\text{blue}) = 0.3147$ and $P(\text{red}) = 0.2142$.
- (c) $P(X|Y)$ given $P(Y) = \frac{1}{2}$ and $P(Y \cap X) = \frac{1}{3}$.
- (d) $P(M|N)$ given $P(\bar{N}) = 0.6137$, $P(N \cap M) = 0.1249$ and $P(M \cup N) = 0.4511$.
- (e) Using (a) to (d), state:

- i. $P(\bar{A}|B)$
- ii. $P(\bar{\text{red}} | \text{blue})$
- iii. $P(\bar{X}|Y)$
- iv. $P(\bar{M}|N)$

2. Given events V and W such that $P(V \cap W) = 0.28$ and $P(W) = 0.4$, find $P(\bar{V}|W)$.

3. Numbered cards of different colours are selected at random from a pack during a game such that the probability of a selected card being *yellow* is 20%, and the probability of it being *yellow and a five* is 4%. Find the probability that a selected card is a five *given that it is a yellow*.

4. A football team knows from looking at their past results that they concede at least two goals in $\frac{2}{5}$ of all games. In $\frac{1}{4}$ of all of their games they concede at least two goals and lose the game. Find the probability that, given that they concede at least two goals in a game, they lose.

5. Given events M and N such that $P(M) = 0.6$, $P(N) = 0.4$ and $P(M \cup N) = 0.9$, find $P(M|N)$.

6. For events V and W , $P(V) = 0.6$, $P(V|W) = 0.8$ and $P(V \cap W) = 0.5$. Calculate $P(V \cup W)$.

2.3 Independent Events

The probability of tossing a coin and it landing tails is $\frac{1}{2}$. The probability of it landing tails *given that it is a weekday*, notated as $P(\text{tails} \mid \text{weekday})$, is *still* $\frac{1}{2}$. Since it being a weekday or not makes no difference to the probability of a coin showing tails, the two events are said to be *independent*.

A and B are Independent if:

$$P(A|B) = P(A)$$

The probability a randomly selected card from a standard pack of playing cards being a Diamond is $\frac{1}{4}$. The probability the card is a Diamond *given that we discover the card picked is red*, notated as $P(\text{Diamond} \mid \text{red})$, is $\frac{1}{2}$. Since knowing the card picked is red changes the probability of it being a Diamond, the two events are said to be *not independent*.

A and B are not Independent if:

$$P(A|B) \neq P(A)$$

Example

Problem: Given $P(D) = 0.25$, $P(E) = 0.6$ and $P(D \cap E) = 0.15$, show that E and D are independent.

Solution:

$$P(D|E) = \frac{P(D \cap E)}{P(E)} = \frac{0.15}{0.6} = 0.25$$

Since $P(D|E) = P(D)$, events D and E are independent.

Exercise 4.3

- For each pair of events X and Y , state whether you would expect them to be independent or not.
 - A person is selected at random in a survey:
 - X: They were born in January.
 - Y: They work in an office.
 - A mechanic is inspecting a car's tyre treads:
 - X: The front tyres have low tread.
 - Y: The rear tyres have low tread.
- For each pair of events, determine whether or not they are independent.
 - A and B given:
 - $P(A) = 0.3$
 - $P(B) = 0.8$
 - $P(A \cap B) = 0.16$
 - X and Y given:
 - $P(X) = \frac{3}{4}$
 - $P(Y) = \frac{7}{12}$
 - $P(X \cap Y) = \frac{7}{16}$
 - M and N given:
 - $P(M \cup N) = 0.67$
 - $P(\overline{M}) = 0.55$
 - $P(M \cap N) = 0.18$
- A music tutor taught 40 students last year, each learning one of violin, trumpet or saxophone, and each have lessons either midweek or at the weekend. A student is chosen at random to take a survey to give feedback. Let V represent that the student chosen took violin lessons, T that they took trumpet lessons and W that they had lessons at the weekend.

	violin	trumpet	saxophone
midweek	12	11	9
weekend	3	4	1

(a) Show that V and W are independent.

(b) Show that T and W are not independent.

2.4 The Multiplication Rule for the Intersection

The conditional probability rule can be rearranged to allow the **intersection** of two events to be calculated using a conditional probability. For independent events, since $P(B|A) = P(B)$, this simplifies further.

Multiplication rule for any A and B :

$$P(A \cap B) = P(A) \times P(B|A)$$

Multiplication rule for Independent A and B :

$$P(A \cap B) = P(A) \times P(B)$$

Example

Problem: Studies of a species of plant in the UK reveal that 8% of the plants have a particular genetic mutation. Amongst those plants with the genetic mutation, 35% produce yellow flowers. If a plant of this species in the UK is selected at random and observed, determine the probability that it will both have the genetic mutation and produce yellow flowers.

Solution:

$$P(\text{mutation} \cap \text{yellow}) = P(\text{mutation}) \times P(\text{yellow} | \text{mutation}) = 0.08 \times 0.35 = 0.028$$

Exercise 4.4

1. Calculate each probability by first *writing out the formula required*:

(a) $P(X \cap Y)$ given:

- $P(X) = 0.4$
- $P(Y) = 0.26$
- $P(Y|X) = 0.18$

(b) $P(\text{blue} \cap \text{yellow})$ given:

- $P(\overline{\text{blue}}) = \frac{2}{3}$
- $P(\text{yellow}) = \frac{1}{2}$
- $P(\text{blue} | \text{yellow}) = \frac{3}{4}$

(c) $P(E \cap F)$ given:

- E and F are independent
- $P(E) = 0.82$
- $P(\overline{F}) = 0.7$

2. Events A and B are such that $P(A) = 0.4$, $P(B) = 0.7$ and $P(A|B) = 0.56$. Determine the value of:

(a) $P(A \cap B)$

(c) $P(B|A)$

(b) $P(A \cup B)$

3. Independent events X and Y are such that $P(X) = 0.4$ and $P(Y) = 0.8$. Determine the value of $P(X \cup Y)$.

4. A large box in a maths classroom contains a number of calculators, with 85% of them made by Arizona Instruments and the rest made by Calculo. 70% of the calculators overall are working, whilst 80% of those made by Calculo are working. Determine the probability that a calculator selected at random from the box is a working Calculo calculator.

5. As part of an analysis of a large company's vulnerability to a cyber attack, it is found that 16% of company-issued employee laptops have out-of-date antivirus protection, whilst 24% are only protected by a weak password. Of those with out-of-date antivirus protection, 60% are only protected by a weak password. If a laptop is selected at random, determine the probability that it both has out-of-date antivirus protection and is only protected by a weak password.

6. Given:

- $P(\overline{\text{pink}}) = 0.5$
- $P(\overline{\text{pink}} | \overline{\text{blue}}) = 0.4$
- $P(\overline{\text{pink}} \cap \overline{\text{blue}}) = 0.32$

Find the value of $P(\text{blue} \cap \text{pink})$.

2.5 Bayes Theorem and Reversing the Condition

In 1763 the Reverend Thomas Bayes, also a statistician, published “*An Essay towards solving a Problem in the Doctrine of Chances*”. In it, he stated a formula that has come to be known as *Bayes’ Theorem*, which can be applied in a particular, specialised way to enable existing knowledge to be updated as new information is collected. In turn, this has resulted in an approach towards statistical inference that has been growing since the mid-20th century called *Bayesian statistics*, substantially made more feasible by modern computing power.

Whilst knowledge of Bayes’ Theorem is required in the Advanced Higher Statistics course, Bayesian *statistics* is *not* part of this course, and so will not be covered here. If you would like to find out more about Bayesian statistics, and whether it may be relevant to you in the future, it is recommended that you discuss this with your teacher at a later point in the course. University modules introducing Bayesian statistics are typically available as part of an undergraduate degree in Mathematics or Statistics.

Bayes’ Theorem can be derived directly from the formulae already introduced in this chapter. Note that:

$$\begin{aligned} P(A \cap B) &= P(A|B) \times P(B) \\ P(B \cap A) &= P(B|A) \times P(A) \end{aligned}$$

Since $P(A \cap B) = P(B \cap A)$:

$$P(A|B) \times P(B) = P(B|A) \times P(A)$$

Dividing both sides by $P(B)$ gives:

Bayes’ Theorem:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Bayes’ Theorem provides a the link between a conditional probability and its *reverse*. Using it to find a conditional probability given its reverse is sometimes referred to as *reversing the condition*.

Example

Problem: Given events V and W such that $P(V|W) = 0.3$, $P(W) = 0.8$ and $P(\bar{V}) = 0.1$, find $P(W|V)$.

Solution:
$$P(W|V) = \frac{P(V|W) \times P(W)}{P(V)} = \frac{0.3 \times 0.8}{0.9} = \frac{4}{15}$$

Careful reading of text may be required to identify when a probability provided is a *conditional* probability.

Example

Problem: In a neighbourhood, 14% of homes have a garage, whilst 62% of homes have a garden. Of those homes with a garage, 87% have a garden. If a home chosen at random has a garden, determine the probability that it has a garage.

Solution:
$$P(\text{garage} | \text{garden}) = \frac{P(\text{garden} | \text{garage}) \times P(\text{garage})}{P(\text{garden})} = \frac{0.87 \times 0.14}{0.62} = 0.1965$$

Exercise 4.5

1. For each, first write out the formula for Bayes' Theorem explicitly:

(a) Events X and Y are such that:

- $P(X) = 0.7$
- $P(Y) = 0.6$
- $P(Y|X) = 0.5$

Calculate $P(X|Y)$.

(b) For events V and W :

- $P(W) = \frac{4}{5}$
- $P(\overline{V}) = \frac{1}{3}$
- $P(W|V) = \frac{3}{4}$

Determine $P(V|W)$.

2. A bag contains numbered counters of various colours, a quarter of which are red. Three-eighths of the counters are numbered 1, of which two-fifths are red. If a counter is selected at random and it is red, determine the probability that it is numbered 1.

3. Research has shown that 1.7% of people possess a specific gene. A prototype test for the presence of the gene in a person is being developed, and it shows as *positive* for the gene for 7% of everyone people tested. It is later established that the probability a test shows as positive for a person known to possess the gene is 0.9. If a person is selected at random to take the test, and it shows positive, determine the probability that they actually do possess the gene.

4. Given that $P(M) = \frac{2}{3}$, $P(N) = \frac{3}{4}$ and $P(N|M) = \frac{3}{5}$:

(a) Calculate $P(M|N)$.

(b) Calculate $P(\overline{M}|N)$.

5. An MOT test is an annual examination of cars and other vehicles required to check they are roadworthy. 83% of the vehicles a local garage performs MOT tests on are cars, whilst the rest are vans. Considering only the first MOT test performed on each vehicle, before any repairs are carried out:

- 15% of all tests result in a fail
- Cars account for 86% of all failed tests.

Determine the proportion of cars that *pass* their first MOT.

6. A region has access to a local radio stations, *Pacific 525*, as well as a number of national stations. When a resident of the region directs their car radio to automatically find a station, the probability that it will connect to *Pacific 525* is 0.64, otherwise it will instead connect to a national station. 72% of the time, when their car automatically connects there will be music playing. When their car radio connects to *Pacific 525* the probability music will be playing is 0.84. If the car radio connects to a station and music is playing, determine the probability that the station is *Pacific 525*.

7. Events E and F are such that:

- $P(\overline{F}|E) = 0.23$.
- $P(F) = 0.87$.
- $P(\overline{E}) = 0.09$.

Calculate $P(\overline{E}|F)$.

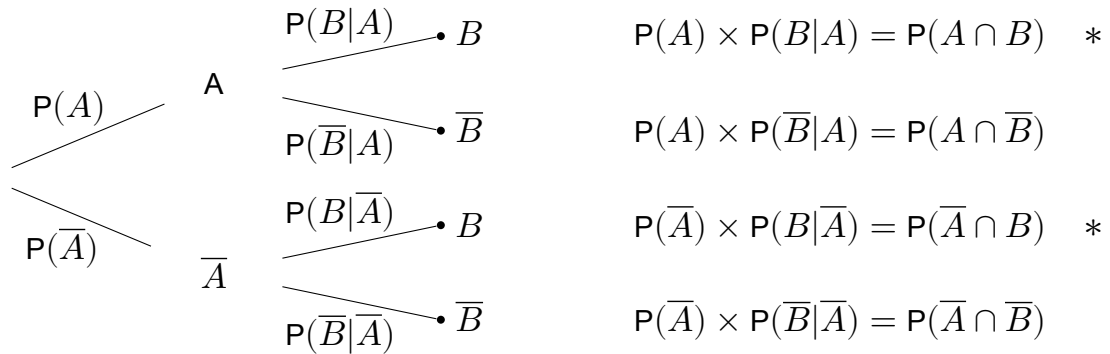
2.6 Total Probability and Tree Diagrams

The *law of total probability* for X conditional on n mutually exclusive, exhaustive events Y_i is:

$$P(X) = \sum_{i=1}^n P(Y_i) \times P(X|Y_i)$$

It may help to recognise the right hand side of the equation as containing the *multiplication rule for the intersection*: the total probability of X is obtained by adding together the probabilities of all (disjoint) intersections in which X occurs.

The tree diagram below shows the appropriate notation for each stage, in which first event A and then event B are considered:



The *total probability* of B , $P(B)$, can be calculated as:

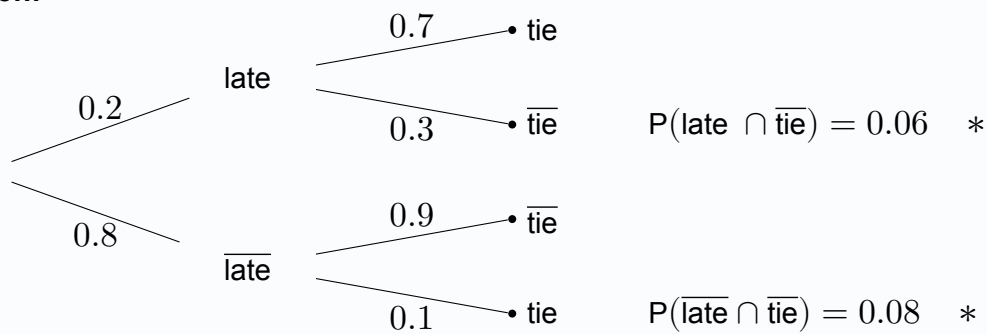
$$\begin{aligned}
 P(B) &= P(A \cap B) + P(\bar{A} \cap B) \\
 &= P(A) \times P(B|A) + P(\bar{A}) \times P(B|\bar{A})
 \end{aligned}$$

The advantage of the use of tree diagrams, for questions where they are suitable, is that they can offer a more intuitive and visual approach to tackling problems than solely using formulae. If they are used, they must be annotated carefully with events and probabilities, and accurate probability notation must still be used when giving answers.

Example

Problem: A teacher observes that one of their pupils is late to school 20% of the time, and when this happens they have their tie with them 70% of the time. When they are on time they have their tie 90% of the time. Construct a tree diagram and calculate the probability that, on a randomly chosen day, they have no tie.

Solution:



$$P(\overline{\text{tie}}) = 0.06 + 0.08 = 0.14$$

Exercise 4.6

Construct a tree diagram for each question.

1. A UK-based car company makes a special edition model with a colour choice of blue or yellow. It sells cars both within the UK as well as exporting to Europe.
 - 72% of the special edition models manufactured by the company are blue.
 - The rest are yellow.
 - 84% of the blue cars are sold within the UK and 95% of the yellow cars are sold within the UK.

Calculate the probability that a randomly selected special edition model is sold within the UK.

2. A large number of trees are planted as part of a tree-planting project.
 - Three-quarters of trees are oak.
 - The rest are alder trees.
 - The probability an oak tree failing is 0.356.
 - The probability of failure for an alder tree is 0.234.

Calculate the probability that a tree randomly selected for observation survives.

3. The star striker for a football team has a probability of 0.7 of scoring in any given game. When they do, the probability that the team goes on to win is 0.8, whilst when they don't the probability of a win drops to 0.45. Calculate the probability that the team win a randomly chosen game.
4. When Betty's hears a song on the radio:
 - The probability that it is a song she knows is 0.8.
 - When she knows it, the probability that it is a song she likes is 0.6.
 - When she doesn't know it, the probability she will like it is 0.3.

Determine the proportion of songs Betty hears on the radio that she likes.

5. An obstacle course consists of three parts, with each part needed to be completed successfully for a competitor to be allowed to move on to the next. 30% of competitors fail to complete the first part, and of those who make it onto the second part only 40% manage to make it onto the third part. Of those that do make it on to the third part, however, 65% successfully complete it. Calculate the probability that a competitor makes it onto the third part.
6. As part of a study examining the survival rates of hedgehogs over winter hibernation periods, a number of hedgehogs are captured, weighed, tagged and then released. Depending on their weight they are classed as having a "good weight", an "average weight" or being "underweight", with 18%, 76% and 6% observed respectively in each category. From the study it is found that the probability of survival for each category is 91%, 72% and 35% respectively. Assuming that the distribution of hedgehog weights and probabilities of survival can be taken as representative of hedgehog populations in general, find the proportion of hedgehogs that do not survive winter hibernation.
7. The probability of being dealt a pair (such as two fives) in an opening hand in a game of poker is 0.06. If this happens, a particular player knows that based on their game history they have a probability of 0.83 of winning the hand, whilst otherwise they have a probability of 0.36. Calculate the probability they win a hand.

2.7 Bayes Theorem and Tree Diagrams

In this chapter, Bayes' Theorem has already been used to calculate the *reverse* of a conditional probability: calculating $P(B|A)$ when $P(A|B)$ is known. However, this process can be made more intuitive and more readily extended to more complex problems by using tree diagrams combined with the formula for conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A key misunderstanding often made is that a high value for $P(B|A)$ must lead to a similarly high value for $P(A|B)$. The following example demonstrates that this is not always the case, and gives an insight into a common problem that must be considered whenever medical testing is being undertaken, especially for rare conditions or diseases:

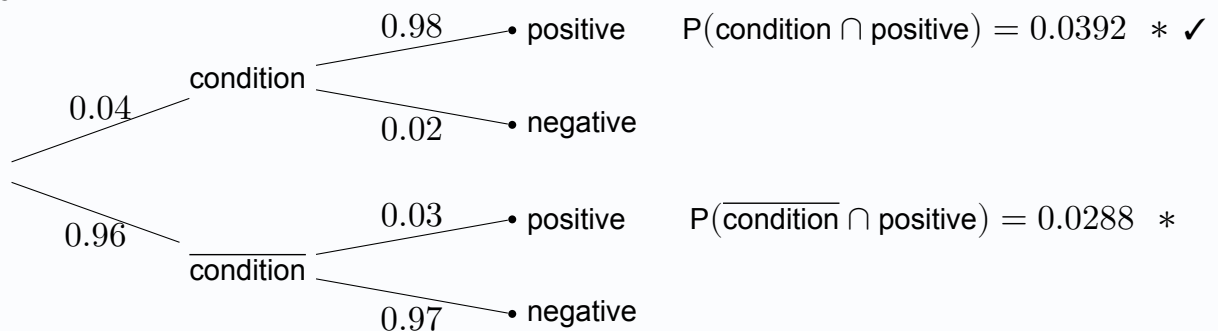
Example

Problem: A medical test for a condition has a *sensitivity* of 98%, meaning that when the patient *has the condition*, the test will correctly return a *positive* result with a probability of 0.98. This means there is a *false negative* rate of 2%. The test has *specificity* of 97%, meaning that when the patient *does not* have the condition, the test will correctly return a *negative* result with a probability of 0.97. This means there is a *false positive* rate of 3%. The *prevalence* in the general population is 4%, meaning only 4% of people have the condition.

If a person is selected at random from the general population to be tested for the condition:

- Calculate the probability that test comes back *positive*.
- Given that the test comes back positive, determine the probability that they do actually *have* the condition.

Solution:



(a) $P(\text{positive}) = 0.0392 + 0.0288 = 0.068$

(b) $P(\text{condition} | \text{positive}) = \frac{P(\text{condition} \cap \text{positive})}{P(\text{positive})} = \frac{0.0392}{0.068} = 0.5765$

✓
* + *

It may be surprising that a test which appears so effective can lead to such a high proportion of false positives, and calculations such as this should be considered when healthcare professionals establish testing procedures. In practice, the false positive rate can be drastically reduced by deciding when a test may be appropriate. GPs, for example, will ask questions about a patient's symptoms and medical history to determine whether they have reason to suspect that the chance that they have the condition is significantly greater than a random member of the population, and therefore whether a test is appropriate.

Exercise 4.7

Construct a tree diagram for each question.

1. A company sends customers' orders by either regular or express delivery.
 - 35% of orders are sent by express delivery.
 - Of those sent by express delivery, 98% arrive undamaged,
 - 91% of the orders sent by regular deliveries arrive undamaged.Customers are requested to submit a form online to confirm whether their order arrived damaged or undamaged.
 - (a) Determine the probability that a form received by the company states that an order arrived damaged.
 - (b) Given that an order arrived damaged, calculate the probability that the order was sent by express delivery.
2. Anti-virus software is being developed to detect whether attachments sent by email are malicious or safe, and flag those that are malicious. In its current state it can correctly flag that an attachment is malicious 92% of the time. It incorrectly flags safe attachments 2% of the time. A company is interested in using the software, and 1.2% of all attachments emailed to their employees are actually malicious.
 - (a) Determine the proportion of attachments that are flagged as malicious.
 - (b) Given that an attachment is flagged, calculate the probability that is malicious.
3. A pupil sitting a multiple choice test is able to answer $\frac{3}{4}$ of the questions and guesses the rest. They know from trying practice tests that the probability that are correct in the questions they are able to answer is 0.9, whilst they have a one-in-five chance of getting a correct answer by guessing. Determine the probability that a question which they get wrong was one that they guessed.
4. A marketing campaign for a new product is estimated to have been seen by three times as many people through its social media advert as by its promotional email. The probability that seeing the social media advert leads to a person visiting their website is 0.037, whilst for the promotional email that figure drops to 0.021.
 - (a) Determine the probability that someone who sees the marketing campaign in either form visits the website.
 - (b) Determine the probability that a visit to the website due to the marketing campaign came from that person seeing the social media advert.
5. A sports team is given an extra rest day half of the time after winning a game, a quarter of the time after drawing a game and only a tenth of the time when they lose a game. They win 55% of their games, draw 25% of their games and lose the rest. Given that they are given an extra rest day, find the probability that they lost the last game.
6. All precision-engineered parts at a small factory undergo two inspections as part of quality control. During Inspection A, 8% of parts are flagged for a more thorough inspection during Inspection B. Of those that are more thoroughly inspected during Inspection B, 15% are scrapped due to flaws in the manufacturing process. 3% of those that had not been flagged initially are also then scrapped during Inspection B as flaws are spotted that were initially missed. Finally, of those parts that have not been scrapped, 1% are set aside for stress-testing of the final products and never leave the factory. Given that a part doesn't leave the factory, find the probability that it was due to a flaw.
7. For medical tests for conditions that are known to have the given specificity and sensitivity, as well as the prevalence of the condition, calculate the probability that a randomly tested person has the condition, given that they test positive:
 - (a) Sensitivity = 94%, Specificity = 89%, Prevalence = 9%
 - (b) Sensitivity = 98%, Specificity = 99%, Prevalence = 1%

Review Exercise

1. Events E and F are such that:

- $P(F \cap E) = 0.7$
- $P(\overline{E}) = 0.2$
- E and F are exhaustive.

Calculate $P(E|F)$.

2. A second-hand online computing retailer has 40 laptops in stock, broken down by whether they use Windows or Mac, and the grades A , B and C depending on the condition they are in:

	A	B	C
Windows	7	18	5
Mac	4	5	1

A laptop is selected at random.

W is the event that the laptop selected uses Windows.

A is the event that the laptop selected is in grade A condition.

B is the event that the laptop selected is in grade B condition.

(a) Find:

i. $P(W \cup A)$.

ii. $P(\overline{W} \cap B)$.

80% of the Windows laptops and 90% of the Mac laptops come with the original power supply.

(b) Calculate the probability that a randomly selected laptop comes with its original power supply.

(c) Determine the probability that a randomly selected laptop is a Mac laptop, given that it does not come with its original power supply.

3. At the start of each round of a board game played by four players, a random process determines who gets to go 1st, who goes 2nd, who goes 3rd and who goes 4th. Sam knows that when she plays against her two friends, she wins the round two-thirds of the time when she goes 1st, half of the time when she goes 2nd and only a fifth of the time when she goes either 3rd or 4th.

(a) Assuming that each round is independent of the result of previous rounds, calculate the proportion of rounds that Sam wins.

(b) Given that Sam lost the last round, calculate the probability that she went 1st.

4. For the events M and B : $P(M \cap B) = \frac{3}{25}$, $P(M) = \frac{1}{5}$ and $P(\overline{B}) = \frac{2}{5}$. Show that events M and B are independent.

5. As part of a machine learning project, a program is created that tries to correctly identify whether it is being shown a picture of a dog, a cat or a human. It has been tested and found to correctly identify a human with the probability 0.8, correctly identify a dog with the probability 0.6 and correctly identify a cat with the probability 0.55. When subsequently put to use by the public, the program is shown a picture of a human 60% of the time, a dog 25% of the time and a cat 15% of the time. Assuming the probabilities of successful detection are the same when used by the public as when under testing, calculate:

(a) the proportion of times that the program is incorrect

(b) the probability that the program was shown a cat, given that it is incorrect.

Mann-Whitney Rank Sum Test

Mann-Whitney Rank Sum Test

The parametric two-sample t -test for population means relies on an assumption that each population is normally distributed. Where this assumption cannot be safely made, the **non-parametric Mann-Whitney Rank Sum test** may be used to explore a difference in population **medians**. It is an example of a *permutation test*, which are those that can be approached by considering equally-likely possibilities.

Consider random samples of two mice of species A and four of species B, which will be ordered from lightest to heaviest, and then ranks from 1 to 6 assigned. Below are all 15 possible arrangements of ranks for the six mice, and the *sum of the ranks* obtained by the two mice of species A:

Arrangement	Rank Sum	Arrangement	Rank Sum	Arrangement	Rank Sum
AABBBB	$1 + 2 = 3$	ABBBAB	$1 + 5 = 6$	BBABAB	$3 + 5 = 8$
ABABBB	$1 + 3 = 4$	ABBBBA	$1 + 6 = 7$	BBABBA	$3 + 6 = 9$
ABBABB	$1 + 4 = 5$	BABBAB	$2 + 5 = 7$	BBBAAB	$4 + 5 = 9$
BAABBB	$2 + 3 = 5$	BBAABB	$3 + 4 = 7$	BBBABA	$4 + 6 = 10$
BABABB	$2 + 4 = 6$	BABBBA	$2 + 6 = 8$	BBBBAA	$5 + 6 = 11$

If the population distributions of mass for mice from species A and mice from species B are *identical* then the probability of obtaining for A a rank sum less than or equal to 4 (for example) by random chance is:

$$P(\text{rank sum obtained} \leq 4) = \frac{2}{15} = 0.1333$$

If it is assumed that the two population distributions of mice masses have the *same shape and variance* then hypotheses may be formed which focus on a possible difference in the *medians*, η_A and η_B , of the distributions. Hypotheses for this two-sample test may be one-tailed or two-tailed. Here, a one-tailed example is given:

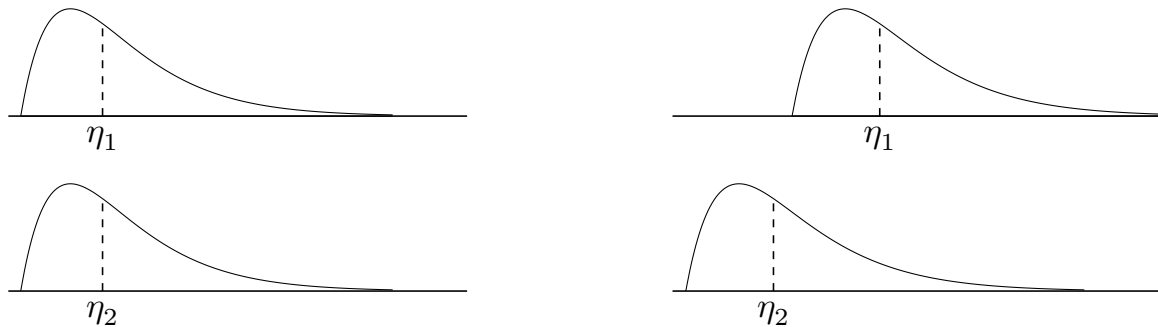
$$\left. \begin{array}{l} H_0 : \eta_A = \eta_B \\ H_1 : \eta_A < \eta_B \end{array} \right\} \begin{array}{l} \alpha = 0.05 \\ \text{One-tailed} \end{array}$$

The p -value of 0.1333 obtained about would not provide evidence to suggest, at the 5% level of significance, that the median mass of a mouse of species A is less than the median mass of a mouse of species B (since $0.1333 > 0.05$).

Since the process of listing possible arrangements would become excessively time-consuming for anything other than small samples, Mann-Whitney tests will be typically conducted using *critical values* from the Data Booklet, and *test statistics*. However it is also necessary to be able to perform the test *from first principles* as above, covered in more detail later in this chapter.

3.1 Mann-Whitney Rank Sum Test

Given two independent samples, the non-parametric **Mann-Whitney Rank Sum test** can be used to assess whether the two population distributions have different *medians*. It is typically used in place of a two-sample *t*-test where the two distributions are not *normal* and sample sizes are small. Instead a required assumption is that the population distributions have the *same shape and variance*, to test whether there is evidence of a difference in *locations*:



Same locations: $\eta_1 = \eta_2$

Different locations: $\eta_1 \neq \eta_2$

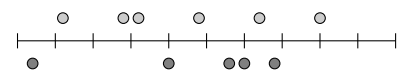
Conditions for valid use of the Mann-Whitney Rank Sum Test:

- The two distributions of the underlying populations have **the same shape and variance**.
- The data were obtained through **random and independent samples**

Consider two independent samples taken from populations *A* and *B* and the following hypotheses:

$$\left. \begin{array}{l} H_0 : \eta_A = \eta_B \\ H_0 : \eta_A \neq \eta_B \end{array} \right\} \begin{array}{l} \alpha = 0.05 \\ \text{Two-tailed} \end{array}$$

<i>A</i>	2.8	3.2	3.3	3.7	4.1	4.5
<i>B</i>	2.6	3.5	3.9	4.0	4.2	



To perform the test using a test statistic and critical value approach, first the samples are combined, ordered and ranked, with **rank sums** W_A and W_B obtained:

Values	2.6	2.8	3.2	3.3	3.5	3.7	3.8	4.0	4.1	4.2	4.5
Sample	B	A	A	A	B	A	B	B	A	B	A
Ranks	1	2	3	4	5	6	7	8	9	10	11

$$\begin{array}{l} W_A = 35 \\ W_B = 31 \end{array}$$

The test statistic, W , is the *smallest possible rank sum that can be obtained for the smaller sample*, here *B*. A time-consuming approach would be to reorder the values and rerank them to check if this produces a smaller rank sum W_B .

Values	4.5	4.2	4.1	4.0	3.8	3.7	3.5	3.3	3.2	2.8	2.6
Sample	A	B	A	B	B	A	B	A	A	A	B
Ranks	1	2	3	4	5	6	7	8	9	10	11

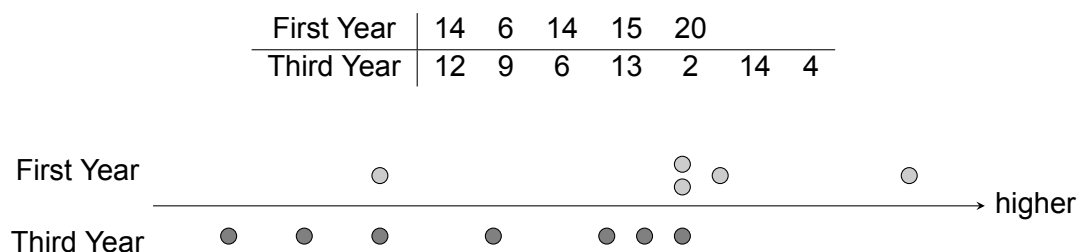
$$\begin{array}{l} W_A = 37 \\ W_B = 29 \end{array}$$

Hence, the test statistic can now be confirmed as $W = 29$. However this value can be obtained without reordering using the formula $m(m + n + 1) - W_m$, where m and n represent the sizes of the smaller and larger samples respectively:

$$5(5 + 6 + 1) - 31 = 29$$

Exercise 17.1

1. One year on World Maths Day (14th March) pupils across all the year groups in a school attempt a times tables speed challenge. A teacher checks the scores obtained by five randomly selected First Year pupils and seven randomly selected Third Year pupils. Their results are shown below.



A Mann-Whitney test is performed to assess whether the median score for First Year pupils is greater than that of Third Year pupils.

- (a) Use the data provided in the table to verify the test statistic of 20.5.
 - (b) Conclude the test at the 5% level of significance.
2. A highways official wants to compare two brands of paint used for road markings. Twenty similar locations are selected and at ten of them brand A is used and at the others brand B. The number of months the markings last for were recorded and the results shown below.

Brand A	35.6	36.1	37.0	35.8	34.9	34.9	36.0	37.8	36.6	36.5
Brand B	37.2	36.4	39.7	37.5	37.2	40.5	38.8	38.2	37.7	36.6

Perform a Mann-Whitney test, using the rank sum for Brand A of 65.5, to determine whether there is a difference between the two brands' durability.

3. The durability of two brands of power lawn mowers are to be compared. For independent random samples from each brand, the number of years each lawn mower lasted before breaking down beyond repair is recorded:

Brand X	2.3	3.7	1.9	6.8	3.5	4.4
Brand Y	5.9	3.8	6.4	5.6	4.9	5.9

Test at the 10% level of significance whether the data indicates that there is a difference in median number of year that power lawn mowers from each brand last.

4. The contents of a random sample of six pots of 'Extrafrute' jam were measured. The results, in grams, were:

342 354 349 350 347 356

The results for a random sample of eight pots of 'Jambo' jam were:

340 341 348 342 346 347 342 344

Use a non-parametric test to assess the evidence that the pots of the 'Extrafrute' brand have more jam in them.

5. A farmer decides to test a new diet for egg production. A group of twelve hens is randomly divided into two groups of six. One group is fed the new diet and the other the old. The egg yields recorded in the first year of the test are displayed in the stem-and-leaf diagram below.

Old Diet		New Diet	
7	22		
0	23		
4 2	24	5 8	
0	25	5 8 9	
1	26	0	

24|8 represents 248 grams

Use a Mann-Whitney test at the 5% level to test the hypothesis that the new diet leads to improved egg production.

6. The times (in minutes) it took six randomly selected field mice to learn to run a simple maze and the times it took six randomly selected house mice to learn to run the same maze are given.

Field Mice	House Mice
18	24
24	16
12	10
20	19
13	14
15	16

Stating an assumption required, use a non-parametric test to determine if there is a difference in learning rate between the two types of mice.

7. An investigation was conducted into the dust content in the flue gases of two types of solid-fuel boilers. Thirteen boilers of type A and nine boilers of type B were used under identical fuelling and extraction conditions. Over a similar period, the following quantities, in grams, of dust were deposited in similar traps inserted in each of the twenty-two flues.

Type A	73.1	56.4	82.1	67.2	55.2	78.7	75.1	48.0	60.6	63.1	53.3	55.5	61.5
Type B	53.0	39.3	55.8	46.0	56.4	58.8	41.2	66.6	58.9				

A non-parametric test for a difference in population medians was performed at the 5% level of significance.

Mann-Whitney rank sum test

data: TypeA and TypeB

alternative hypothesis: true location shift is not equal to 0

W for TypeB = 134.5 p-value = 0.0416

- State the suitable null and alternative hypotheses the the test conducted.
- By performing the usual check and the rank sum for Type B given in the computer output, obtain the required test statistic.
- Use the answer from (b) to complete the hypothesis test and interpret the result.
- State three assumptions required for the test performed to be valid.

3.2 Normal Approximation to the Mann-Whitney Rank Sum Test

The Data Booklet only provides Mann-Whitney critical values for sample sizes where $n \leq 20$ and $m \leq 20$. When dealing with larger samples, where $m, n > 20$, a Mann-Whitney Rank Sum Test may still be performed using a **normal approximation**. Page 16 of the Data Booklet provides the required properties of the distribution of the test statistic, W :

$$E(W) = \frac{1}{2}m(m+n+1) \quad \text{and} \quad V(W) = \frac{1}{12}mn(m+n+1)$$

Since here a *discrete* distribution is being approximated by the *continuous* normal distribution, a **continuity correction** is required. If the test statistic W is taken as the lowest possible rank sum W_m for the smaller sample, a continuity correction of $+0.5$ should be used.

Example

Problem: 26 Atlantic herrings and 24 North Sea herrings were caught and their lengths measured in cm. The 50 lengths were ordered from smallest to largest and the ranks summed:

$$W_{\text{Atlantic}} = 503 \quad W_{\text{North Sea}} = 772$$

Assess at the 5% significance level whether Atlantic and North Sea herrings differ in median length.

Solution:

$$\left. \begin{array}{l} H_0 : \eta_{\text{Atlantic}} = \eta_{\text{North Sea}} \\ H_1 : \eta_{\text{Atlantic}} \neq \eta_{\text{North Sea}} \end{array} \right\} \begin{array}{l} \alpha = 0.05 \\ \text{Two-tailed} \end{array}$$

Find the smallest possible rank sum for the smaller sample:

$$m = 24, n = 26, W_m = 772$$

$$24 \times (24 + 26 + 1) - 772 = 452$$

Note that $452 < 772$

Hence $W = 452$

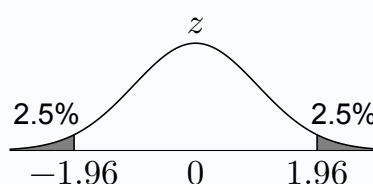
$$E(W) = \frac{1}{2} \times 24 \times (24 + 26 + 1) = 612$$

$$V(W) = \frac{1}{12} \times 24 \times 26 \times (24 + 26 + 1) = 2652$$

$$W \approx N(612, 2652)$$

$$\text{Test statistic: } z = \frac{452.5 - 612}{\sqrt{2652}} = -3.10$$

Critical value: -1.96



Since $-3.10 < -1.96$, reject H_0 at the 5% level of significance. There is evidence to suggest that the median length of an Atlantic herring differs from that of a North Sea herring.

Exercise 17.2

1. A commuter made a number of car journeys, on weekday mornings, from Perth to Glasgow. She believed that journeys on wet days would take longer, on average, than journeys on dry days. Journey length (in minutes) and weather conditions (either *dry*, D, or *wet*, W) for 48 randomly chosen days throughout the Autumn months, of which 21 were classified as *dry*. The results are shown below.

40	D	50	D	55	D	59	D	61	W	70	W
44	D	51	W	56	W	59	W	62	W	72	W
45	W	51	W	56	D	59	W	63	W	73	W
45	D	52	D	56	D	59	D	63	W	74	D
47	D	52	D	57	D	60	W	63	W	77	W
47	W	54	W	57	W	61	W	68	W	77	W
48	D	54	W	58	W	61	W	68	W	82	D
50	D	55	D	58	W	61	D	70	W	86	W

The data were ranked with the shortest time given rank 1, and the rank sum for the *dry* days was 399.

- (a) Perform a Mann-Whitney test at the 1% significance level to determine whether or not her belief is supported by the data.
- (b) State an assumption required for the Mann-Whitney test conducted in part (a) to be valid.
2. Archaeologists determined the silver content (%Ag) of random samples of 30 coins discovered in Cyprus from each of the first and second mintages of the reign of King Manuel I. The data were ranked and ranks sums obtained:

First mintage rank sum = 1015

Second mintage rank sum = 815

Carry out a hypothesis test to compare the medians of the silver content for the two mintages.

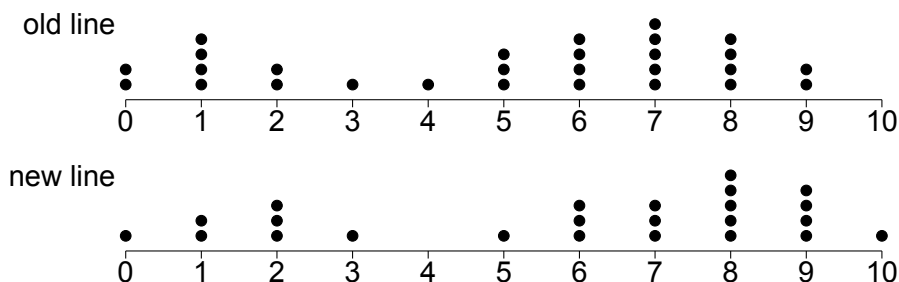
3. During the development of a new adhesive, a research team tested the tensile strength (N/m^2) of random samples of 21 bonds made with Formulation A and 25 bonds made with Formulation B. The results were ranked and ranks sums were:

$$W_A = 583$$

$$W_B = 498$$

Perform a non-parametric hypothesis test to determine whether or not the data provide evidence that Formulation A makes stronger bands than Formulation B.

4. A factory which makes electric heaters has two manufacturing lines in operation, one older and one newer. A quality control inspector wishing to compare the performance of each line selects a random sample of 28 heaters from the old line and 24 heaters from the new line. A quality rating is given for each heater from 0 (terrible) to 10 (flawless), and the results displayed in the dot plots below.



- (a) Explain why the dot plots suggest a Mann-Whitney test would be more appropriate for this data than a t -test.

The rank sum for the new line is found to be 562.5.

- (b) Carry out a Mann-Whitney test to determine if there is evidence, at the 10% level of significance, that the median quality rating for heaters produced by the new line is higher than that of the old line.

3.3 Mann-Whitney from First Principles

In Chapter 10, hypothesis tests were initially introduced using p -values, which can be compared directly to the chosen significance level to conclude a test. Exact p -values can also be calculated directly for Mann-Whitney tests using an approach introduced on the first page of this chapter. The conditions required for a Mann-Whitney test using the *first principles* method are the same as when using critical values, but with the addition that the data must contain *no tied values*, and hence no shared ranks.

The standard steps involved in performing a one-tailed test in this manner are, for samples of size m and n (where $m \leq n$):

- Use ${}^{m+n}C_m$ to calculate the number of possible arrangements of the $m + n$ values.

- Use the observed data to obtain the rank sum for the smaller sample (of size m) as usual.

- Typically by listing all relevant *subsets* of ranks for the smaller sample, determine the number of possible such subsets that results in a rank sum equal to or less than the rank sum obtained from the data.

- The p -value for the test can be calculated using the values from the previous steps as
$$\frac{\text{relevant arrangements}}{\text{possible arrangements}}.$$

Example

Problem: Three randomly selected cameras from Brand A and seven from Brand B are tested for reliability, with the number of *shutter actuations* before failure recorded. When ranked with 10 as the most reliable (greatest number of actuations) to 1 as the least reliable (least number of actuations), a rank sum of 9 is obtained for Brand A.

By considering possible arrangements of rankings, and stating any assumptions required, perform a Mann-Whitney test at the 10% significance level to assess whether the data suggests Brand A's cameras are less reliable.

Solution:

Assume that the population distributions of number of actuations for each brand's cameras have the same shape and variance.

$$\begin{array}{l} H_0 : \eta_A = \eta_B \\ H_1 : \eta_A < \eta_B \end{array} \quad \left. \begin{array}{l} \alpha = 0.1 \\ \text{One-tailed} \end{array} \right\}$$

Number of possible arrangements = ${}^{10}C_3 = 120$.

Subsets of relevant arrangements with a rank sum for A ≤ 9 :

Ranks for A	Rank Sum
1, 2, 3	6
1, 2, 4	7
1, 2, 5	8
1, 2, 6	9
1, 3, 4	8
1, 3, 5	9
2, 3, 4	9

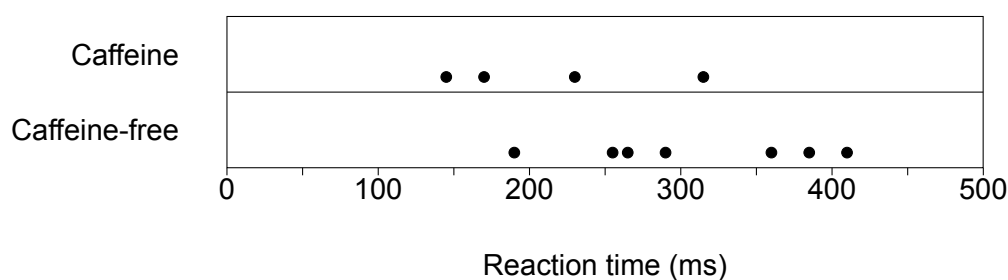
$$p\text{-value} = P(\text{rank sum} \leq 9) = \frac{7}{120} = 0.0583$$

Since $0.0583 < 0.1$, reject H_0 at the 10% level of significance. There is evidence to suggest that the median number of shutter actuations before failure for Brand A is less than that for Brand B, and so Brand A's cameras are less reliable.

Exercise 17.3

1. A Mann-Whitney test is to be performed using two independent samples of sizes 3 and 5. Once data for the samples have been collected and ordered, a rank sum will be obtained for the smaller sample.
 - (a) Determine the number of possible arrangements of the 8 values.
 - (b) List all the possible subsets of rankings for the smaller sample which would result in a rank sum of 8 or lower.
 - (c) Hence determine $P(\text{rank sum} \leq 8)$.
2. A Mann-Whitney test is to be performed using two independent samples of sizes 4 and 5. Once data for the samples have been collected and ordered, a rank sum will be obtained for the smaller sample.
 - (a) Determine the number of possible arrangements of the 9 values.
 - (b) List all the possible subsets of rankings for the smaller sample which would result in a rank sum of 12 or lower.
 - (c) Hence determine $P(\text{rank sum} \leq 12)$.
3. To investigate the effects of caffeine on reaction times, four students from a group of eleven are randomly chosen to be given a soft drink containing caffeine, whilst the remaining seven are given the caffeine-free version of the soft drink. The students will be unaware which they have consumed.

The reaction time of each student is then tested using an online test, with the results shown on the dot plot below.



- (a) Show the calculation required to obtain the rank sum of 15 for the caffeine group.
- (b) List all of the subsets of ranks for the caffeine group which would produce a value equal to or less than 15.

A non-parametric hypothesis test is to be conducted under the following hypotheses:

$$H_0 : \eta_{\text{caffeine}} = \eta_{\text{caffeine-free}}$$

$$H_1 : \eta_{\text{caffeine}} < \eta_{\text{caffeine-free}}$$

- (c) State an assumption required for the test.
 - (d) Under the null hypothesis, calculate the probability of the rank sum for the caffeine group being equal to or less than 15.
 - (e) Use your answer to part (c) to conclude the test at the 5% level of significance.
4. A 800m race between four hockey players and eight rugby players results in the hockey players finishing first, second, fourth and sixth. Assume that the population distributions of times for each group have the same shape and are equal in variance.
 - (a) By considering possible arrangements and relevant subsets, calculate the p -value for a Mann-Whitney test with an alternative hypothesis that hockey players are faster over 800m than rugby players.
 - (b) Use your answer from (a) to conclude the test at the 10% level of significance.
 - (c) State an assumption required for the test to be valid.

Review Exercise

1. A wildlife enthusiast sets up a remote camera which is configured to take a photo when movement is detected. Sometimes it is placed in some local woods, whilst at other times it is positioned beside a pond. The photos are later reviewed and the number of animals spotted each day by the camera is recorded. The number of animal sightings at each location over the last two weeks is shown below.

Woods	7	3	6	2	10	8			
Pond	6	4	1	7	3	1	4	3	

Perform a non-parametric hypothesis test at the 5% level of significance to determine whether there is a difference in the median number of wildlife sightings between the two camera locations.

2. 23 red wines and 23 white wines in stock at a large supermarket are randomly chosen to have their alcohol content (%ABV) recorded. The results are displayed in the stem-and-leaf diagram below.

white										red									
										15	0	0							
									0	14	0	2	5	5	8				
		5	5	3	0	0				13	0	0	5	5	5	5	5	8	
9	8	5	5	5	5	0	0			12	0	5	5	8	8				
		8	5	5	3	0	0			11	0	5							
				5	5	2				10	5								

12 | 8 represents 12.8%

- (a) Comment on what the stem-and-leaf diagram shows.

The data are ranked from lowest to highest, and a rank sum for the white wines of 401.5 is obtained.

- (b) Perform a Mann-Whitney test to assess at the 10% level of significance whether there is evidence to suggest the median alcohol content differs between white and red wines.

A t -test was also considered for a comparison of the alcohol content of the two types of wine.

- (c) Comment on what the stem-and-leaf diagram may suggest about the suitability of both a t -test and a Mann-Whitney test.

3. Independent samples, of sizes 4 and 9 respectively, are taken from populations A and B . A non-parametric hypothesis test is performed at the 2% significance level, with truncated computer output from the test as follows:

```
Mann Whitney test
data: sample A and sample B
alternative hypothesis: true location shift is not equal to zero
W = 12
```

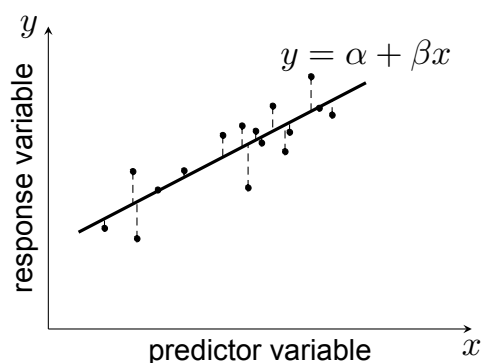
- (a) State the null and alternative hypotheses used for the test.
- (b) Determine the critical value, and hence make a suitable conclusion.
- (c) State two assumptions required for the test performed to be valid.
- (d) Explain the steps required in order to obtain the test statistic of $W=12$.

Further Linear Regression

In Chapter 5 the concept of using *linear regression* to explore straight-line relationships between variables, and to make predictions, was introduced. Linear regression will now be explored in much more detail.

The Method of Least Squares Regression

Given bivariate data, it is x is generally selected as the **predictor** (or *independent*) variable, whilst Y is a *random variable* known as the **response** (or *dependent*) variable. In other words, it is typical to want to be able to predict an unknown Y value given a known x value.



The theoretical model for linear regression is that of a straight line of form $y = \alpha + \beta x$ describing the *best fit* for a relationship between variables. However, the population of bivariate data will not perfectly lie along this line due to chance variation.

The Linear Regression Model

$$Y_i = \alpha + \beta x_i + \varepsilon_i$$

ε_i ("epsilon") are random variables called **errors** which account for observed values of Y_i being greater than or less than those predicted by the regression line for any given value of x_i .

The true values of **intercept parameter** α and **slope parameter** β cannot be known exactly without access to the full *population* of bivariate data. However the sample observations (x_i, y_i) can be used to give estimates, a and b , of α and β . The method of least squares gives formulae for a and b derived by considering how $\sum \varepsilon_i^2$ might be minimised. These rely on several assumptions about the population of errors given on Page 5 of the Data Booklet, along with the formulae:

$$b = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad a = \bar{y} - b\bar{x}$$

This leads to the relationship $\hat{Y}_i = a + bx_i$, where $\hat{Y}_i$ ("y-hat") denotes the expected value of Y_i for a given x_i , and also the best 'fitting' prediction.

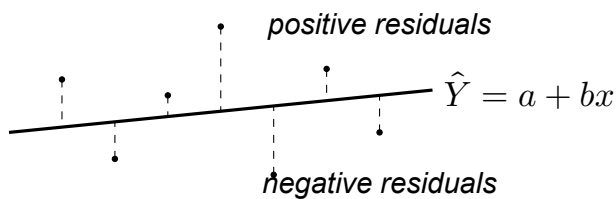
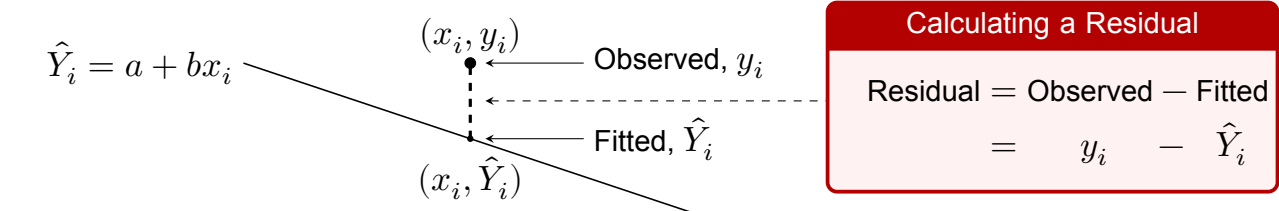
Assumptions for the Errors

- ε_i are independent.
- $E(\varepsilon_i) = 0$
- $V(\varepsilon_i) = \sigma^2$ where $V(Y) = \sigma^2$

4.1 Calculating the Value of a Residual

For any given x_i , the equation of the least squares regression line $\hat{Y}_i = a + bx_i$ gives a *predicted* value $\hat{Y}_i$, also referred to as a **fitted** value. For any data point (x_i, y_i) , the value of y_i may be called the **observed** value.

The difference between an observed value y_i and a corresponding fitted value $\hat{Y}_i$ is called a **residual**:



Data points *above* the regression line have a *positive* residual: the observed y_i was *greater* than that predicted by the regression line for the value of x_i .

Data points *below* the regression line have a *negative* residual: the observed y_i was *less* than that predicted by the regression line for the value of x_i .

To calculate a residual the fitted value, $\hat{Y}_i$, needs to first be found using the equation of the regression line.

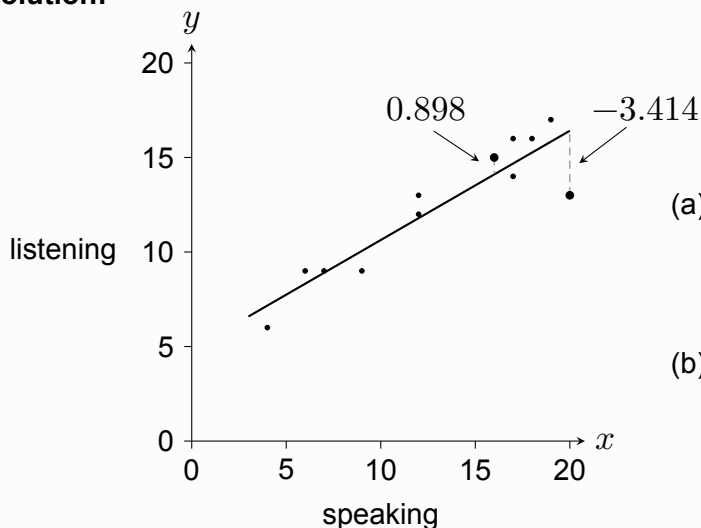
Example

Problem:

Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the y on x regression line is calculated as $\hat{Y} = 4.854 + 0.578x$. Calculate the residual for a student who gains:

- (a) 20 in the speaking test and 13 in the listening test.
- (b) 16 in the speaking test and 15 in the listening test.

Solution:



(a) $\hat{Y} = 4.854 + 0.578 \times 20 = 16.414$
Residual = $13 - 16.414 = -3.414$

(b) $\hat{Y} = 4.854 + 0.578 \times 16 = 14.102$
Residual = $15 - 14.102 = 0.898$

Exercise 18.1

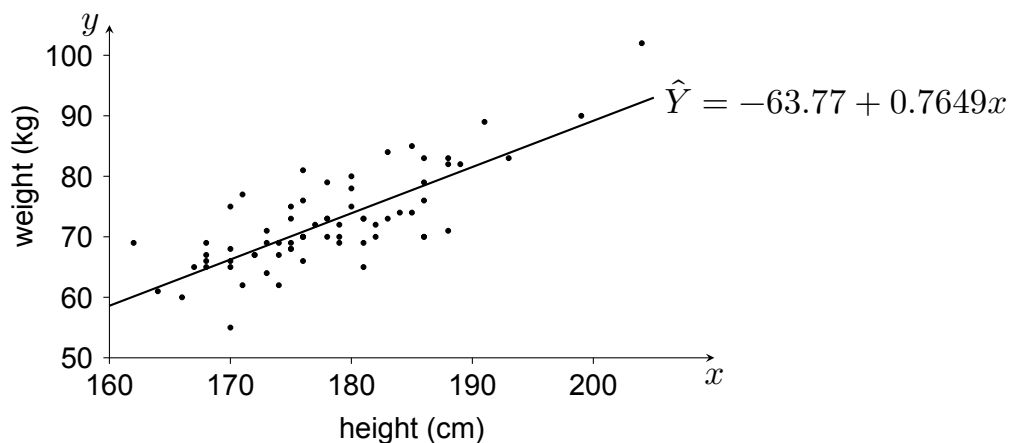
1. Copy and complete the table below:

Regression Line $\hat{Y} = ax + b$	Observed x	Observed y	Fitted $\hat{Y}$	Residual ϵ
$\hat{Y} = 5.3 + 0.1x$	7	8		
$\hat{Y} = -0.7 + 0.3x$	6.1	1.04		
$\hat{Y} = 180 - 2.5x$	40	178		
$\hat{Y} = -45 + 3.5x$	200			-30
$\hat{Y} = 100 + 20x$	-15			50

2. The relationship between the number of calories (x) and the carbohydrate content (y grams) of different types of drinks available in a city's coffee shops can be described using the least squares regression line:

$$\hat{Y} = 5.5572 + 0.6361x$$

- (a) Calculate the expected carbohydrate content for a small cappuccino, which contains 50 calories.
- (b) Determine the value of the residuals for each of the following:
- An extra large white chocolate mocha (450 calories and 310 grams of carbohydrates).
 - A large caramel macchiato (240 calories and 150 grams of carbohydrates).
 - A double espresso (10 calories and 1 gram of carbohydrates).
3. The heights (x cm) and weights (y kg) are recorded for a sample of male professional footballers.



A linear model is fit to the data as shown with a regression line equation of:

$$\text{weight} = -63.77 + 0.7649 \times \text{height}$$

(a) State whether each of the following have positive or negative residuals:

- The tallest footballer shown.
- The lightest footballer shown.

(b) Find the value of the residual for a footballer with:

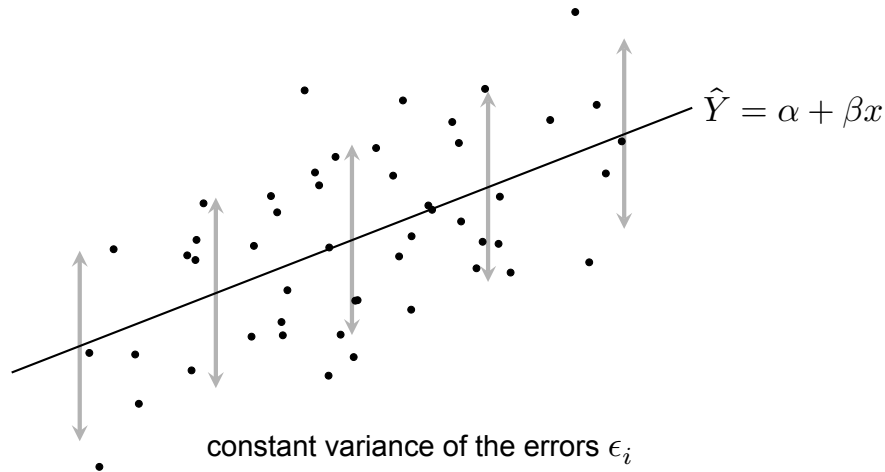
- Height 186cm and weight 70kg.
- Height 191cm and weight 89kg.

4.2 Estimating the Error Variance

The least squares regression model assumes that the variance of the errors is a constant for all x_i (and Y_i):

$$V(\varepsilon_i) = \sigma^2$$

The *error variance* is the variance of the population of data *relative to the regression line's predicted values*, loosely indicated below by the series of arrows below.



The error variance σ^2 is estimated by:

Estimating the
Error Variance

$$s^2 = \frac{\text{SSR}}{n - 2}$$

Where the (always positive) *sum of squared residuals*, SSR, is calculated as: $\text{SSR} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$

Both the formula for estimating the error variance σ^2 and the more convenient second formula for the sum of squared residuals SSR are given on page 5 of the Data Booklet.

Example

Problem:

Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students: $S_{xx} = 344.9$, $S_{yy} = 132.9$, $S_{xy} = 193.6$.

- Calculate the sum of squared residuals for this data set.
- Obtain an estimate of the error variance.

Solution:

$$(a) \text{ SSR} = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 132.9 - \frac{193.6^2}{344.9} = 24.23$$

$$(b) s^2 = \frac{\text{SSR}}{n - 2} = \frac{24.23}{12 - 2} = 2.423$$

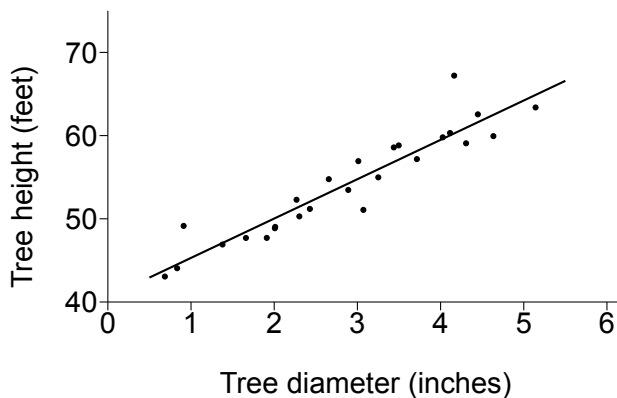
Exercise 18.2

1. A bivariate random sample has summary statistics:

$$S_{xx} = 28.92 \quad S_{yy} = 495.0 \quad S_{xy} = 105.3 \quad n = 14$$

It is determined that linear relationship appears to exist between the variables.

- Obtain an estimate s^2 for the error variance, σ^2 .
 - Hence determine the value of s for this bivariate data.
2. 26 randomly selected trees in a forest have their diameter (x) at a set distance from the ground measured, in inches, and the height (y) of each tree also measured, in feet. The equation of the least squares regression line for y on x is obtained, and shown on the scatterplot below.



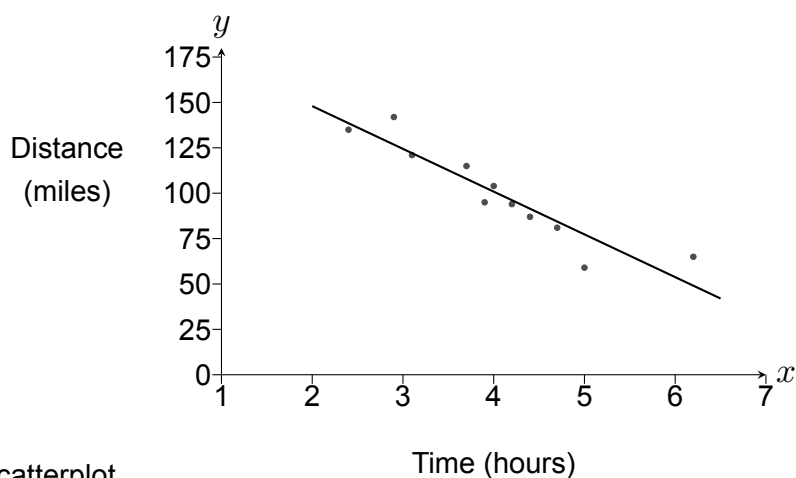
Summary statistics:

$$\begin{aligned} \sum x &= 74.76 \\ \sum x^2 &= 253.76 \\ \sum y &= 1408.3 \\ \sum y^2 &= 77277 \\ \sum xy &= 4233.1 \\ n &= 26 \end{aligned}$$

- Obtain an estimate s^2 for the error variance, σ^2 .
 - Hence determine the value of s for this bivariate data.
3. A running website randomly selected eleven amateur runners who recently completed a marathon and asked them to complete a survey which included their finishing time in the marathon and information about their training routine. The scatterplot below shows, for each runner, their finishing time (x , in hours) and the total distance they ran in the final four weeks of their training (y , in miles).

Summary statistics:

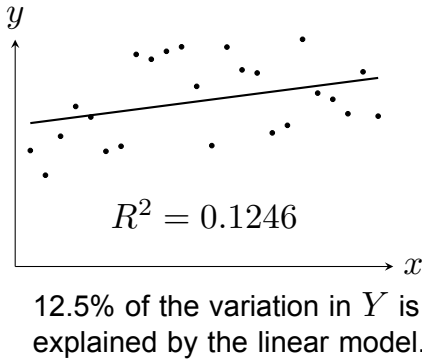
$$\begin{aligned} \sum x &= 44.5 \\ \sum x^2 &= 191.21 \\ \sum y &= 1098 \\ \sum y^2 &= 116768 \\ \sum xy &= 4179.2 \\ n &= 11 \end{aligned}$$



- Make a comment on the scatterplot.
- Obtain an estimate s^2 for the error variance, σ^2 .
- Determine the equation of the least squares regression line.
- Calculate the correlation coefficient for the data and comment on its value.
- A runner, not shown, completed the marathon in 3 hours and 30 minutes, after running a total of 110 miles in the four weeks before the race. Calculate the residual for this runner.

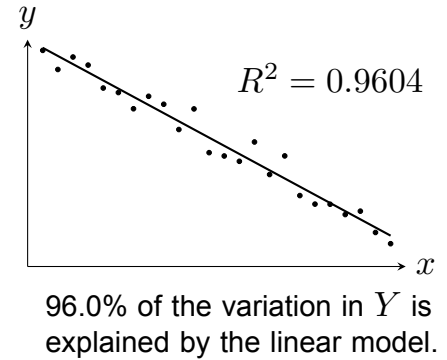
4.3 The Coefficient of Determination

The *coefficient of determination*, R^2 , is the proportion of the total variation in the response variable, Y , that is explained by the linear regression model. The higher the value of R^2 , the less variation remains for Y_i relative to the total variation of Y . This means that the coefficient of determination is a very useful measure of how ‘good’ a model is for the data.



Coefficient of Determination

$$R^2 = \frac{\text{explained variation}}{\text{total variation}}$$



For the linear regression models covered in this course, the coefficient of determination is connected to the *correlation coefficient*, r , by a simple relationship:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \implies \boxed{R^2 = r^2}$$

Both the formula above and the interpretation of the coefficient of determination implies that $0 \leq R^2 \leq 1$.

Example

Problem:

Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the following summary statistics are obtained:

$$S_{xx} = 334.9 \quad S_{yy} = 132.9 \quad S_{xy} = 193.6 \quad n = 12$$

A linear regression model is proposed, and an equation for the regression line of y on x is obtained:

$$\hat{Y} = 4.854 + 0.578x$$

Calculate the coefficient of determination and interpret this value in context.

Solution:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{193.6}{\sqrt{334.9 \times 132.9}} = 0.9177$$

$$R^2 = r^2 = 0.9177^2 = 0.842$$

This suggests that 84.2% of the variation in listening marks can be explained by the linear model.

The conclusion from the example above could also be phrased as:

“87.4% of the variation in listening marks can be explained by the speaking marks.”

Exercise 18.3

1. A bivariate random sample has summary statistics:

$$S_{xx} = 28.92 \quad S_{yy} = 495.0 \quad S_{xy} = 105.3 \quad n = 14$$

Determine the value of the coefficient of determination and interpret it.

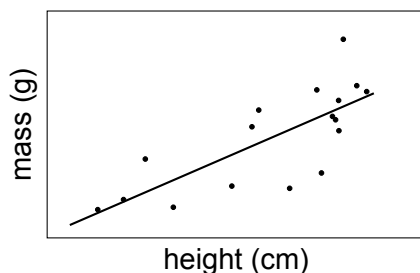
2. For a series of studies, computer output giving equations of least square regression lines and some summary statistics are provided. For each, determine the coefficient of determination and interpret it.

- (a) The salary and time employed is recorded for a sample of 70 employees at a company.



y on x regression line
 $\text{salary} = 20359 + 1506 \times \text{years_employed}$
 sample estimate for r
 0.857

- (b) For a random sample of 17 red squirrels, the mass and height are recorded.



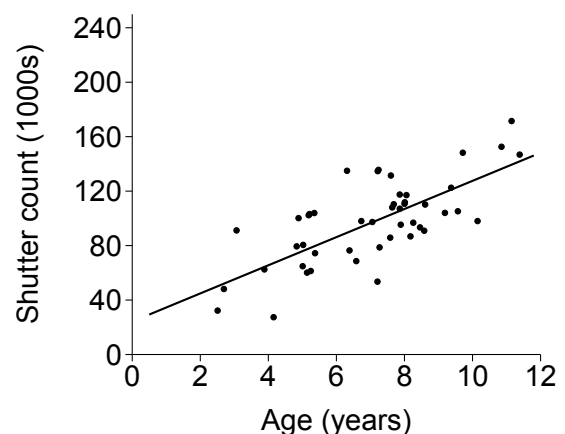
y on x regression line
 $\text{mass} = 45.5 + 6.50 \times \text{height}$
 sample correlation coefficient
 0.757

- (c) The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track.

y on x regression line	n	r	s
$\text{top_speed} = 269.2 - 1.72 \times \text{lap_time}$	34	-0.670	8.49

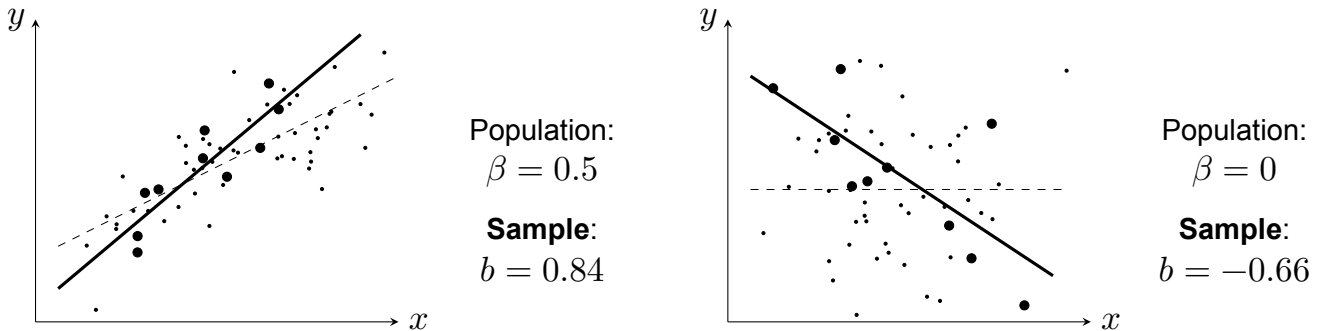
3. A seller on an online auction site has a stock of used digital cameras, all aged somewhere between 2 and 12 years. For each camera, the total number of photos taken (the *shutter count*) is determined using special software. A y on x regression line is shown on the scatterplot of the data below. A correlation coefficient is calculated for the data of 0.718.

- Make one comment on the scatterplot.
- Interpret the correlation coefficient.
- Calculate the coefficient of determination, and interpret it in context.
- The seller is considering expanding their business to also sell a range of film cameras, produced between 1976 and 1984. Give two reasons why they should not use the regression line shown to make predictions about the shutter counts of the any film cameras they sell.



4.4 Test for the Slope Parameter β

The true value of the *slope parameter*, β , cannot be known without obtaining a full population. The *sample statistic* b is an estimator for β , and may differ slightly or substantially from it, as shown below.



The slope parameter β can be *positive*, *negative* or *zero*. Where $\beta = 0$ it can be said that knowing x_i has no value in predicting y_i . In other words, a linear regression model is only **useful for prediction** in cases where the slope parameter is *non-zero*, or $\beta \neq 0$. The **parametric Test for Beta** assesses whether data from a sample provide evidence to suggest that this is the case. The test statistic below follows a t -distribution with $n - 2$ degrees of freedom, and in this course the test will always be *two-tailed* with hypotheses $H_0 : \beta = 0$ and $H_1 : \beta \neq 0$.

Test Statistic

$$t_{n-2} = \frac{b\sqrt{S_{xx}}}{s}$$

Conditions

- ε_i are independent and identically distributed as $N(0, \sigma^2)$.
- The data was obtained through random sampling.

Example

Problem: A bivariate sample has summary statistics:

$$S_{xx} = 157.875 \quad S_{yy} = 188 \quad S_{xy} = -133.5 \quad n = 8$$

Test at the 5% significance level whether a linear regression model would be useful for prediction.

Solution:

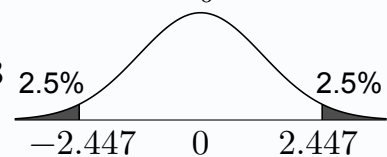
$$\begin{array}{l} H_0 : \beta = 0 \\ H_1 : \beta \neq 0 \end{array} \quad \left. \begin{array}{l} \alpha = 0.05 \\ \text{Two-tailed} \end{array} \right\}$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{-133.5}{157.875} = -0.8456$$

$$SSR = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 188 - \frac{(-133.5)^2}{157.875} = 75.11, \quad s = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{75.11}{8-2}} = 3.538$$

$$\text{Test statistic: } t = \frac{b\sqrt{S_{xx}}}{s} = \frac{-0.8456 \times \sqrt{157.875}}{3.538} = -3.003$$

$$\text{Critical value: } t_{6,0.975} = -2.447$$



Since $-3.003 < -2.447$, reject H_0 at the 5% level of significance. There is evidence to suggest that the slope parameter is not zero, so evidence that a linear model would be useful for prediction.

Exercise 18.4

1. A random sample of bivariate data, sample size 27, yields the following summary statistics:

$$S_{xx} = 87.8 \quad S_{yy} = 415.5 \quad S_{xy} = -129.6 \quad SSR = 224.2$$

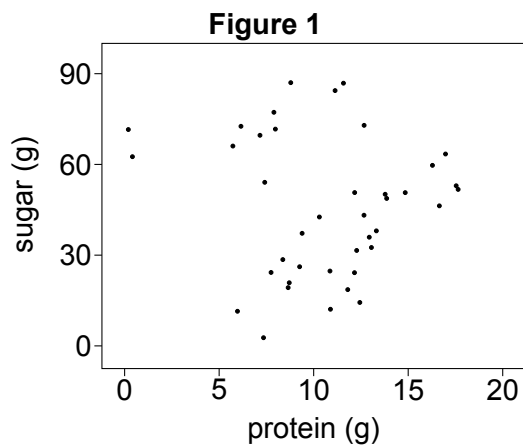
A test for β is to be performed on the data.

- State suitable null and alternative hypotheses.
 - Show the calculations required to reach a test statistic of -4.618 .
 - State the degrees of freedom, ν , of the t -distribution required for the test.
 - Conclude the test at the 5% level of significance.
2. The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track. Computer output related to a linear model constructed using this data is below.

y on x regression line	n	s	sum of squares for x
top_speed = 269.2 - 1.72 × lap_time	34	8.49	638.6

Perform a test for the slope parameter at the 10% level of significance to assess whether there is evidence that a linear model is useful for prediction for this data.

3. A nutritionist purchases a sample of drinks sold from vending machines inside a selection of gyms and health centres in his area. The protein content in grams and the sugar content in grams is recorded for each drink. **Figure 1** shows a scatterplot of the data and **Output 1** shows results of a hypothesis test conducted (where $\alpha = 0.05$) to assess whether a linear model would be useful for prediction.

**Output 1**

```

model: sugar = 4.01 - 0.124 × protein
t = -0.42295
df = 38
p-value = 0.6747
alternative: true slope parameter is not
equal to zero

```

- State suitable null and alternative hypotheses for the test conducted.
 - State the sample size.
 - Determine the critical value required, and hence write a conclusion for the hypothesis test.
 - State the required assumption about the population of errors for this test.
4. A sample of bivariate data with summary statistics below is proposed to be used for a linear model:

$$S_{xx} = 75.6 \quad S_{xy} = -235.2 \quad s^2 = 4.04 \quad n = 70$$

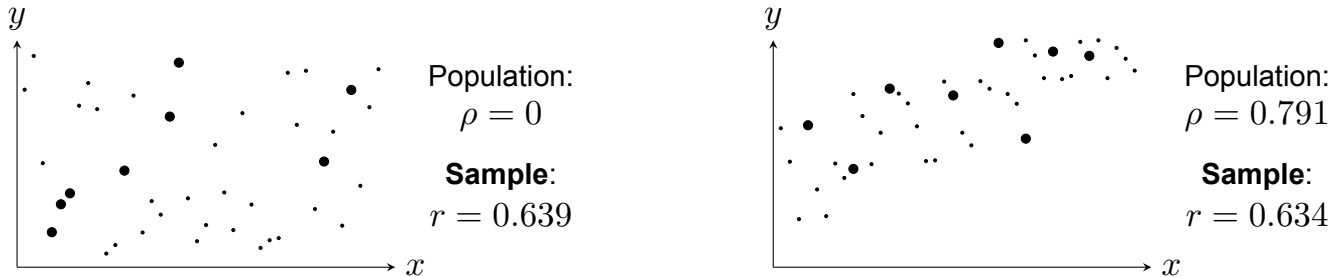
Conduct a parametric test at the 1% level of significance to assess whether there is evidence that the slope parameter is non-zero.

5. Test whether the sample data below provides evidence that a linear model would be useful for prediction.

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

4.5 Test for the Correlation Coefficient ρ

The previously introduced correlation coefficient, r , is a sample statistic which measures the strength of the linear association between two variables. It is an estimate of the population parameter ρ ("rho"), where $-1 \leq \rho \leq 1$. If $\rho = 0$ then there is no *linear* relationship between the variables. Given a sample of bivariate data for which there exists *no* linear relationship, it is regardless common for the sample to give a *non-zero* value for r :



The *parametric Test for Rho* assesses a sample of bivariate data for evidence of a **linear association** between the variables, or $\rho \neq 0$. The test statistic below follows a t_{n-2} distribution, and in this course the test will always be *two-tailed* with hypotheses $H_0 : \rho = 0$ and $H_1 : \rho \neq 0$.

Test Statistic

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Conditions

- The variables are independent and follow an approximately bivariate normal distribution.
- The data was obtained through random sampling.

Any conclusion should comment on the evidence for a linear association between the two variables.

Example

Problem: A bivariate sample has summary statistics:

$$S_{xx} = 157.875 \quad S_{yy} = 188 \quad S_{xy} = -133.5 \quad n = 8$$

Test at the 5% significance level for evidence of a linear association between x and y .

Solution:

$$\begin{array}{l} H_0 : \rho = 0 \\ H_1 : \rho \neq 0 \end{array} \quad \left. \begin{array}{l} \alpha = 0.05 \\ \text{Two-tailed} \end{array} \right\}$$

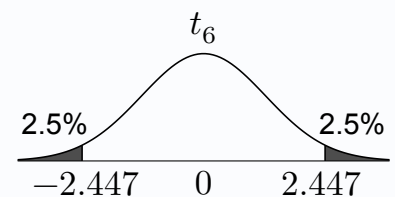
$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{-133.5}{\sqrt{157.875 \times 188}} = -0.7749$$

$$\text{Test statistic: } t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.7749\sqrt{8-2}}{\sqrt{1-(-0.7749)^2}} = -3.003$$

$$\text{Critical value: } t_{6,0.975} = -2.447$$

Since $-3.003 < -2.447$, reject H_0 at the 5% level of significance.

There is evidence to suggest that there is a linear association between the two variables.



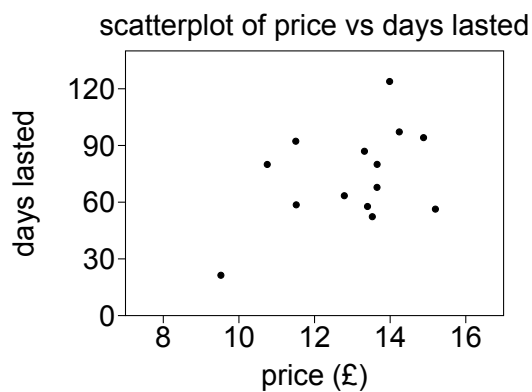
Exercise 18.5

- A random sample of bivariate data, sample size 27, yields a sample correlation coefficient of -0.678 . A test for linear association is to be performed on the data.
 - State suitable null and alternative hypotheses.
 - Show the calculations required to reach a test statistic of -4.612 .
 - State the degrees of freedom, ν , of the t -distribution required for the test.
 - Conclude the test at the 5% level of significance.
- A bivariate random sample has summary statistics:

$$S_{xx} = 28.92 \quad S_{yy} = 495.0 \quad S_{xy} = 105.3 \quad n = 14$$

Perform a parametric test to assess at the 10% level of significance whether there is evidence to suggest a linear association exists between x and y .

- A guitarist records over the course of several years the price of sets of strings that she buys and the number of days that they last without a string breaking. A scatterplot of the data collected is shown below alongside some summary statistics from computer software.



data:
guitar_strings

sample correlation coefficient:
0.437

sample size:
14

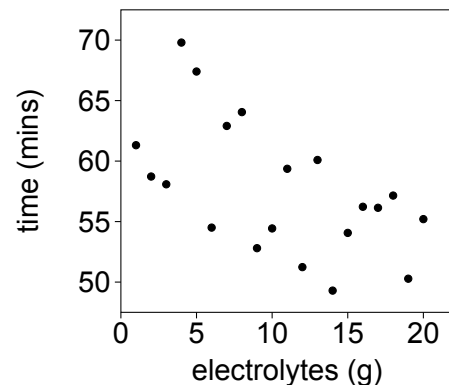
Test at the 1% level of significance whether her data provides evidence of a linear association between the price of a set of strings and the time until a string breaks.

- Electrolyte powders* are often used to help improve hydration during exercise. A keen cyclist records the time (y minutes) it takes them to complete a familiar route during a series of rides over the summer, randomly choosing the mass (x grams) of electrolyte powder to add to their water bottle each time.

Summary statistics for the rides are below:

$$\begin{array}{cccc} S_{xx} & S_{yy} & S_{xy} & n \\ 665 & 567.2 & -346.4 & 20 \end{array}$$

Test whether the data supports a claim from the cyclist that a linear association exists between the amount of electrolyte powder they use and their cycling performance, stating one assumption required.



- Test whether the sample data below provides evidence of a linear association between x and y .

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

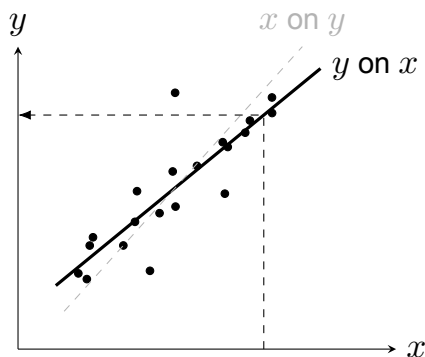
4.6 Regression Line of x on y

Each least squares regression line met so far has been an y on x regression line, calculated with the aim of minimising errors when *predicting y values*. However, a y on x regression line model cannot be used to predict x values.

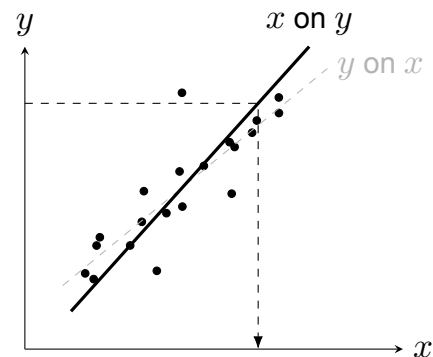
If the prediction of x values based on y values is of interest, it is necessary to obtain the equation of an x on y regression line. This takes a different form and different values of a and b as shown below:

Calculating an x on y Regression Line

$$\hat{X} = a + by \quad \text{where} \quad b = \frac{S_{xy}}{S_{yy}} \quad \text{and} \quad a = \bar{x} - b\bar{y}$$



Prediction with a y on x model



Prediction with an x on y model

Example

Problem:

Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the following summary statistics are obtained:

$$\begin{aligned} \Sigma x &= 157 & \Sigma x^2 &= 2389 & \Sigma y &= 149 & \Sigma y^2 &= 1983 & \Sigma xy &= 2143 \\ S_{xx} &= 334.9 & S_{yy} &= 132.9 & S_{xy} &= 193.6 & n &= 12 & r &= 0.9177 \end{aligned}$$

Determine the equation for the least squares regression line of x on y , and hence predict the speaking mark for a student who has only completed the listening component and obtained a mark of 17 in it.

Solution:

$$b = \frac{S_{xy}}{S_{yy}} = \frac{193.6}{132.9} = 1.457$$

$$a = \bar{x} - b\bar{y} = \frac{157}{12} - 1.457 \times \frac{149}{12} = -5.008$$

$$\hat{X} = -5.008 + 1.457y$$

For a student with a listening mark (y) of 17:

$$\hat{X} = -5.008 + 1.457 \times 17 = 15.39$$

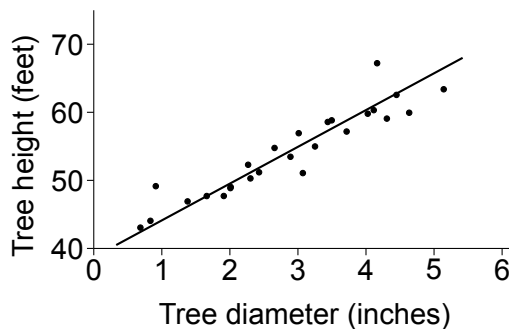
Exercise 18.6

1. A bivariate sample has summary statistics:

$$\sum x = 117 \quad \sum x^2 = 1869 \quad \sum y = 660 \quad \sum y^2 = 54638 \quad \sum xy = 9519 \quad n = 8$$

It is determined from inspecting a scatterplot that there appears to be a linear relationship between the variables. Obtain the equation of the least squares regression line of x on y .

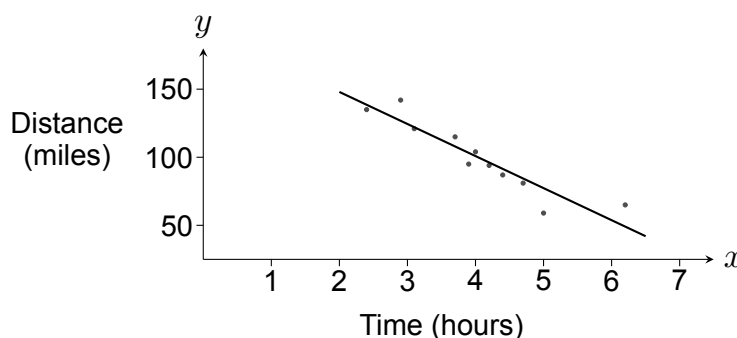
2. 26 randomly selected trees in a forest have their diameter at a set distance from the ground measured, in inches, and the height of each tree also measured, in feet. The equation of the least squares regression line for x on y is obtained, and shown on the scatterplot below.



Summary statistics:

$$\begin{aligned} \sum x &= 74.76 \\ \sum x^2 &= 253.76 \\ \sum y &= 1408.3 \\ \sum y^2 &= 77277 \\ \sum xy &= 4233.1 \\ n &= 26 \end{aligned}$$

- (a) Calculate the equation of the least squares regression line of x on y .
 (b) Hence estimate the diameter of a tree which is 65 feet tall.
3. A running website randomly selected eleven amateur runners who recently completed a marathon and asked them to complete a survey which included their finishing time in the marathon and information about their training routine. The scatterplot below shows, for each runner, their finishing time (x , in hours) and the total distance they ran in the final four weeks of their training (y , in miles).



Summary statistics:

$$\begin{aligned} \sum x &= 44.5 \\ \sum x^2 &= 191.21 \\ \sum y &= 1098 \\ \sum y^2 &= 116768 \\ \sum xy &= 4179.2 \\ n &= 11 \end{aligned}$$

A least squares regression line for y on x is shown on the graph, with equation $\hat{Y} = 40.6 + 4.73x$.

The website wishes to allow users the ability to input the number of miles in their training plan for the four weeks leading up to the race, and receive in response an estimate for their finishing time.

- (a) Explain why a *new* regression line will need to be determined for this tool on the website.
 (b) Calculate the equation of this new regression line.
 (c) Determine an estimate of the finishing time for a runner who completed 70 miles.
 (d) Explain why the new regression line obtained in part (b) is not suitable for predicting the finish time of a runner who completed 162 miles.
4. Determine the equation of an x and y regression line for the bivariate data below.

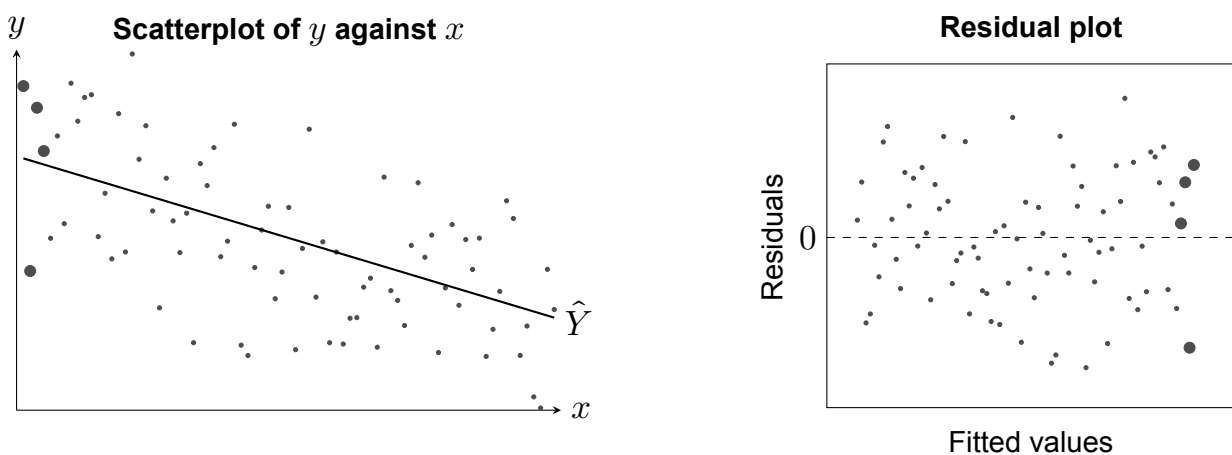
x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

4.7 Residual Plots

When considering the use of a linear regression model for bivariate data, it is recommended to produce a plot of *residuals* (ε_i) against *fitted values* ($\hat{Y}_i$), which have been calculated using the sample regression line. The purpose of this is both to ascertain whether a linear regression model is *the most appropriate* model, as well as to judge the validity of several commonly required assumptions, including those required to obtain the estimates for α and β :

$$E(\varepsilon_i) = 0 \quad \text{and} \quad V(\varepsilon_i) = \sigma^2$$

Below is an illustration of a standard scatterplot showing a sample of bivariate data on the left, with a y on x regression line included, and a residual plot for the same data on the right. The data points highlighted with larger circles on the y against x scatterplot have their residuals similarly highlighted on the residual plot.

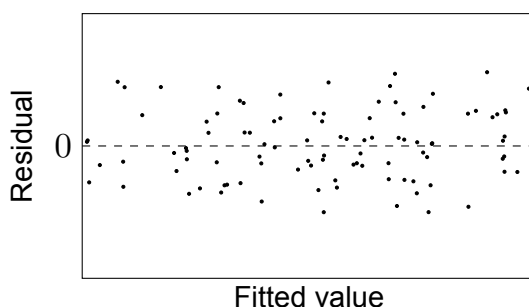


It should be observed that the residual plot broadly appears to show:

Interpreting Residual Plots	
A random scatter of points...	
...centred on zero...	$E(\varepsilon_i) = 0$
...with a constant variance.	$V(\varepsilon_i) = \sigma^2$

This means that the residual plot would give no reason to doubt the assumptions made regarding the errors. This *ideal* residual plot would suggest that the linear regression model is *appropriate*.

The following residual plots are accompanied with relevant comments and suitable conclusions.

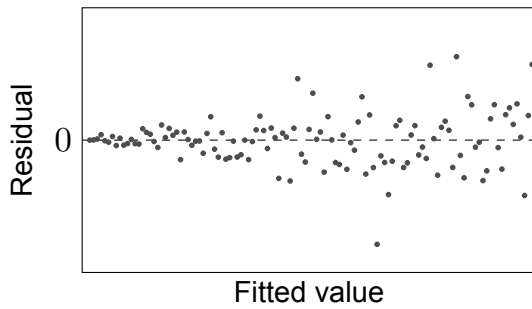


The residual plot appears to show a random scatter of points centred on zero, with a constant variance as the fitted value increases.

$$E(\varepsilon_i) = 0$$

$$V(\varepsilon_i) = \sigma^2$$

This suggests the that linear model is **appropriate**.

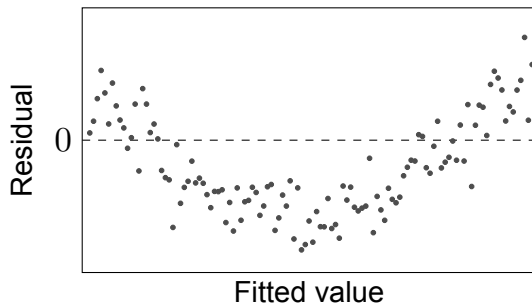


The residual plot appears to show a 'funnel' shaped pattern of points centred on zero, with variance increasing as the fitted value increases.

$$E(\varepsilon_i) = 0$$

$$V(\varepsilon_i) \neq \sigma^2$$

This suggests the that linear model is **not appropriate**.

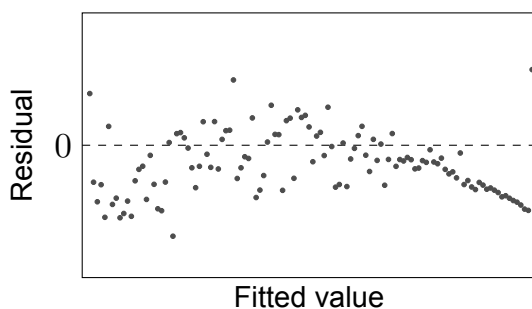


The residual plot appears to show a pattern of points following a curve, with constant variance.

$$E(\varepsilon_i) \neq 0$$

$$V(\varepsilon_i) = \sigma^2$$

This suggests that the linear model is **not appropriate**.



The residual plot appears to show a pattern of points following a curve, with variance decreasing as the fitted value increases.

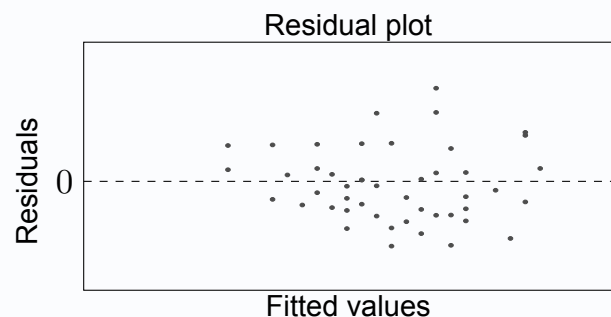
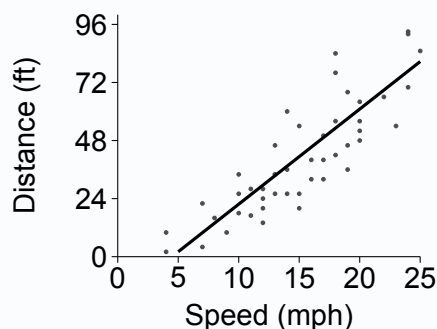
$$E(\varepsilon_i) \neq 0$$

$$V(\varepsilon_i) \neq \sigma^2$$

This suggests that the linear model is **not appropriate**.

Example

Problem: A 1920s study on automobile safety involved recording the stopping distances (in feet) from selected speeds (in miles per hour) for a random sample of 47 cars. Below is a scatterplot of the results with a regression line obtained from a linear model, and a residual plot.

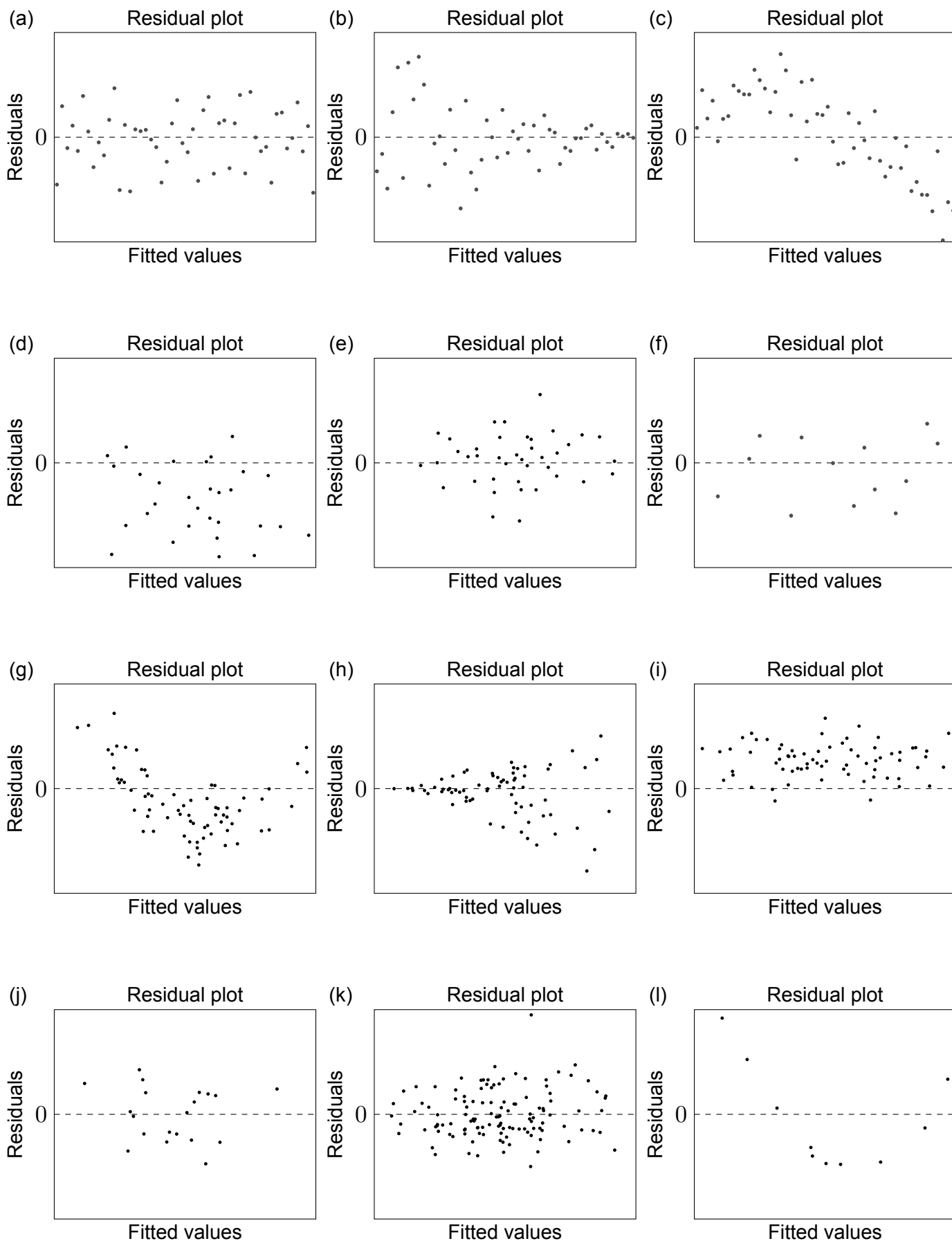


Use the residual plot to comment on the validity of the linear model.

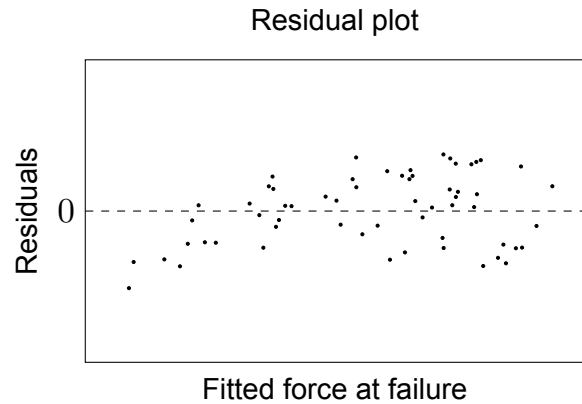
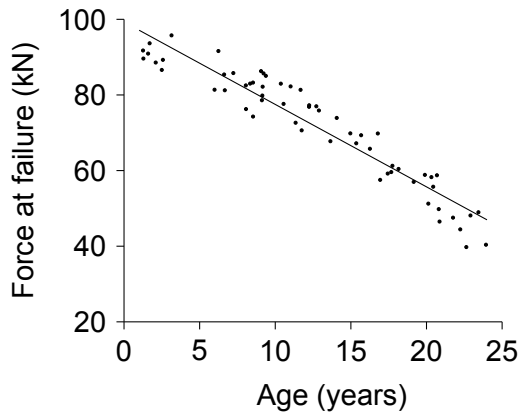
Solution: The residual plot appears to show a random scatter of points centred on zero, with a broadly constant variance as the fitted value increases. This suggests the linear model between stopping distance and speed could be appropriate.

Exercise 18.7

1. For each, comment on the validity of the linear model from which they were produced.

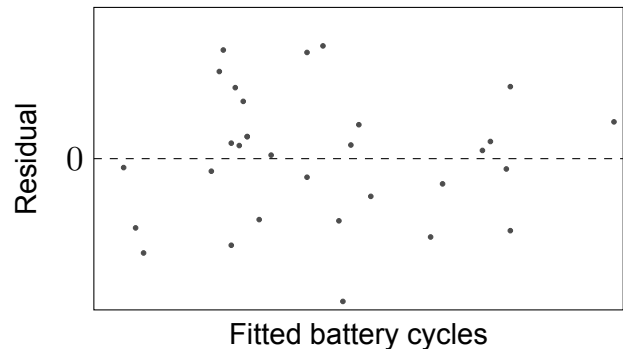
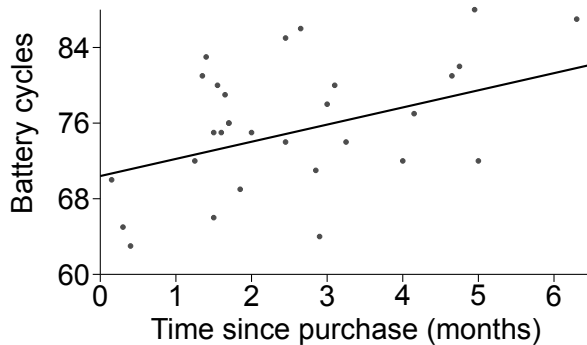


2. Suspension clamps are used to hold electricity cables in place on pylons. Over time these clamps are subject to wear, making them more likely to fail. A random sample of clamps are subject to an increasing force, until they fail. A proposed linear regression line is shown on the scatterplot of force at failure (kN) against age in years below, as well as a residual plot.



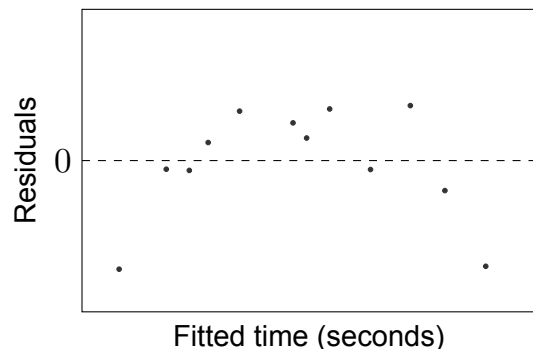
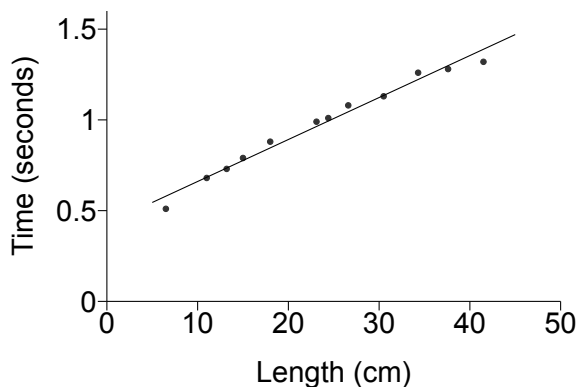
Comment on the suitability of the linear model fitted.

3. A number of randomly selected mobile phones returned as faulty within a six month warranty period each have their time since purchase (x) and number of completed battery cycles (y) recorded. The equation of the least squares regression line for y on x is obtained, and shown on the scatterplot below alongside a residual plot.



Explain why the residual plot suggests that a linear model would be suitable for the data.

4. A student investigating the relationship between the length of a pendulum and the time taken for one complete swing to occur made a simple pendulum from string and a metal disc, suspended from a clamp over the edge of a table. The time taken for one complete swing (y seconds) was recorded for 12 different string lengths (x cm).



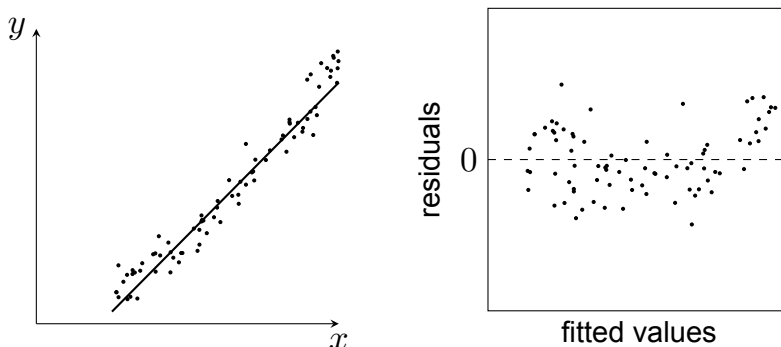
Comment on the suitability of the linear model fitted.

4.8 Transforming Data

Where a *non-linear* relationship seems to exist between two variables then it would not be appropriate to use a model which describes a *linear* relationship between them. The presence of a non-linear relationship may be immediately clear from an initial scatterplot, or it may only be revealed once a residual plot for a proposed linear model is created.

A linear model might initially seem to describe well relationship between x and y , as might the result of $R^2 = 0.9228$.

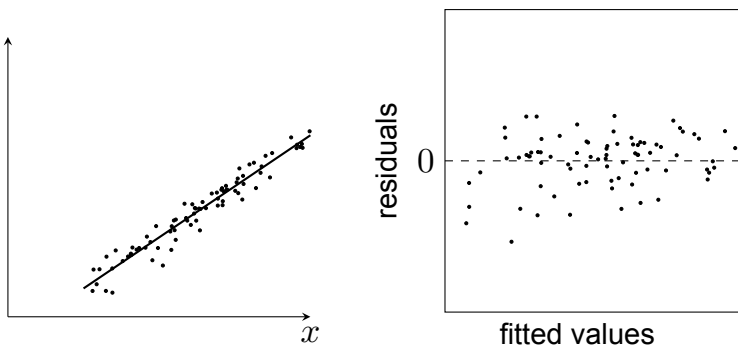
However, the residual plot suggests that a linear model between x and y is *not* appropriate, since the residuals follow a curved pattern.



The simulated data above was actually generated using a *quadratic* relationship between x and y . A scatterplot of $\sqrt{y}$ against x for this data is shown below, alongside a residual plot:

Both the scatterplot and the residual plot for this new model of $\sqrt{y}$ on x suggest that it is appropriate for the data.

This new, better-fitting model also produces an improved value for the coefficient of determination, of $R^2 = 0.9402$.



Creating a model such as this, in which one or both variables have some operation performed on them (such as taking the *square root*) is called applying a *transformation* to the data. In practice, knowing which transformation(s) to try will generally be guided by a background understanding of the variables, or previous studies. In this course, any required transformations will be clearly specified.

Familiar statistics may still be calculated using transformed variables:

Statistic	For an y on x model	New model	New formula
Sum of Squares for x	$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$	y on $\sqrt{x}$	$S_{\sqrt{x}\sqrt{x}} = \sum (\sqrt{x})^2 - \frac{(\sum \sqrt{x})^2}{n}$
Slope parameter estimate	$b = \frac{S_{xy}}{S_{xx}}$	$\log_a y$ on x	$b = \frac{S_{x \log_a y}}{S_{xx}}$
Correlation coefficient	$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$	y^3 on x	$r = \frac{S_{xy^3}}{\sqrt{S_{xx}S_{y^3y^3}}}$

Care should be taken when interpreting these statistics and when calculating using regression line equations.

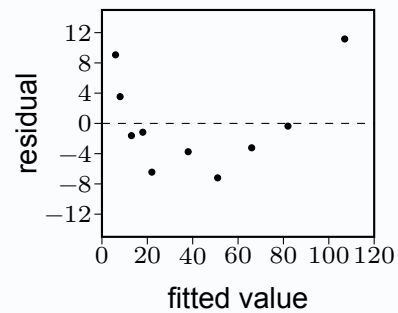
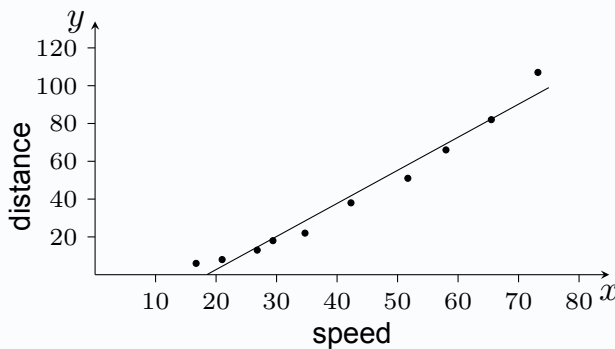
The example opposite illustrates this for the coefficient of determination, and calculating a predicted value.

Example

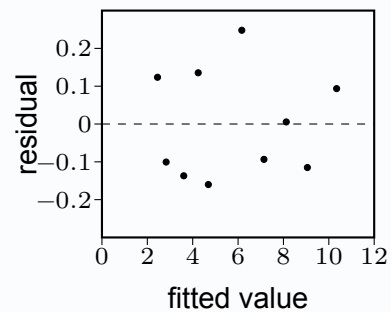
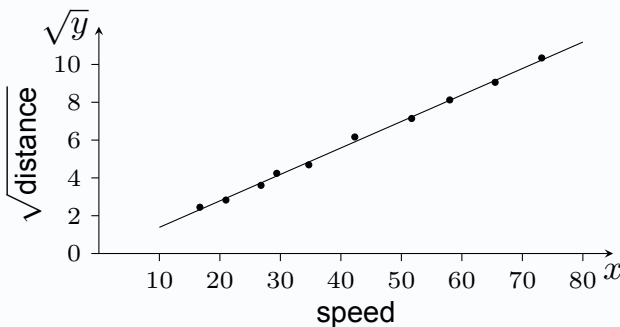
Problem: Data was recorded for a sample of 10 cars of their stopping distance (y metres) when travelling at a range of speeds (x miles per hour). The table below shows the results.

Speed (x)	16.7	21.0	26.8	29.4	34.7	42.3	51.7	58.0	65.5	73.2
Distance (y)	6	8	13	18	22	38	51	66	82	107

The line $\hat{Y} = -32.3 + 1.75x$ was fitted, and a scatterplot and a residual plot drawn.



A new linear model was created between speed (x) and the square root of the stopping distance ($\sqrt{y}$). For this new model, a scatterplot and residual plot were produced, and a least squares regression line was computed as: square root of distance = $-0.016 + 0.14 \times \text{speed}$



- Comment on the appropriateness of the linear model of y on x .
- Comment on the appropriateness of the new model of $\sqrt{y}$ on x .
- By first creating a new table for x and $\sqrt{y}$ and finding summary statistics for this data, calculate the coefficient of determination for this new model and interpret its value.
- Estimate the stopping distance at a speed of 50 miles per hour.

Solution:

- The residual plot appears to show a curved pattern, so this linear model is not appropriate.
- The residual plot for the new model appears to show a random scatter of points centered on zero, with a constant variance as fitted values increase, so the new model seems appropriate.

$$(c) \quad r = \frac{S_{x\sqrt{y}}}{\sqrt{S_{xx}S_{\sqrt{y}\sqrt{y}}}} = \frac{3400.0}{\sqrt{10758.9 \times 5951.5}} = 0.9987 \text{ and } R^2 = 0.9987^2 = 0.997.$$

99.7% of the variation of $\sqrt{\text{distance}}$ is explained by the model.

- $\sqrt{\text{distance}} = -0.016 + 0.14 \times 50 = 6.984$, so distance = $6.984^2 = 48.8$ metres

Exercise 18.8

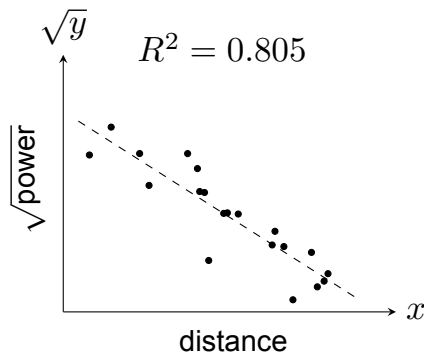
1. For each of the following scatterplots, showing transformed data, in context:

- Describe any relationship shown by the scatterplot of the transformed data.
- Interpret the coefficient of determination of the proposed linear model shown by the regression line.

(a) Data before transformation:

x = distance

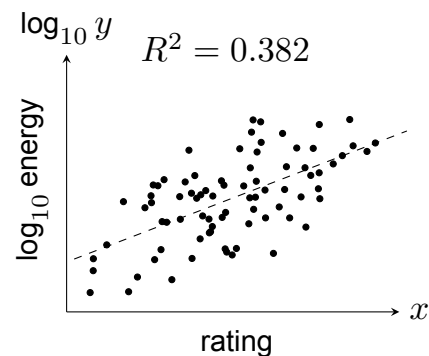
y = power



(b) Data before transformation:

x = rating

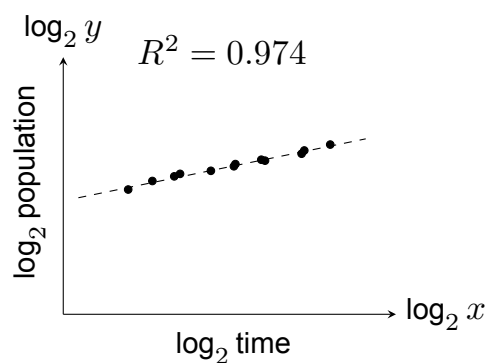
y = energy



(c) Data before transformation:

x = time

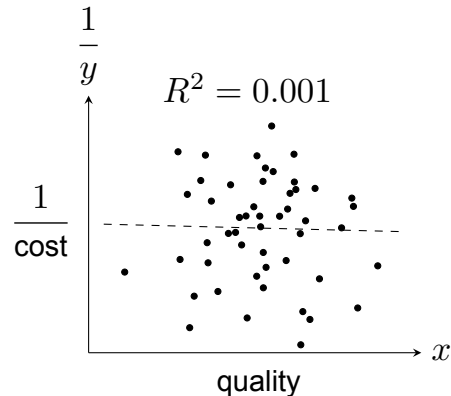
y = population size



(d) Data before transformation:

x = quality

y = cost



2. It is determined that a linear relationship exists between x and $\frac{1}{y}$.

Summary statistics for this model are as follows:

$$\Sigma x = 35.9 \quad \Sigma \frac{1}{y} = 133.4 \quad \Sigma x^2 = 142.3 \quad \Sigma \frac{1}{y^2} = 1922.4 \quad \Sigma x \frac{1}{y} = 517.8 \quad n = 10$$

- (a) Determine the equation of the least squares regression line for the model of $\frac{1}{y}$ on x .
- (b) Calculate an estimate for the value of y when $x = 5$.

3. It is determined that a linear relationship exists between $\log x$ and $\log y$.

Summary statistics for this model are as follows:

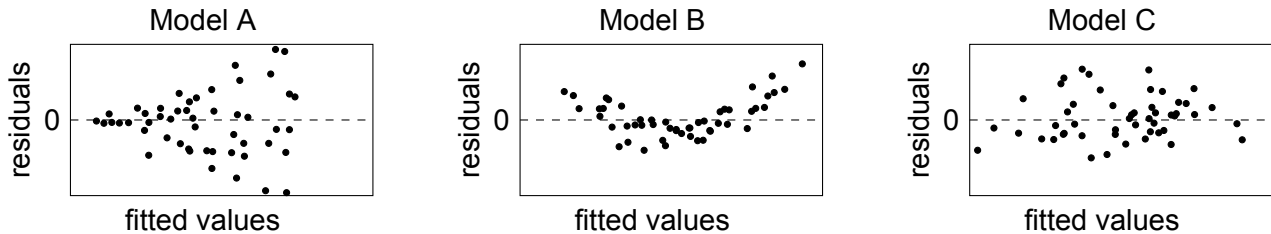
$$\begin{aligned} \Sigma \log x &= 46.3 & \Sigma \log y &= 1870.4 & \Sigma (\log x)^2 &= 308.4 \\ \Sigma (\log y)^2 &= 350761.2 & \Sigma (\log x)(\log y) &= 8370.3 & n &= 10 \end{aligned}$$

Calculate the correlation coefficient for the linear model and interpret it.

4. A linear model created by a statistician for variables x and y is used to produce a residual plot. After studying the residual plot, the statistician concludes that the assumption for the population of errors that $V(\epsilon_i) = 0$ is not supported.

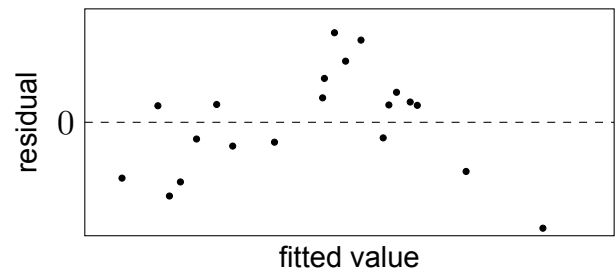
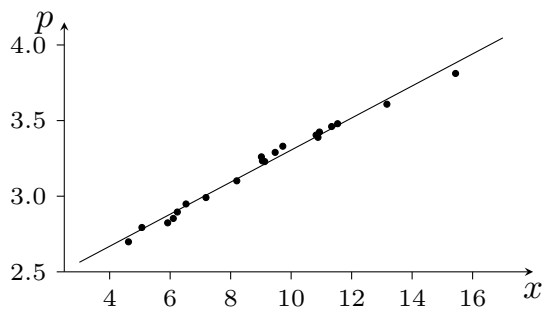
(a) Suggest a feature of the residual plot which could have led to their conclusion.

The statistician tries three different transformations, producing three new models (A, B and C). Residual plots for each model are below:



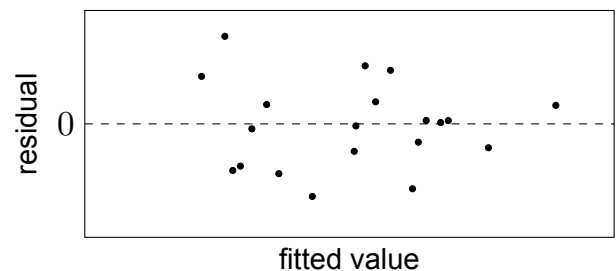
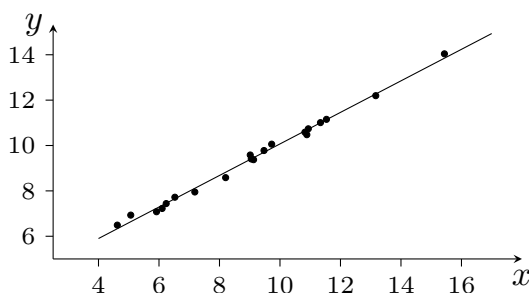
(b) State which model the statistician should use, with justification.

5. Data was collected on each weekday across four weeks of a university computer's peak processing speed (p GHz) and the amount of power consumed (x kWh). The regression line $\hat{U} = 2.245 + 0.106x$ is fitted to the data.



(a) Comment on the suitability of the linear model of p on x .

A transformation using $y = 2^p$ was applied, and a new least squares regression line of y on x found, with the following scatterplot and residual plot.



(b) Comment on the suitability of this new model of y on x .

The new model has summary statistics:

$$\Sigma x = 180.4 \quad \Sigma y = 187.8 \quad \Sigma x^2 = 1779 \quad S_{xy} = 105.9 \quad n = 20$$

(c) Find the new least squares regression line of y on x .

(d) Estimate the peak processing speed corresponding to a power consumption of 145 kWh.

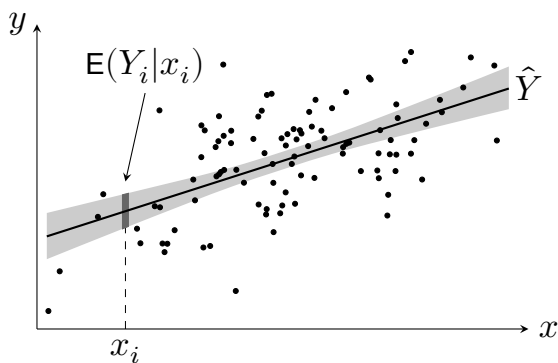
(Note: $t = a^b$ can be written as $b = \log_a t$).

4.9 Confidence Intervals and Prediction Intervals

One purpose for adopting a regression approach to examine the relationship between two variables is the wish to make predictions related to the response variable (Y) when given a particular value for the predictor (x).

The least squares regression line equation of $\hat{Y}_i = a + bx_i$ gives an *estimate* for the **mean** value of the response variable for any given x_i , whilst the *true* mean response value for a given x_i , which is denoted $E(Y_i | x_i)$, is unknown.

It can be desirable to obtain a **confidence interval** for $E(Y_i | x_i)$.



It can be shown that the *point estimate* for the mean response, $\hat{Y}_i$, has a sample standard error of:

$$s \sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}}$$

If the errors, ε_i , are identically and independently distributed as $N(0, \sigma^2)$ then the formula below, given on Page 5 of the Data Booklet, can be derived.

A Confidence Interval for the Mean Response

A $100(1-\alpha)\%$ CI for $E(Y_i | x_i)$ is given by: $\hat{Y}_i \pm t_{n-2, 1-\alpha/2} \times s \sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}}$

With all confidence intervals, the level of confidence relates to the success rate of the *procedure used to generate the interval*, rather than the probability of a calculated interval containing the mean response value $E(Y_i | x_i)$.

Example**Problem:**

The lengths in centimetres (x) and masses in kilograms (y) of 12 salmon were measured, with the results:

$$\Sigma x = 30 \quad \Sigma y = 540 \quad S_{xx} = 12 \quad S_{yy} = 4800 \quad S_{xy} = 150$$

Construct a 95% confidence interval for the mean length of salmon with mass 3kg.

Solution:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{150}{12} = 12.5, \quad a = \bar{y} - b\bar{x} = \frac{540}{12} - 12.5 \times \frac{30}{12} = 13.75 \quad \Rightarrow \quad \hat{Y}_i = 13.75 + 12.5x_i$$

$$\hat{Y} = 13.75 + 12.5 \times 3 = 51.75$$

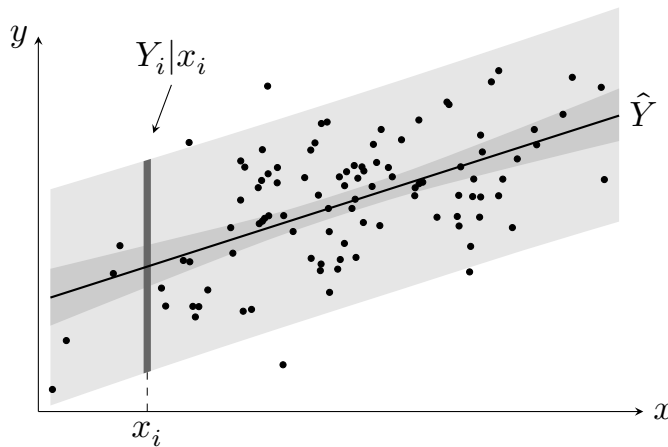
$$\text{SSR} = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 4800 - \frac{150^2}{12} = 2925, \quad s^2 = \frac{\text{SSR}}{n-2} = \frac{2925}{10} = 292.5, \quad s = \sqrt{292.5} = 17.1$$

$$\text{A 95\% CI for } E(Y \mid x = 3) \text{ is } 51.75 \pm 2.228 \times 17.1 \times \sqrt{\frac{1}{12} + \frac{(3 - 2.5)^2}{12}} = 51.75 \pm 12.30$$

So a 95% confidence interval for the mean length of a 3kg salmon is (39.45cm, 64.05cm).

Instead of an interval for the value of *mean* response, it may be desired to obtain an interval for **individual** response values for a given x_i , denoted as $Y_i | x_i$.

The desired interval is called a **prediction interval**.



Since the value of a randomly sampled individual can be expected to have more variability than the value of a sample mean, a wider interval around $\hat{Y}_i$ is needed.

Note the more darkly shaded region lying closer to the regression line, showing the narrower confidence interval for the same data.

In the long-run, a 95% prediction interval for $Y_i | x_i$, such as indicated in the diagram, will tend to capture 95% of all observed response values for any given x_i .

A Prediction Interval for an Individual Response

A $100(1 - \alpha)\%$ PI for $Y_i | x_i$ is given by: $\hat{Y}_i \pm t_{n-2, 1-\alpha/2} \times s \sqrt{1 + \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}}$

This formula, also given on Page 5 of the Data Booklet, relies also on the errors, ε_i , being identically and independently distributed as $N(0, \sigma^2)$

Example**Problem:**

The lengths in centimetres (x) and masses in kilograms (y) of 12 salmon were measured, with the results:

$$\Sigma x = 30 \quad \Sigma y = 540 \quad S_{xx} = 12 \quad S_{yy} = 4800 \quad S_{xy} = 150$$

Construct a 95% prediction interval for the length of salmon with mass 3kg.

Solution:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{150}{12} = 12.5, \quad a = \bar{y} - b\bar{x} = \frac{540}{12} - 12.5 \times \frac{30}{12} = 13.75 \quad \Rightarrow \quad \hat{Y}_i = 13.75 + 12.5x_i$$

$$\hat{Y} = 13.75 + 12.5 \times 3 = 51.75$$

$$\text{SSR} = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 4800 - \frac{150^2}{12} = 2925, \quad s^2 = \frac{\text{SSR}}{n-2} = \frac{2925}{10} = 292.5, \quad s = \sqrt{292.5} = 17.1$$

$$\text{A 95\% PI for } Y | x = 3 \text{ is } 51.75 \pm 2.228 \times 17.1 \times \sqrt{1 + \frac{1}{12} + \frac{(3 - 2.5)^2}{12}} = 51.75 \pm 40.03$$

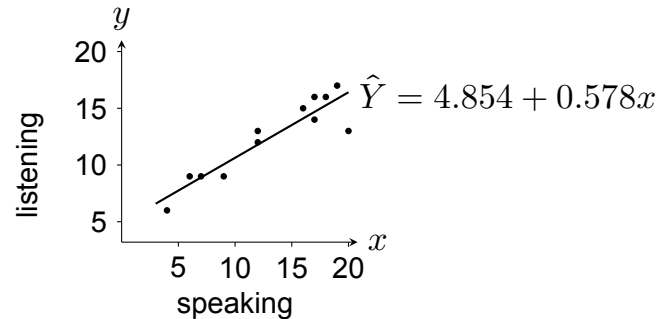
So a 95% prediction interval for the observed length of a 3kg salmon is (11.72cm , 91.78cm).

Exercise 18.9

1. Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). Data is gathered from a random sample of 12 students, and a regression line obtained.

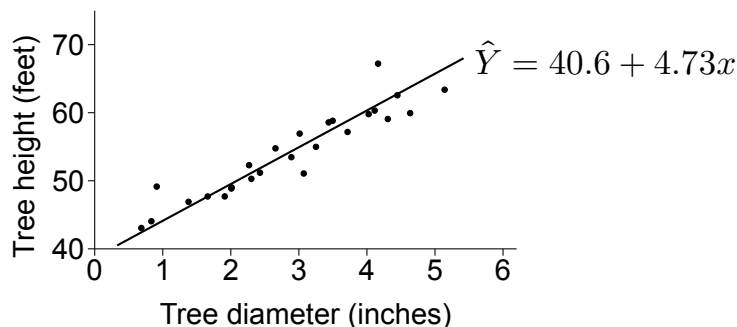
Summary statistics are calculated as:

$$\begin{aligned}\Sigma x &= 157 \\ S_{xx} &= 334.9 \\ s &= 1.45 \\ n &= 12\end{aligned}$$



A pupil achieves a speaking score of 15. They are yet to complete the listening component.

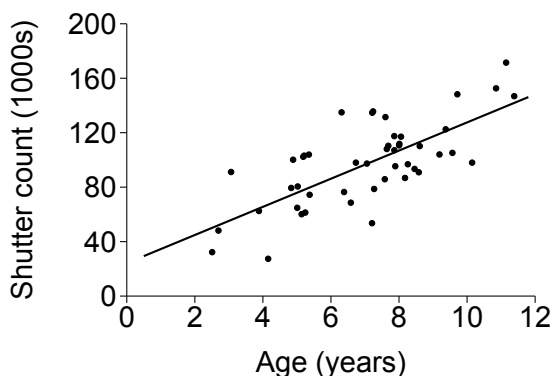
- Obtain a 95% confidence interval for the mean listening score given a speaking score of 15.
 - Obtain a 95% prediction interval for the listening score given a speaking score of 15.
2. 26 randomly selected trees in a forest have their diameter at a set distance from the ground measured, in inches, and the height of each tree also measured, in feet. The equation of the least squares regression line for x on y is obtained, and shown on the scatterplot below.



Summary statistics:

$$\begin{aligned}\Sigma x &= 74.76 \\ S_{xx} &= 38.8 \\ s^2 &= 5.09 \\ n &= 26\end{aligned}$$

- Obtain a 99% confidence interval for the mean height of a tree with diameter 4 inches.
 - Obtain a 99% prediction interval for the height of a tree with diameter 4 inches.
3. A seller on an online auction site has a stock of used digital cameras, all aged somewhere between 2 and 12 years. For a random sample of 48 cameras, the total number of photos taken (the *shutter count*) is determined using special software. A scatterplot with a least squares regression line is below.



Summary statistics for the data are as follows:

$$\begin{aligned}\Sigma x &= 337.7 & S_{xx} &= 221.8 \\ S_{yy} &= 44085.2 & S_{xy} &= 2192.8\end{aligned}$$

The equation for regression line was found:

$$\text{shuttercount} = 24.22 + 10.36 \times \text{age}$$

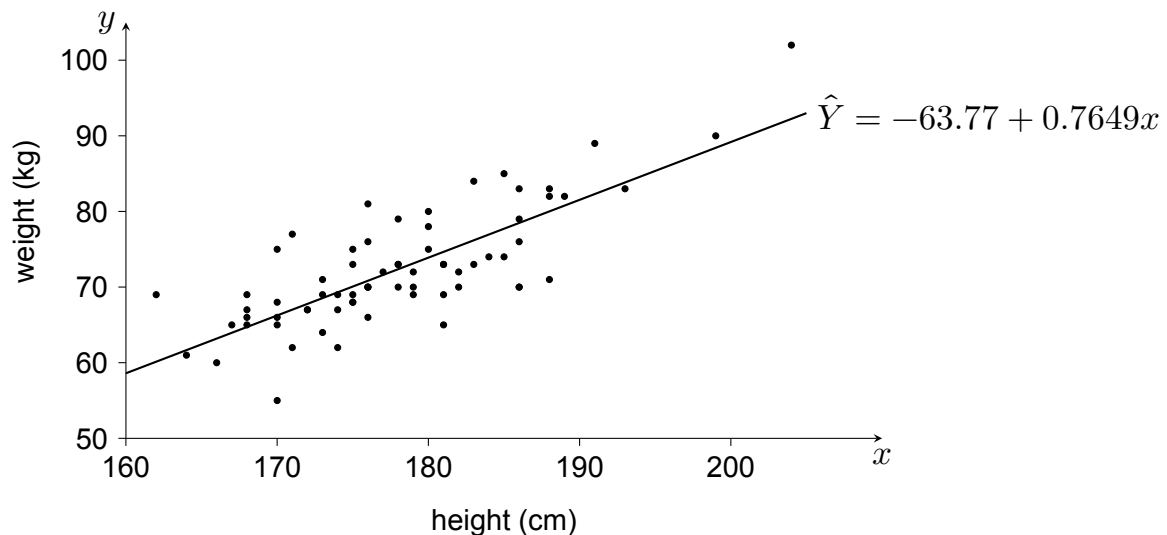
- Obtain a 90% prediction interval for the shutter count of a 6 year old camera.
- Obtain a 90% confidence interval for the mean shutter count of a 6 year old camera.

4. The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track. Computer output related to a linear model constructed using this data is below.

y on x regression line	n	s	sum of squares for x	sigma x
speed = 269.2 - 1.72 × time	34	8.49	638.6	72.05

One driver completed their lap in a time of 52 seconds, but their top speed was not recorded. They wish to estimate a range of values for their top speed. Obtain a suitable interval at the 95% level.

5. The heights (x cm) and weights (y kg) are recorded for a random sample of professional footballers.



A linear model is fit to the data as shown with a regression line equation of:

$$\text{weight} = -63.77 + 0.7649 \times \text{height}$$

A footballer, whose weight is unknown, has a height of 190 centimetres.

- (a) Use the equation of the regression line to calculate an estimate for the footballer's weight.

Computer output is used to generate a 95% prediction interval for the weight of a footballer with a height of 190 centimetres.

$$\begin{aligned} &95\% \text{ PI} \\ &(71.59, ****) \end{aligned}$$

- (b) Determine the missing upper value for the prediction interval above, indicated by '****'.
 (c) State the assumptions about the population of errors required for the predication interval above.

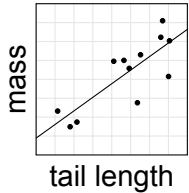
Summary statistics are obtained for the data, as shown below:

$$S_{xx} = 4319.0 \quad S_{yy} = 4122.4 \quad S_{xy} = 3303.6 \quad n = 69$$

- (d) Calculate a confidence interval for the weight of a footballer of height 172cm.
 (e) Explain why the method used above may not be appropriate to help estimate the mean weight of a footballer with a height of 158 centimetres.

Review Exercise

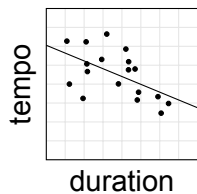
1. Biologists studying populations of European hamsters record the tail length (x centimetres) and body mass (y grams) of a random sample of 12 wild hamsters found living in central Vienna. The linear regression equation was calculated for the data shown in the scatterplot as $\hat{Y} = -15.24 + 71.88x$.



$$\Sigma x = 62.25, \quad \Sigma y = 4291.5, \quad \Sigma x^2 = 325.9, \quad \Sigma y^2 = 1558042, \quad \Sigma xy = 22475$$

- Calculate the residual of a 410 gram hamster with a tail length of 5.2cm.
- Use a suitable hypothesis test to assess the usefulness of the linear model for prediction at the 10% level of significance.

2. Data is collected on the tempo (T beats per minute) and duration (d minutes) of a random sample of pop-punk songs. The regression line $\hat{T} = 256.65 - 20.65d$ is obtained for the data.



$$\begin{array}{lll} \Sigma d = 43.6 & \Sigma T = 3719.5 & \Sigma dT = 8947.2 \\ \Sigma d^2 = 108.5 & \Sigma T^2 = 772792 & n = 18 \end{array}$$

- Calculate the coefficient of determination and interpret its value.
- Perform a hypothesis test at the 1% level of significance to check for evidence of a linear association between the tempo and duration of pop-punk songs.

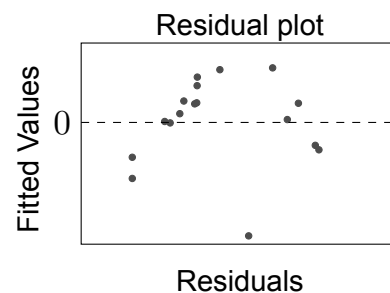
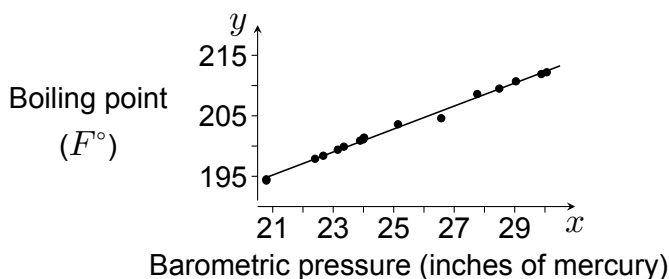
3. The creators of a brand new sports website collect data on the annual social media marketing spend (£ x thousand) and new daily visitors (y) of similar sites, to plan their own budget. The results are shown below, including a least squares regression line of equation $\hat{Y} = 29.5 + 5.6x$.



$$\Sigma x = 230.6 \quad \Sigma x^2 = 2244.6 \quad S_{xy} = 1118.6 \quad S_{yy} = 20640.9 \quad n = 26$$

- Calculate an estimate of the error variance.
- Calculate a 95% prediction interval for the daily new visitors of a site with an annual social media marketing spend of £15,000.
- Explain why a prediction interval is more likely to be of interest to the magazine's creators than a confidence interval.

4. The Scottish physicist James Forbes was interested in finding a way to estimate altitude from measurements of the boiling temperature of water. Some of his observations on boiling point (y , in F°) and barometric pressure (x , in inches of mercury) are summarised below. After plotting a scatterplot for this data, a least squares regression line was obtained with the equation $\hat{Y} = 155.29 + 1.9018x$, from which a residual plot was also created.



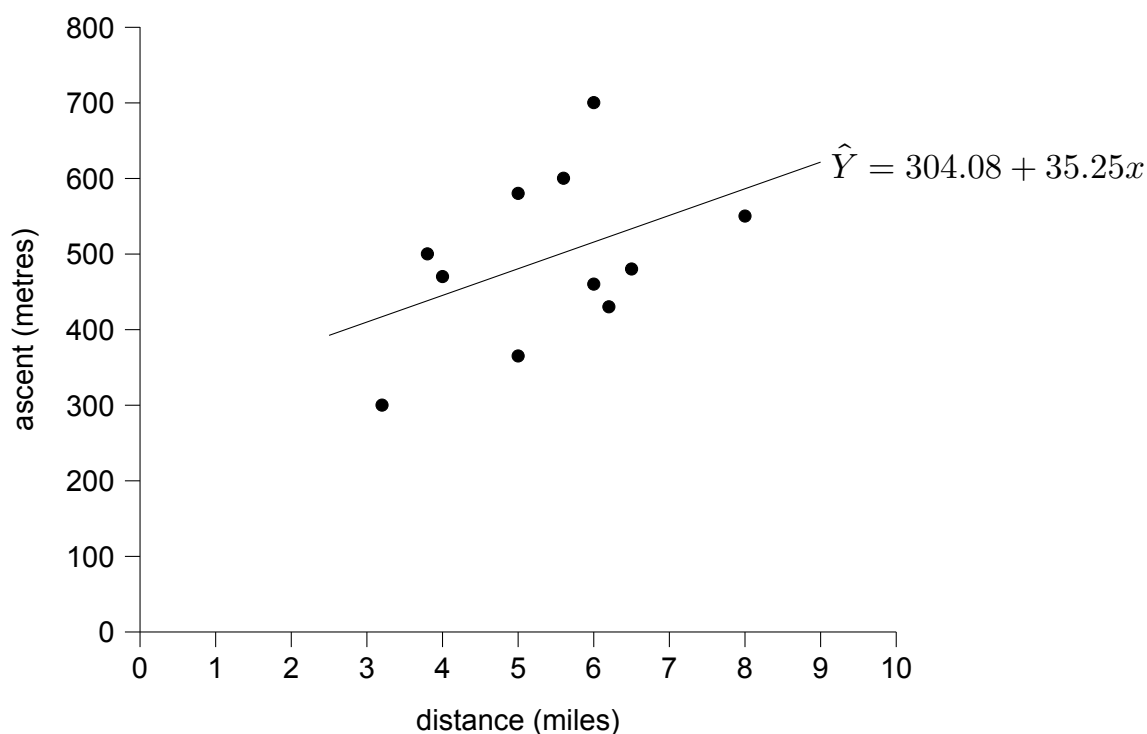
- Comment on the suitability of the linear model.
- Suggest what next steps might be taken on the basis of the residual plot for the linear model.

5. The website *Fife Walking* contains an extensive collection of walking routes grouped by area. The length (x miles) and total vertical ascent (y metres) is shown in **table 1** below for a random sample of eleven routes in the *Ochil Hills* area.

Table 1

Route Name	Distance (miles)	Ascent (metres)
Beld Hill and Craig Rossie	6	460
Dumyat	3.8	500
Myretoun Hill and Balquharn Glen	4	470
Colsnaur Circuit	5	580
Core Hill	6.5	480
Eastbow Hills and Steele's Knowe	6.2	430
King's Seat Circuit	5.6	600
Burn of Sorrow Circular	6	700
Hilfoot Hill and Castle Campbell	3.2	300
Innerdownie, Whitewisp and Glenquey Circuit	8	550
Ben Shee	5	365

The data is displayed below in a scatterplot below (**figure 1**), with the equation of a y on x least squares regression line calculated as $\hat{Y} = 304.08 + 35.25x$.

Figure 1

- Calculate the correlation coefficient and interpret it, with reference to the context.
- Explain why the regression line obtained would not be suitable for estimating the distance a route covers if it is known that the total ascent is 430 metres.
- Determine the equation of a more suitable regression line, and use it to estimate the distance covered for a route with an ascent of 430 metres.
- A walker completes a route in the Lomond Hills area, which also features on the *Fife Walking* website. Explain why it would may not be appropriate for them to use either of the regression lines obtained to predict their total ascent if their distance covered is known.

Answers

Chapter 1 Answers - Exploratory Data Analysis

Exercise 1.1

- (a) Quantitative (b) Qualitative (c) Quantitative (d) Qualitative (e) Quantitative (f) Qualitative
- (a) Discrete (b) Continuous (c) Discrete (d) Discrete (e) Continuous (f) Continuous
- Categorical: *activity level* and *smoker*. Continuous: *height* and *mass*. Discrete: *pulse*.

Exercise 1.2

- Population parameter (population mean)
- Sample statistic (sample mean)
- Sample statistic (sample mean)

Exercise 1.3

- $\bar{x} = 35.9, s = 15.74$
- $\bar{f} = 5.8, s = 2.145$
- $\bar{x} = 29.125, s = 7.586$
- median = 40.5, IQR = 27.5

Exercise 1.4

- dot plot of golfer ages
- stem-and-leaf diagram
(ice cream cost)

0	90
1	40 50 50 70 80 90
2	00 10 20 50
3	60

$n = 12$

2|50 represents £2.50
- box plot of luggage masses

Exercise 1.5

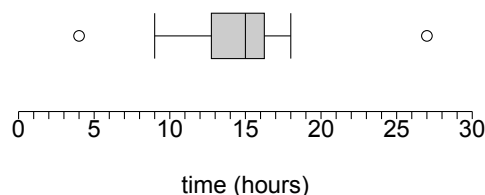
- $\frac{23}{94}$ or 0.2447
- A clear key/legend linked to the plot, numbers/scale on the y axis.
- The proportions of pupils in year group taking the bus seem to be broadly similar.
- Approximately 15 minutes.
- 7

Exercise 1.6

1. Lower fence = 7.5 and upper fence = 51.5. Since all values lie within this interval, there are no outliers.

(b)

box plot of studying time



2. (a) Lower fence = 6.5 and upper fence = 22.5. Since both 4 and 27 lie outwith this interval, they are outliers.

3. (a) Upper fence = 26.5. Since both circled points shown on the box plot are greater than 26.25, they are outliers.
 (b) If the lengths of their stays were incorrectly recorded in the hospital's records, it would be valid to remove them from the data set before conducting any further analysis. *Other valid reasons are possible.*

Exercise 1.7

- The population of hazelnuts masses appear to follow a positively skewed distribution.
- The population of tree heights appear to be normally distributed.
- The population of waiting times appear to be uniformly distributed.
- The population of number of successful passes appears to follow a bimodal distribution.

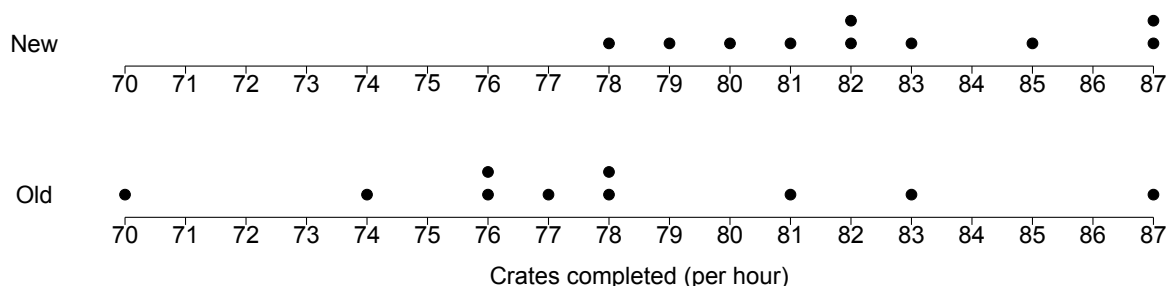
Exercise 1.8

1. (a) $\bar{x}_A = 301.3$, $s_A = 4.57$
 $\bar{x}_B = 302.3$, $s_B = 12.3$

- (b) The median volume of coffee dispensed by Model B is greater.
 The amount of coffee dispensed by Model B is more varied.

2. The median score awarded by judge A was higher.
 The higher interquartile range for judge A shows that their scores were more varied.
3. (a) The daily 2pm temperature was more varied in January than in June, as shown by the slightly higher standard deviation.
 The median 2pm temperature in June was higher.
- (b) Consistent scales have not been used for the box plots, making the location and spread of each less easily comparable.
4. (a)

Production line data



- (b) $\bar{x}_{\text{new}} = 82.4$, $s_{\text{new}} = 3.13$
 $\bar{x}_{\text{old}} = 78$, $s_{\text{old}} = 4.76$

- (c) The mean number of crates completed per hours was higher for the new production line.
 The number of crates completed per hour was more consistent for the new production line.

Chapter 1 Review Exercise

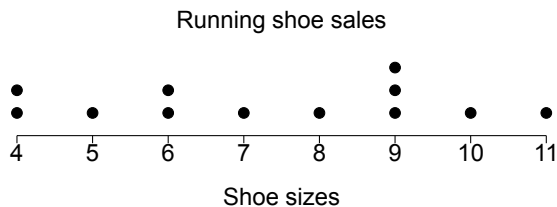
1. (a) i. Track; artist. ii. Release year; no. of streams. iii. Length.

(b) Fences could be calculated to determine whether it is an outlier.

2. (a) (b) median = 7.5 and interquartile range = 3.25

(c) mean = 7.33 and sd = 2.35

(d) Shoe sizes are discrete data.



3. (a) A legend should be provided for the diagram, such as $3|2 = 3.2$ metres.
 (b) The stem-and-leaf diagram shows that trees in Woodland A were taller, on average.
4. (a) The fences are 1.13 and 1.85. Since both the minimum value of 1.00 and the maximum of 1.90 lie outwith this interval, the *iris setosa* data contains at least two outliers.
 (b) The median length of *iris versicolor* petals appears to be greater than the length of *iris setosa* petals. The length of *iris versicolor* petals appears to be more varied than those of the *iris setosa* species, as seen by their interquartile ranges of 0.18 and 0.60 centimetres respectively.

Chapter 2 Answers - An Introduction to Probability Theory

Exercise 2.1

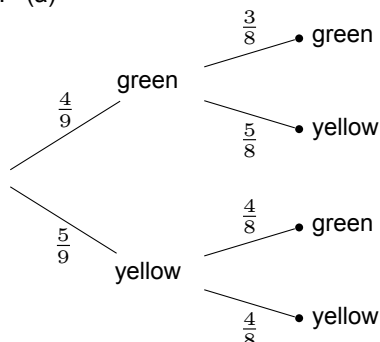
1. (a) $P(\text{vowel}) = \frac{5}{26}$
 (b) $P(\text{consonant}) = \frac{21}{26}$
 (c) $P(\text{statistics}) = \frac{5}{26}$
2. (a) $P(\text{blue}) = \frac{4}{15}$
 (b) $P(\text{green or red}) = \frac{11}{15}$
 (c) $P(\text{not red}) = \frac{3}{5}$
3. (a) $P(\text{orange}) = \frac{1}{3}$
 (b) $P(\text{orange and strawberry}) = 0$
 (c) $P(\text{not caramel}) = \frac{7}{12}$
4. (a) $P(\text{Scotland}) = \frac{2}{19}$
 (b) $P(\text{Wales or NI}) = \frac{13}{76}$
 (c) $P(\text{not England}) = \frac{21}{76}$
5. (a) $P(\text{blue and 5}) = \frac{1}{18}$
 (b) $P(5) = \frac{1}{9}$
 (c) $P(\text{blue or 5}) = \frac{7}{18}$
6. (a) $P(G) = \frac{9}{10}$
 (b) $P(G \text{ and } A) = \frac{27}{80}$
 (c) $P(\text{not } A) = \frac{5}{8}$
7. (a) $P(\text{nurse}) = \frac{25}{64}$
 (b) $P(\text{not ward 2}) = \frac{39}{64}$
 (c) $P(\text{ward 2 and not doctor}) = \frac{21}{64}$

Exercise 2.2

1. (a) {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}
 (b) $P(\text{even and tails}) = \frac{1}{4}$
2. (a) $P(\text{tea and toast}) = \frac{1}{12}$
 (b) $P(\text{museli and not coffee}) = \frac{1}{6}$
 (c) $P(\text{not porridge and not tea}) = \frac{1}{2}$
3. (a) $P(\text{tails twice in a row}) = \frac{3}{8}$
 (b) $P(\text{tails at least twice}) = \frac{1}{2}$
4. $P(\text{sum} = 3) = \frac{1}{20}$
 5. $P(\text{product} < 5) = \frac{2}{9}$
 6. $P(\text{at least one number} \leq 2) = \frac{2}{3}$
 7. (a) $P(20\text{p first}) = \frac{1}{5}$
 (b) $P(20\text{p selected}) = \frac{2}{5}$
 (c) $P(\text{total} \geq 15\text{p}) = \frac{1}{2}$
 (d) $P(\text{total} < 12\text{p}) = \frac{2}{5}$
8. (a) 56
 (b) i. $P(A \text{ board and C pens}) = \frac{1}{56}$
 ii. $P(\text{neither chosen}) = \frac{15}{28}$
 iii. $P(\text{at least one chosen}) = \frac{13}{28}$

Exercise 2.3

1. (a)



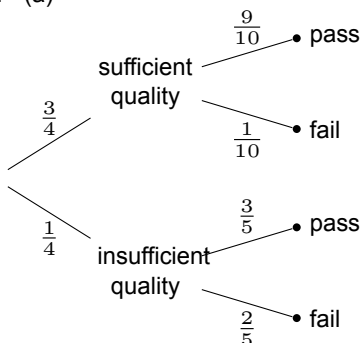
(b) $P(\text{both green}) = \frac{1}{6}$

(c) $P(\text{at least one green}) = \frac{13}{18}$

2. $P(\text{both same colour}) = \frac{9}{19}$

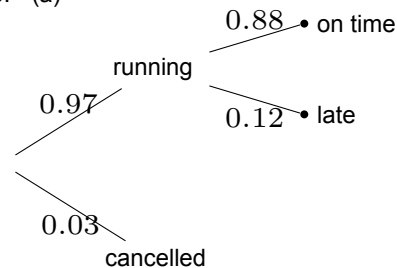
3. $P(\text{neither } O_+) = 0.4281$

4. (a)



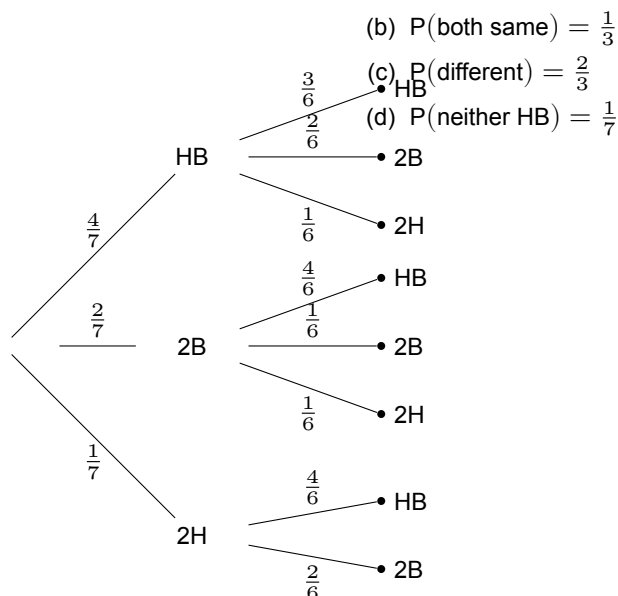
(b) $P(\text{both pass}) = 0.825$

5. (a)



(b) $P(\text{on time}) = 0.8536$

6. (a)



(b) $P(\text{both same}) = \frac{1}{3}$

(c) $P(\text{different}) = \frac{2}{3}$

(d) $P(\text{neither HB}) = \frac{1}{7}$

7. (a) $P(\text{no stops}) = 0.224$

(b) $P(\text{at least one stop}) = 0.776$

(c) $P(\text{exactly two stops}) = 0.252$

Exercise 2.4

1. (a) $P(A) = \frac{5}{9}$

(b) $P(B) = \frac{4}{9}$

(c) $P(\bar{A}) = \frac{4}{9}$

(d) $P(\bar{B}) = \frac{5}{9}$

(e) $P(A \cap B) = \frac{1}{3}$

(f) $P(A \cup B) = \frac{2}{3}$

(g) $P(\bar{A} \cap B) = \frac{1}{9}$

(h) $P(A \cap \bar{B}) = \frac{2}{9}$

2. (a) $x = 0.35$

(b) i. $P(N) = 0.55$

ii. $P(\bar{M}) = 0.5$

iii. $P(M \cup N) = 0.7$

3. (a) $P(\text{prime}) = \frac{3}{8}$

(b) $P(\text{odd}) = \frac{1}{2}$

(c) $P(\overline{\text{prime}}) = \frac{5}{8}$

(d) $P(\overline{\text{odd}}) = \frac{1}{2}$

(e) $P(\text{prime} \cap \text{odd}) = \frac{5}{16}$

(f) $P(\text{prime} \cup \text{odd}) = \frac{9}{16}$

(g) $P(\overline{\text{prime}} \cap \text{odd}) = \frac{3}{16}$

(h) $P(\text{prime} \cap \overline{\text{odd}}) = \frac{1}{16}$

4. (a) $P(C \cap D) = 0.34$

(b) $P(C \cup D) = 0.91$

(c) $P(\bar{C} \cap \bar{D}) = 0.09$

(d) $P(\bar{C} \cup \bar{D}) = 0.66$

(e) $P(C \cap \bar{D}) = 0.45$

(f) $P(\bar{C} \cap D) = 0.12$

(g) $P(C \cup \bar{D}) = 0.88$

(h) $P(\bar{C} \cup D) = 0.55$

5. $P(\text{biology}) = \frac{13}{24}$

6. $P(\text{PM} \cap \text{NO}_2) = \frac{1}{10}$

Exercise 2.5

1. $P(X \cup Y) = 0.6$
2. $P(A \cup B) = \frac{29}{30}$
3. $P(E \cup F) = 0.61$
4. $P(C \cup D) = \frac{11}{14}$
5. $P(A \cap B) = 0.163$
6. $P(X \cap Y) = \frac{9}{80}$
7. $P(V \cap W) = 0.2$
8. $P(\bar{J}) = \frac{7}{72}$
9. $P(X \cup Y) = 0.9$
10. $P(\bar{B}) = \frac{1}{6}$
11. $P(\text{fish}) = 0.85$
12. $P(\bar{N}) = \frac{3}{8}$
13. $P(\bar{R}) = 0.19$
14. (a) $P(X \cup Y) = 0.9 \neq 1$
 $\therefore X, Y$ not exhaustive
 (b) $P(R \cup T) = 1$
- $\therefore R, T$ exhaustive
- (c) $P(E \cup F) = 0.85 \neq 1$
 $\therefore E, F$ not exhaustive
15. (a) $P(W \cup V) = 0.3 \neq 0$
 $\therefore W, V$ not mutually exclusive
 (b) $P(X \cap Y) = 0$
 $\therefore X, Y$ mutually exclusive
 (c) $P(A \cap B) = 0$
 $\therefore A, B$ mutually exclusive

Chapter 2 Review Exercise

1. (a) $P(\overline{\text{yellow}} \cup 1) = \frac{2}{3}$
 (b) $P(\text{neither yellow}) = \frac{5}{18}$
2. (a) $P(\text{English} \cap N5) = \frac{21}{64}$
 (b) $P(\text{History} \cup \text{Higher}) = \frac{43}{64}$
3. $P(F) = 0.9$
4. $P(\text{takes payment}) = 0.8926$
5. (a) $P(\text{not a tie}) = \frac{5}{6}$
 (b) $P(\text{attacker wins}) = \frac{7}{12}$
- (c) $P(\text{bonus point}) = \frac{1}{8}$
6. $P(N \cup \bar{A}) = \frac{7}{25}$
7. $P(B \cap C) = 0.2$

Chapter 3 Answers - Sampling Theory

Note: Whilst the answers included here are intended to include an appropriate level of detail, variations in the amount and type of information given in exam questions will require a flexible and common-sense approach to be used in any exam. Past exam questions and marking schemes should be used as a guide, as well as course reports and relevant CPD events.

Exercise 3.1

1. (a) i. People who use his local supermarket.
 ii. All shoppers who use the self-checkout tills on that Sunday morning.
- (b) The sampling frame excludes people who prefer to shop at other times, such as during the evening or midweek. It also only includes those using self-checkout tills, likely excluding those who have a large trolley of shopping. Both of these issues are likely to make the sample unrepresentative of the target population, and may introduce bias.
- (c) The sampling methods is non-random and so will likely introduce bias to the data, making it unsuitable.
2. (a) There may be so many flowers that it is too time-consuming and impractical to make a sampling frame comprised of individual flowers.
- (b) A better choice of sampling unit for the sampling frame may be square metres of field, forming a numbered grid.
- (c) A random number generator will help avoid possible bias in the selection of square metres to include in the sample, by ensuring it is random and not influenced by the botanist. It should help produce a sample that is more representative of the population of wildflowers.

Exercise 3.2

1. (a) One-stage cluster sampling.
 (b) Stratified sampling.
 (c) Systematic sampling.
 (d) Simple random sampling.
 (e) Quota sampling.
 (f) Convenience Sampling.
2. (a) Each of the different types of room may have their own distinct characteristics in terms of ageing or the quality of maintenance. Stratified sampling will ensure that a proportionate number from each type of room will be included in the sample.
- (b) 30 out of 360 cabins means 1 in every 12 cabins should be sampled, so the sample should contain: 2 Deluxe cabins; 6 Premium cabins and 22 Standard cabins. To sample the Deluxe cabins, a numbered list of those cabins should be obtained as a suitable sampling frame, and then a random number generator used to produce 2 different random numbers. Those cabins corresponding to the numbers selected should be included in the sample. This process should then be repeated for the Premium cabins and the Standard cabins, using a numbered sampling frame for each and randomly selecting the appropriate number of cabins of each type. Combining all of the selected cabins will make a stratified sample.

3. (a) Convenience sampling.
(b) He only samples coral reefs close to his town, ignoring that those near other towns or villages, or rural locations, may have a different level of health. This means the sample is unlikely to be representative of the population of coral reefs along the West Coast of Scotland. (Alternatively, the non-random selection method may be discussed.)
4. (a) Using cluster sampling means less physical travelling to a larger number of petrol stations, which would cost more and take more time.
(b) To sample 40 pumps, 5 petrol stations are needed if one-stage cluster sampling is used. A sampling frame of all the petrol stations numbered from 1 to 126 should be drawn up. A random number generator can then be used to select 5 numbers, and the petrol stations corresponding to those numbers should be included in the sample. From each of those petrol stations, all of the pumps should be sampled.
5. (a) Systematic sampling.
(b) She didn't use a random starting point, so it is not a true random sampling process that is being used.
6. (a) Quota sampling.
(b) This is a non-random sampling method which is liable to produce a sample not representative of the target population, and may introduce bias into the data gathered.

Chapter 3 Review Exercise

1. (a) i. Convenience sampling.
ii. This is a non-random sampling method which may produce a sample that is not representative of the target population, and may introduce bias to the data.
(b) i. Simple random sampling.
ii. With the 150 employees in an ordered list, the random cell selection tool could be used to select the first employee to be included in the sample. From there, every 5th employee on the list could be selected, looping back around to the start of the list if needed until the sample of size 30 has been obtained.
(c) i. A numbered list of the restaurants in the chain could be obtained. A random number generator could be used to pick several numbers, and those restaurants corresponding to the numbers could be selected to be sampled. For a one-stage cluster sample, all employees at those restaurants should be sampled.
ii. One advantage of this method is that it will require less restaurants to be visited for an equivalent sample size, reducing travel costs. One disadvantage is that visiting less restaurants may make it more likely than any restaurants in the chain experiencing problems have none of their employees included in the sample.
2. For an overall sample size of 150 out of 5000 (3%), the sample needs to include 3 people from Management, 12 from Sales and 135 from the Shop Floor. For each strata, an ordered sampling frame should be obtained, and numbers assigned. For the Management sample, a random number generator should be used to choose 3 numbers, and those staff corresponding to those numbers should be selected to be in the sample. The same process should be repeated to choose 12 from the Sales sampling frame, and 135 from the Shop Floor sampling frame. Put together, this will make a sample of size 150 with proportions from each strata matching those of the target population.
3. The pizzas sampled will only be those from takeaway restaurants, and may not be representative of pizzas sold in the UK as a whole. Additionally, only those takeaways with a presence on social media will be included.
4. (a) It is unlikely to be possible for the researcher to obtain a list of all pupils in Scotland studying Higher Mathematics.
(b) From an ordered list of all centres in Scotland which presents pupils for Higher Maths, simple random sampling could be used to select a number of schools. For each of those schools selected, an ordered list of all pupils taking Higher Maths should be obtained, and used to select a random number of pupils from each selected school. Each of those pupils selected should be included in the overall sample.
5. A random letter selector and a random number generator could be used together to select a letter and a number, to correspond to grids, such as 'C' and '5' to select grid C5. This process could be repeated to create a list of grids to for the botanist to visit and study the health of flowers in those grids.

Chapter 4 Answers - Further Probability Theory

Exercise 4.1

- $P(X) = \frac{5}{9}$
 - $P(X|Y) = \frac{3}{4}$
 - $P(X \cap Y) = \frac{1}{3}$
 - $P(X \cup Y) = \frac{2}{3}$
 - $P(Y) = \frac{4}{9}$
 - $P(Y|X) = \frac{3}{5}$
 - $P(Y \cap X) = \frac{1}{3}$
 - $P(Y \cup X) = \frac{2}{3}$
- $P(\text{scored}) = \frac{73}{135}$
 - $P(\text{scored} | 3\text{-pointer}) = \frac{15}{41}$
- $P(3\text{-pointer}) = \frac{41}{135}$
 - $P(3\text{-pointer} | \text{scored}) = \frac{15}{73}$
- $P(\text{rainy}) = 0.3$
 - $P(\text{not rainy}) = 0.7$
 - $P(\text{leaves} | \text{rainy}) = 0.4$
 - $P(\text{does not leave} | \text{rainy}) = 0.6$
 - $P(\text{leaves} | \text{not rainy}) = 0.9$
 - $P(\text{does not leave} | \text{not rainy}) = 0.1$
- $P(A) = \frac{2}{5}$
 - $P(B) = \frac{2}{5}$
- $P(C) = \frac{1}{3}$
 - $P(A \cap B) = \frac{2}{15}$
 - $P(B \cup C) = \frac{3}{5}$
 - $P(A \cap B \cap C) = \frac{1}{15}$
 - $P(A|C) = \frac{3}{5}$
 - $P(C|B) = \frac{1}{3}$
 - $P(\overline{B}|C) = \frac{3}{5}$
 - $P(C|\overline{A}) = \frac{2}{9}$
 - $P((A \cap B)|C) = \frac{1}{5}$

Exercise 4.2

- $P(A|B) = 0.2$
 - $P(\text{red} | \text{blue}) = 0.4849$
 - $P(X|Y) = \frac{2}{3}$
 - $P(M|N) = 0.3233$
 - $P(\overline{A}|B) = 0.8$
 - $P(\overline{\text{red}} | \text{blue}) = 0.5151$
 - $P(\overline{X}|Y) = \frac{1}{3}$
 - $P(\overline{M}|N) = 0.6767$
- $P(\overline{V}|W) = 0.3$
- $P(5 | \text{yellow}) = 0.2$
- $P(\text{lose} | \text{concede} \geq 2) = \frac{5}{8}$
- $P(M|N) = 0.25$
- $P(V \cup W) = 0.725$

Exercise 4.3

- Independent.
 - Not independent.
- Not independent.
 - Independent.
 - Independent.
- Show e.g. $P(V) = P(V|W)$
 - Show e.g. $P(T) \neq P(T|W)$

Exercise 4.4

- $P(X \cap Y) = 0.072$
 - $P(\text{blue} \cap \text{yellow}) = \frac{3}{8}$
 - $P(E \cap F) = 0.246$
- $P(A \cap B) = 0.392$
 - $P(A \cup B) = 0.708$
- $P(X \cup Y) = 0.88$
- $P(\text{working} \cap \text{Calculo}) = 0.12$
- $P(\text{out-of-date} \cap \text{weak}) = 0.096$
- $P(\text{blue} \cap \text{pink}) = 0.38$

Exercise 4.5

- $P(X|Y) = \frac{7}{12}$
 - $P(V|W) = 0.625$
- $P(1 | \text{red}) = 0.6$
- $P(\text{gene} | \text{positive}) = 0.2186$
- $P(M|N) = \frac{8}{15}$
 - $P(\overline{M}|N) = \frac{7}{15}$
- $P(\text{pass} | \text{car}) = 0.8446$
- $P(\text{Pacific 525} | \text{music}) = 0.7467$
- $P(\overline{E}|F) = 0.1946$

Exercise 4.6

- $P(\text{UK}) = 0.8708$
- $P(\text{survives}) = 0.6745$
- $P(\text{win}) = 0.695$
- $P(\text{likes}) = 0.54$
- $P(\text{not make second part}) = 0.28$
- $P(\text{do not survive}) = 0.268$
- $P(\text{win}) = 0.3882$

Exercise 4.7

1. (a) $P(\text{damaged}) = 0.0655$
(b) $P(\text{express} \mid \text{damaged}) = 0.1069$
2. (a) $P(\text{flagged}) = 0.0308$
(b) $P(\text{malicious} \mid \text{flagged}) = 0.3584$
3. $P(\text{guessed} \mid \text{wrong}) = 0.7273$
4. (a) $P(\text{visits}) = 0.033$
(b) $P(\text{social media} \mid \text{visits}) = 0.8409$
5. $P(\text{lost} \mid \text{rest day}) = 0.0559$
6. $P(\text{flaw} \mid \text{doesn't leave}) = 0.8048$
7. (a) $P(\text{has condition} \mid \text{positive}) = 0.4580$
(b) $P(\text{has condition} \mid \text{positive test}) = 0.4975$

Chapter 4 Review Exercise

1. $P(E \mid F) = \frac{7}{9}$
2. (a) i. $P(W \cup A) = \frac{17}{20}$
ii. $P(\overline{W} \cap B) = \frac{1}{8}$
- (b) $P(\text{power}) = 0.825$
(c) $P(\text{Mac} \mid \overline{\text{power}}) = \frac{1}{7}$
3. (a) $P(\text{wins}) = 0.3917$
(b) $P(1\text{st} \mid \text{lost}) = 0.1370$
4. $P(M \mid B) = \frac{1}{5} = P(M)$
hence P, M are independent
5. (a) $P(\text{incorrect}) = 0.2875$
(b) $P(\text{cat} \mid \text{incorrect}) = 0.2348$

Chapter 5 - An Introduction to Linear Regression

Exercise 5.1

Exercise 5.2

Exercise 5.3

Chapter 5 Review Exercise

Chapter 6 - Random Variables

Exercise 6.1

Exercise 6.2

Exercise 6.3

Exercise 6.4

Exercise 6.5

Exercise 6.6

Chapter 6 Review Exercise

Chapter 7 - Discrete Distributions

Exercise 7.1

1. $E(X) = 3$ and $V(X) = 2$
2. $E(X) = 8$ and $V(X) = \frac{56}{3}$
3. (a) $E(X) = 4.5$ and $V(X) = 5.25$
(b) $E(2X + 5) = 14$ and $V(2X + 5) = 21$
4. (a) $X \sim U(20)$
(b) $E(X) = 10.5$ and $V(X) = 33.25$
(c) $E(1 - 3X) = -30.5$ and $V(1 - 3X) = 299.25$
5. (a) $E(X - Y) = -1$
(b) $V(X - Y) = \frac{8}{3}$
6. (a) $k = 10$
(b) $P(2 \leq X < 6) = 0.4$
(c) $SD(X) = \frac{\sqrt{33}}{2}$
7. (a) $k = 8$
(b) $E(1 - 2Y) = -8$
8. $X \sim U(11)$ and $Y \sim U(7)$

Exercise 7.2

- | | | |
|-----------|-----------------------------------|------------|
| 1. 0.2503 | 4. 0.0988 | (b) 0.3599 |
| 2. 0.2787 | | |
| 3. 0.0579 | 5. (a) $X \sim B(4, \frac{1}{7})$ | 6. 0.0214 |

Exercise 7.3

- | | | |
|---------------|---------------|---------------|
| 1. (a) 0.9894 | 2. (a) 0.9527 | (c) 0.4829 |
| (b) 0.0367 | (b) 0.2235 | |
| (c) 0.9527 | (c) 0.5756 | 4. (a) 0.1468 |
| (d) 0.0106 | 3. (a) 0.7414 | (b) 0.5033 |
| (e) 0.0473 | (b) 0.0003 | (c) 0.1927 |

Exercise 7.4

- | | | |
|---------------------------------------|-------------------------------------|---------------------------------------|
| 1. (a) $\mu = 0.8, \sigma^2 = 0.64$ | (b) $E(S) = 2.5$ and $V(S) = 1.875$ | 4. (a) $p = \frac{2}{3}$ |
| (b) $\mu = 2, \sigma^2 = \frac{4}{3}$ | (c) 0.0197 | (b) $E(Y) = 8$ |
| (c) $\mu = 0.4, \sigma^2 = 0.36$ | 3. (a) $p = 0.2$ | 5. (a) $n = 20$ and $p = \frac{4}{5}$ |
| (d) $\mu = 12, \sigma^2 = 7.2$ | (b) $SD(X) = 0.96$ | (b) $P(5 < X < 10) = 0.0006$ |
| 2. (a) $S \sim B(10, \frac{1}{4})$ | | |

Exercise 7.5

1. (a) 0.1336
- (b) 0.0821
- (c) 0.2873
- (d) 0.7127
2. (a) $X \sim Po(4)$
- (b) 0.0183

Exercise 7.6

- | | | |
|----------------|----------------|-----------------------|
| 1. (a) 0.01512 | 2. (a) 0.02231 | 3. (a) $X \sim Po(1)$ |
| (b) 0.5543 | (b) 0.0611 | (b) 0.0190 |
| (c) 0.0620 | (c) 0.4412 | |
| (d) 0.3937 | (d) 0.1912 | |

Exercise 7.7

1. 0.9473
2. 0.9389
3. 0.2661
4. 0.1353
5. 0.0144

Chapter 7 Review Exercise

- | | | |
|------------------------------------|-------------------------------------|--|
| 1. (a) $\frac{5}{8}$ | 3. (a) 0.1294 | (c) The germination of the seeds is independent. |
| (b) $E(X) = 4.5$ and $V(X) = 5.25$ | (b) 0.4769 | |
| 2. (a) 0.1954 | 4. 0.4335 | 6. (a) 0.6703 |
| (b) 0.2071 | 5. (a) 0.8298 | (b) 0.1954 |
| | (b) $\mu = 17$ and $\sigma = 1.597$ | 7. (a) $P(M V) = 0.2$ |

- (b) $X \sim B(8, 0.3)$
 (c) 0.0580

8. (a) 0.2
 (b) 0.1528

- (c) 0.0267

Chapter 8 Answers - Continuous Distributions

Exercise 8.1

- | | | |
|---|--|---|
| 1. (a) $\frac{2}{3}$
(b) 0.7667
(c) 0.1 | (b) $E(3X - 1) = 2,$
$V(5 - 2X) = \frac{100}{3}$
(c) 0.64
(d) 0.4 | 8. (a) 8
(b) 4.04
(c) 0.5
(d) 0.5771 |
| 2. (a) $\frac{1}{3}$
(b) 0
(c) 0.6583 | 5. (a) $\frac{5}{6}$
(b) 7.5 mins
(c) 4.33 mins | 9. (a) $X \sim U(2, 14)$
(b) $\sqrt{12}$
(c) $\frac{5}{12}$ |
| 3. (a) $\frac{1}{6}$
(b) $\frac{5}{6}$
(c) $\frac{5}{12}$ | 6. $\frac{4}{7}$
7. (a) 0.4
(b) $X \sim B(20, 0.4), 0.5841$ | 10. (a) $Y \sim U(16, 22)$
(b) 0.4082 |
| 4. (a) $E(X) = 1, SD(X) = 2.89$ | | |

Exercise 8.2

- | | | |
|---|---|----------------------|
| 1. (a) 52, 60, 68 marked
(b) 237, 243, 249 marked
(c) 5.4, 8, 10.6 marked | (d) 12.6, 15, 17.4 marked
2. 3, 7, 11 marked
3. 5, 8, 11 marked | 4. 26, 32, 38 marked |
|---|---|----------------------|

Exercise 8.3

- | | | |
|---|---|--|
| 1. (a) 0
(b) 0.9573
(c) 0.9573
(d) 0.5
(e) 0.6368
(f) 0.9788
(g) 0.0212 | (h) 0.1170
(i) 0.1170
(j) 0.7123
(k) 0.9251
(l) 0.2843
2. (a) 0.0740
(b) 0.2888 | (c) 0.0339
(d) 0.1618
(e) 0.6612
(f) 0.6826
(g) 0.9544
(h) 0.9500
(i) 0.8990 |
|---|---|--|

Exercise 8.4

- | | | |
|---|--|---|
| 1. (a) 0.7475
(b) 0.1587
(c) 0.6306
(d) 0.4719 | (c) 0.9660
(d) 0.3891
3. (a) $X \sim N(67.4, 6^2)$
(b) i. 0.6676
ii. 0.6554
iii. 0.2647 | 4. (a) i. 0.8997
ii. 0.9500
(b) i. 143
ii. 595
iii. 536 |
| 2. (a) 0.1587
(b) 0.4305 | | |

Exercise 8.5

- | | | |
|---|--|---|
| 1. (a) 1.28
(b) 1.64
(c) 3.09
(d) 2.33
(e) 2.58 | (f) -1.96
(g) -2.58
(h) -1.28
(i) -1.96
(j) 3.09 | (k) 0.81
(l) 2.75
2. (a) $X \sim N(2.4, 0.2^2)$
(b) 0.0228
(c) 2.866 litres |
|---|--|---|

3. (a) 8.692 mins
(b) 7.612 to 8.788 mins
4. $\sigma = 9.9$
5. $\mu = 64.1, \sigma = 14.8$

Exercise 8.6

1. (a) $N(34, 13.09)$
(b) 0.6103
2. 0.1867
3. 0.1379
4. 0.0192

Exercise 8.7

1. (a) $np = 16, nq = 24$
 $np, nq > 5$
(b) i. 0.9265
ii. 0.0031
iii. ≈ 0
2. (a) $np = 54, nq = 27$
 $np, nq > 5$
(b) i. 0.3632
ii. 0.0338
iii. 0.6971
3. $X \sim B(600, 0.85)$ 0.1151
4. 0.2578
5. 0.0044

Exercise 8.8

1. (a) 0.6217
(b) 0.0661
(c) 0.5398
2. (a) 0.6736
(b) 0.5640
(c) 0.1357
3. (a) 0.2611
(b) 0.2349
(c) 0.0550
4. 0.8907
5. (a) 0.7384
(b) 3.8
6. (a) 0.4562
(b) 0.1718
(c) 0.8315
7. (a) 0.2389
(b) 0.4641
8. 0.8529

Chapter 8 Review Exercise

1. (a) 4
(b) 3
(c) $\frac{1}{3}$
2. (a) 0.7977
(b) 0.3413
3. (a) $H \sim N(18.1, 0.7^2)$
(b) 0.0033
(c) 19.248 feet
4. (a) 0.1353
(b) 0.0733
5. 22.99
6. (a) 0.3
(b) 0.6172
(c) 0.4562
7. (a) 0.0951

Chapter 9 - Distribution of the Sample Mean**Exercise 9.1**

1. (a) i. 0.3821
ii. 0.4404
(b) i. $N(24, \frac{4^2}{3})$
ii. 0.3015
- ii. 0.3974
2. 0.1922
3. (a) 0.3085
(b) 0.1170
4. 0.0228
5. (a) 0.0082
(b) 0.3974
6. 0.9992

Exercise 9.2

1. (a) $X \approx N(2.3, \frac{5^2}{40})$
(b) $Y \sim N(83, \frac{9}{25})$
(c) $D \sim N(126, \frac{10}{7})$
- (d) Comment.
2. 0.9857
3. 0.0071
4. 0.1038
5. 0.3758
6. (a) 0.0526

Chapter 9 Review Exercise

1. (a) $\bar{X} \sim N(15, \frac{3^2}{8})$
(b) 0.9706
2. 0.9981
3. (a) Comment.
(b) $\bar{X} \approx N(600, \frac{70^2}{40})$
- (c) 0.1489
4. (a) i. $X \sim N(60, 2^2)$
ii. 0.3085
(b) i. $\bar{X} \sim N(60, \frac{2^2}{8})$
ii. 0.1271
- (c) Comment.
5. (a) $X \sim Po(4.5)$
(b) 0.1736
(c) $X \approx N(4.5, \frac{4.5}{30})$
(d) 0.0985

Chapter 10 - An Introduction to Hypothesis Testing

Exercise 10.1

1. $H_0 : \mu = 8.6$
 $H_1 : \mu > 8.6$
2. $H_0 : \mu = 12$
 $H_1 : \mu > 12$
3. $H_0 : \mu = 4.2$
 $H_1 : \mu < 4.2$
4. $H_0 : \mu = 2.5$
 $H_1 : \mu \neq 2.5$
5. $H_0 : \mu = 65$
 $H_1 : \mu > 65$
6. $H_0 : \mu = 15.7$
 $H_1 : \mu \neq 15.7$

Exercise 10.2

}

1. $0.0162 < 0.05 \Rightarrow \text{reject } H_0$
2. $0.1056 > 0.01 \Rightarrow \text{do not reject } H_0$
3. $0.0485 < 0.1 \Rightarrow \text{reject } H_0$
4. $0.0764 > 0.05 \Rightarrow \text{do not reject } H_0$

Exercise 10.3

1. $0.0286 < 0.1 \Rightarrow \text{reject } H_0$
2. $0.0702 > 0.001 \Rightarrow \text{do not reject } H_0$
3. $0.0485 < 0.05 \Rightarrow \text{reject } H_0$
4. $0.0220 < 0.1 \Rightarrow \text{reject } H_0$
5. $0.0294 < 0.05 \Rightarrow \text{reject } H_0$
6. $0.0488 > 0.01 \Rightarrow \text{do not reject } H_0$
7. $0.0023 < 0.05 \Rightarrow \text{reject } H_0$
8. $0.0250 > 0.01 \Rightarrow \text{do not reject } H_0$

Exercise 10.4

1. $1.67 < 1.96 \Rightarrow \text{do not reject } H_0$
2. $-2.5 < -1.64 \Rightarrow \text{reject } H_0$
3. $2.24 < 2.33 \Rightarrow \text{do not reject } H_0$
4. $-0.77 > -2.33 \Rightarrow \text{do not reject } H_0$
5. $2.00 > 1.96 \Rightarrow \text{reject } H_0$

Exercise 10.5

1. $-2.985 < -1.895 \Rightarrow \text{reject } H_0$
2. $0.636 < 1.476 \Rightarrow \text{do not reject } H_0$
3. $2.317 < 2.447 \Rightarrow \text{do not reject } H_0$
4. $-3.531 < -2.821 \Rightarrow \text{reject } H_0$

Exercise 10.6

1. $1.52 < 1.64 \Rightarrow$ *do not reject* H_0
2. $2.42 > 2.33 \Rightarrow$ *reject* H_0
3. $-2.78 < -1.64 \Rightarrow$ *reject* H_0
4. $1.17 < 1.96 \Rightarrow$ *do not reject* H_0
5. $-2.04 > -2.58 \Rightarrow$ *do not reject* H_0
6. $-2.2 < -1.64 \Rightarrow$ *reject* H_0

Chapter 10 Review Exercise

1. $-2.54 < -2.33 \Rightarrow$ *reject* H_0
2. $2.158 < 2.365 \Rightarrow$ *do not reject* H_0
3. $-1.94 < -1.64 \Rightarrow$ *reject* H_0
4. $-3.00 < -2.33 \Rightarrow$ *reject* H_0
5. $-2.170 < -1.895 \Rightarrow$ *reject* H_0

Chapter 11 - Confidence Intervals**Exercise 11.1**

1. (55.18, 58.82)
2. (12.76, 14.84)
3. (14.53, 15.87)
4. (37.38, 40.62)
5. (49.03, 51.29)
6. (5.41, 5.99)
7. (a) (37.64, 38.76)
(b) (37.46, 38.94)
(c) 31
8. 10.0

Exercise 11.2

1. (499.1, 510.9) $\Rightarrow$ within, comment
2. (419.5, 426.5) $\Rightarrow$ within, comment
3. (10.11, 11.69) $\Rightarrow$ outwith, comment
4. (2.439, 2.961) $\Rightarrow$ within, comment
5. (19.95, 20.85) $\Rightarrow$ outwith, comment
6. (28580, 30020) $\Rightarrow$ within, comment)

Exercise 11.3

1. (6.88, 7.87)
2. (4048.5, 4714.4) $\Rightarrow$ within, comment
3. (960.0, 1024.0)
4. (3.636, 4.464) $\Rightarrow$ outwith, comment
5. (2.760, 4.740) $\Rightarrow$ outwith, comment

Exercise 11.4

1. (0.598, 0.736)
2. (0.133, 0.427)
3. (0.327, 0.387)
4. (0.086, 0.149) $\Rightarrow$ within, comment
5. (0.021, 0.199) $\Rightarrow$ outwith, comment
6. (0.026, 0.083) $\Rightarrow$ within, comment
7. $n = 147$
8. (a) Systematic sampling
(b) (0.264, 0.446) $\Rightarrow$ within, comment

Chapter 11 Review Exercise

1. (63.98, 82.77)
2. (0.116, 0.444)
3. (a) (125.8, 126.8)
(b) Outwith, comment
4. (a) 12.5
(b) $s = 25.77$
5. (0.679, 0.888) $\Rightarrow$ outwith, comment

Chapter 12 - Chi-Squared Goodness-of-Fit Test**Exercise 12.1****Exercise 12.2****Exercise 12.3****Exercise 12.4****Chapter 12 Review Exercise**

Chapter 13 - Control Charts

Exercise 13.1

1. The process is out of statistical control, since sample 6 falls below the lower control line. The process should have been stopped after sample 6.
2. The process is out of statistical control, since from samples 4 to 8 at least four of the five samples lie above the upper 1-sigma line. The process should have been stopped after sample 8.
3. The process is under statistical control.
4. The process is out of statistical control, since from samples 2 to 9 eight consecutive samples lie above the centre line. The process should have been stopped after sample 9.

Exercise 13.2

1. (a) $UCL = 82.5$, $+2\sigma = 81.75$, $+1\sigma = 80.75$, target = 80, $-1\sigma = 79.25$, $-2\sigma = 78.5$, $LCL = 77.75$
(b) $UCL = 500.72$, $+2\sigma = 495.81$, $+1\sigma = 490.91$, target = 486, $-1\sigma = 481.09$, $-2\sigma = 476.19$, $LCL = 471.28$
2. $+2\sigma = 1.447$, $-2\sigma = 1.353$
3. (a) $+1\sigma = 231.79$, $-1\sigma = 228.21$
(b) The process is out of statistical control since, from samples 9 to 11, two of the three samples lie above the upper 2-sigma limit. The process should have been stopped after sample 11.
(c) A required assumption is that the samples of bottles are random.
4. (a) $144 - 3 \times \frac{18}{\sqrt{7}} = 123.59$
(b) $130.4 \leq \bar{x} \leq 164.4$
5. (a) The process is out of statistical control since eight consecutive points lie above the centre line, from samples 9 to 16. However, since these values suggest the lenses being produced have a higher resolution than the target value, which is a good thing, they may not be concerned.
(b) Samples of size 9 are being taken.
(c) Two required assumptions are that:
 - The samples of lenses taken are random.
 - The resolution scores of manufactured lenses are normally distributed.
6. (a) $UCL = 121$, $+2\sigma = 119$, $+1\sigma = 117$, target = 115, $-1\sigma = 113$, $-2\sigma = 111$, $LCL = 109$.
The process is out of statistical control since, from samples 2 to 4, two out of three sample means lie above the upper 2-sigma limit.
(b) $\bar{\bar{x}} = 92$
(c) $UCL = 101$, $LCL = 83$

7. (a)

Population mean	72
Standard deviation	15
Sample size	25
Upper Control Limit	81
$+2\sigma$ Limit	78
$+1\sigma$ Limit	75
Target Value	72
-1σ Limit	69
-2σ Limit	66
Lower Control Limit	63

(b)

Population mean	2800
Standard deviation	60
Sample size	9
Upper Control Limit	2860
$+2\sigma$ Limit	2840
$+1\sigma$ Limit	2820
Target Value	2800
-1σ Limit	2780
-2σ Limit	2760
Lower Control Limit	2740

Exercise 13.3

1. (a) Population proportion = 0.6
Sample size = 24

Upper Control Limit	0.9
$+2\sigma$ Limit	0.8
$+1\sigma$ Limit	0.7
Target Value	0.6
-1σ Limit	0.5
-2σ Limit	0.4
Lower Control Limit	0.3

- (b) Population proportion = 0.16
Sample size = 50

Upper Control Limit	0.316
$+2\sigma$ Limit	0.264
$+1\sigma$ Limit	0.212
Target Value	0.16
-1σ Limit	0.108
-2σ Limit	0.056
Lower Control Limit	0.004

2. (a) The samples of mugs would not be random. Also, for appropriate use of a p chart it is required that $np > 5$ and $nq > 5$. Here, $0.08 \times 30 = 2.4 < 5$.
(b) Upper 2-sigma limit = 0.14265 and lower 2-sigma limit = 0.01735.
(c) The process is under statistical control for the samples shown, as none of the WECO rules have been violated.
3. (a) $0.11 + 3 \times \sqrt{\frac{0.11 \times 0.89}{50}} = 0.2427$
(b) The calculated lower limit would be -0.0227. However, a negative sample proportion is not possible, so the lower control limit is set as 0.
(c) The process is out of statistical control since, from samples 3 to 7, four out of five samples lie below the lower 2-sigma limit.
(d) Since these sample proportions suggest less seeds are failing to germinate than expected, which is a good thing, the company may not be concerned.
4. (a) UCL = 0.9549, LCL = 0.6451.
(b) The process is out of statistical control since, from February to September, eight consecutive sample proportions fall below the target value.
(c) The sample size needs to be at least 30 to ensure both $np > 5$ and $nq > 5$.
5. (a) $\hat{p} = 0.12$
(b) Upper 2-sigma limit = $0.12 + 2 \times \sqrt{\frac{0.12 \times 0.88}{50}} = 0.212$ and lower 2-sigma limit = $0.12 - 2 \times \sqrt{\frac{0.12 \times 0.88}{50}} = 0.028$
(c) The process is under statistical control for the samples shown, as none of the WECO rules have been violated.
6. (a) The population proportion is $p = 0.6$.
(b) UCL = 0.9 and LCL = 0.3.
(c) The sample size is $n = 24$.
(d) The sample size and population proportion are appropriate since both of $24 \times 0.6 = 14.4$ and $24 \times 0.4 = 9.6$ are greater than 5.

Chapter 13 Review Exercise

1. (a) Upper 2-sigma limit = 60.12 and lower 2-sigma limit = 55.88.
(b) The process is out of statistical control since the sample mean is below the lower control limit at sample 14.
2. (a) Upper 1-sigma limit = 0.926 and lower 1-sigma limit = 0.834.
(b) $0.88 \leq \hat{p} \leq 1$

Chapter 14 - Chi-Squared Test for Association**Exercise 14.1****Exercise 14.2****Chapter 14 Review Exercise****Chapter 15 - Two-Sample Parametric Tests****Exercise 15.1****Exercise 15.2****Exercise 15.3****Exercise 15.4****Chapter 15 Review Exercise****Chapter 16 - Wilcoxon Signed Rank Test****Exercise 16.1**

1. $9 > 3 \Rightarrow$ do not reject H_0
2. $5.5 < 3 \Rightarrow$ reject H_0
3. $8 > 5 \Rightarrow$ do not reject H_0
4. $7.5 > 5 \Rightarrow$ do not reject H_0
5. $10 \leq 10 \Rightarrow$ reject H_0
6. $9 < 10 \Rightarrow$ reject H_0
7. $6.5 > 5 \Rightarrow$ do not reject H_0

Exercise 16.2

1. $8.5 < 13 \Rightarrow$ reject H_0
2. $11 > 5 \Rightarrow$ do not reject H_0
3. $46 > 21 \Rightarrow$ do not reject H_0
4. $1.5 < 2 \Rightarrow$ reject H_0
5. $4 < 8 \Rightarrow$ reject H_0

Exercise 16.3

1. $-0.34 > -1.96 \Rightarrow$ do not reject H_0
2. $-1.67 < -1.64 \Rightarrow$ reject H_0
3. $-0.27 > -1.64 \Rightarrow$ do not reject H_0
4. (a) $H_0 : \eta = 110, H_1 : \eta < 110$
(b) $-2.16 > -1.64 \Rightarrow$ reject H_0
(c) Assume that time taken to complete the hike is symmetrically distributed.

Chapter 16 Review Exercise

1. $5 \leq 5 \Rightarrow$ reject H_0
2. $14 > 8 \Rightarrow$ do not reject H_0
3. $-2.40 > -1.96 \Rightarrow$ reject H_0
4. $3 > 2 \Rightarrow$ do not reject H_0

Chapter 17 - Mann Whitney Rank Sum Test

Exercise 17.1

- (a) Rank sum of 20.5, or rank sum of 44.5 leading to smaller value of 20.5.
(b) $20.5 < 21 \Rightarrow$ reject H_0
- $65.5 < 78 \Rightarrow$ reject H_0
- $28 \leq 28 \Rightarrow$ reject H_0
- $27.5 < 31 \Rightarrow$ reject H_0
- $29 > 28 \Rightarrow$ do not reject H_0
- Assume distributions of learning times for each type of mouse have the same shape and variance.
 $38.5 > 26 \Rightarrow$ do not reject H_0
- (a) $H_0 : \eta_A = \eta_B$ and $H_1 : \eta_A \neq \eta_B$
(b) $72.5 > 73 \Rightarrow$ reject H_0
(c) Assume the distributions of mass of dust deposited for each boiler type have the same shape and variance. Assume the samples of boilers are random. Assume the samples are independent.

Exercise 17.2

- (a) $-2.39 < -2.33 \Rightarrow$ reject H_0
(b) Assume that the distributions of driving times for each weather condition have the same shape and variance.
- $-1.47 > -1.96 \Rightarrow$ do not reject H_0
- $-1.96 > -1.64 \Rightarrow$ do not reject H_0
- (a) The distributions of gradings for each line appear to have the same shape and variance, to a Mann-Whitney test is appropriate. The distributions do not appear to be normal, so a t -test would not be appropriate.
(b) $-1.34 < -1.28 \Rightarrow$ reject H_0

Exercise 17.3

- (a) 56
(b) $\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{1, 3, 4\}$
(c) $\frac{4}{56} = \frac{1}{14}$ or 0.0714
(a) 126
(b) $\{1, 2, 3, 4\}, \{1, 2, 3, 5\}, \{1, 2, 3, 6\}, \{1, 2, 4, 5\}$
(c) $\frac{4}{126} = \frac{2}{63}$ or 0.0317
- (a) $1 + 2 + 4 + 8 = 15$
(b) $\{1, 2, 3, 4\}, \{1, 2, 3, 5\}, \{1, 2, 3, 6\}, \{1, 2, 3, 7\}, \{1, 2, 3, 8\}$
 $\{1, 2, 3, 9\}, \{1, 2, 4, 5\}, \{1, 2, 4, 6\}, \{1, 2, 4, 7\}, \{1, 2, 4, 8\}$
 $\{1, 2, 5, 6\}, \{1, 2, 5, 7\}, \{1, 3, 4, 5\}, \{1, 3, 4, 6\}, \{1, 3, 4, 7\}$
 $\{1, 3, 5, 6\}, \{2, 3, 4, 5\}, \{2, 3, 4, 6\}$
(c) Assume that the population distributions of reactions times for each group have the same shape and variance.
(d) $\frac{18}{330} = \frac{3}{55}$ or 0.0545
(e) $0.0545 > 0.05 \Rightarrow$ do not reject H_0
- (a) $\frac{7}{495} = 0.0141$
(b) $0.0141 < 0.1 \Rightarrow$ reject H_0
(c) Assume that the samples of hockey and rugby players are random.

Chapter 17 Review Exercise

- $34 > 29 \Rightarrow$ do not reject H_0
- (a) The stem-and-leaf diagram shows that red wines seem to contain more alcohol on average than white wines.
(b) $-3.04 < -1.64 \Rightarrow$ reject H_0
(c) Since the stem-and-leaf diagram seems to suggest that the distributions of alcohol content for the two groups have the same shape and variance, a Mann-Whitney test would be appropriate. Since the distributions both appear to be normal-like, a t -test would also be appropriate.

- (d)
- i. $H_0 : \eta_A = \eta_B$ and $H_0 : \eta_A \neq \eta_B$
 - ii. $CV = 13$. Since $12 < 13$ reject H_0 at the 2% significance level, so there is evidence to suggest that the two populations do not have the same median.
 - iii. Assuming that the samples are random, and assume that the populations have the same shape and variance.
 - iv. The data for samples A and B will have been combined and ordered, with ranks assigned to each value. The rank sum for sample A will have been found, W_A , and then the value $4 \times (4 + 9 + 1) - W_A$ will have also been calculated. $\bar{w} = 12$ will have been the smaller of those two values.

Chapter 18 - Further Linear Regression

Exercise 18.1

1. Completed table:

Regression Line $\hat{Y} = ax + b$	Observed x	Observed y	Fitted $\hat{Y}$	Residual ϵ
$\hat{Y} = 5.3 + 0.1x$	7	8	6	2
$\hat{Y} = -0.7 + 0.3x$	6.1	1.04	1.13	-0.09
$\hat{Y} = 180 - 2.5x$	40	178	80	98
$\hat{Y} = -45 + 3.5x$	200	625	655	-30
$\hat{Y} = 100 + 20x$	-15	-150	-200	50

2. (a) 37.36 grams

(b) i. Fitted = 291.8
Residual = 18.2

ii. Fitted = 158.2
Residual = -8.2

iii. Fitted = 11.9
Residual = -1.9

3. (a) i. positive
ii. negative

(b) i. Fitted = 78.5
Residual = 8.5

ii. Fitted = 82.3
Residual = -6.7

Exercise 18.2

1. (a) $s^2 = 9.3$

(b) $s = 3.05$

2. (a) $s^2 = 5.257$

(b) $s = 2.29$

3. (a) The scatterplot suggests there is a strong negative linear association between the distance a runner covers in training in the four weeks before a marathon and their finishing time.

(b) $s^2 = 10.97$

(c) $\hat{Y} = 194.8 - 23.48x$

(d) $r = -0.928$

The correlation coefficient also suggests there is a strong negative linear association between the distance a runner covers in training in the four weeks before a marathon and their finishing time.

(e) Fitted = 112.6, residual = -2.6

Exercise 18.3

1. $R^2 = 0.775$. This suggests that 77.5% of the variation in Y is explained by the linear model.

2. (a) $R^2 = 0.734$. This suggests that 73.4% of the variation in the employees' salaries is explained by the linear model. (Or: ...by the number of years they have been employed for).

(b) $R^2 = 0.573$. This suggests that 57.3% of the variation in red squirrel mass is explained by the linear model. (Or: ...by the height of the red squirrels).

(c) $R^2 = 0.449$. This suggests that 44.9% of the variation in the top speed reached is explained by the linear model. (Or: ...by their lap time).

3. (a) The scatterplot appears to show a moderate/strong, positive linear association between a camera's shutter count and age.

(b) The correlation coefficient also suggests there is a moderate/strong, positive linear association between a camera's shutter count and its age.

(c) $R^2 = 0.516$. This suggests that 51.6% of the variation in a camera's shutter count is explained by the linear model. (Or: ...by its age).

(d) Since the data collected only includes digital cameras from within the last 12 years, any results from it may not apply to much older film cameras.

Also, the regression line is y on x , suitable only for predicting shutter count based on a camera's age.

Exercise 18.4

- $H_0: \beta = 0$
 $H_1: \beta \neq 0$
 - $s^2 = \frac{224.2}{27-2} = 8.968$, $s = \sqrt{8.987} = 2.995$, $b = \frac{-129.6}{87.8} = -1.476$, $t = \frac{-1.476 \times \sqrt{87.8}}{2.995} = -4.618$
 - $\nu = 27 - 2 = 25$
 - Critical value for $\nu = 25$ and a 5% two tailed test (and a negative test statistic) is $t_{25,0.975} = -2.060$.
Since $-4.618 < -2.060$ reject H_0 at the 5% level of significance.
There is evidence to suggest that the slope parameter $\beta \neq 0$, and so the linear model is useful for prediction.
- Test statistic $t = -5.120$, critical value $t_{32,0.95} = -1.694$, reject H_0 since $-5.120 < -1.694$.
- $H_0: \beta = 0$
 $H_1: \beta \neq 0$
 - $n = 40$
 - Critical value $t_{0.975,38} = -2.024$. Since $-0.42295 > -2.024$, do not reject H_0 at the 5% significance level.
There is insufficient evidence to suggest that the slope parameter β is not equal to zero, and so no evidence that the linear model is useful for prediction.
 - Assume that the population of errors are independent and identically distributed as $N(0, \sigma^2)$.
- Test statistic $t = -13.458$, critical value $t_{68,0.995} = -2.576$, reject H_0 since $-13.458 < -2.576$.
- Test statistic $t = 0.013$, critical value $t_{6,0.975} = 2.447$, do not reject H_0 at the 5% level since $0.013 < 2.447$.

Exercise 18.5

- $H_0: \rho = 0$
 $H_1: \rho \neq 0$
 - $$t = \frac{-0.678\sqrt{27-2}}{\sqrt{1-(-0.678)^2}} = -4.612$$
 - $\nu = 27 - 2 = 25$
 - Critical value for $\nu = 25$ and a 5% two tailed test (and a negative test statistic) is $t_{25,0.975} = -2.060$.
Since $-4.612 < -2.060$ reject H_0 at the 5% level of significance.
There is evidence to suggest that there is a linear association between the variables.
- Test statistic $t = 6.418$, critical value $t_{12,0.95} = 1.782$, reject H_0 at the 10% level since $6.418 > 1.782$.
- Test statistic $t = 1.683$, critical value $t_{12,0.995} = 3.055$, do not reject H_0 at the 1% level since $1.683 < 3.055$.
- Test statistic $t = -2.898$, critical value $t_{18,0.975} = -2.101$, reject H_0 at the 5% level since $-2.898 < -2.101$.
Assume that electrolytes amount and time are independent and follow approximately bivariate normal distributions.
- Test statistic $t = 0.013$, critical value $t_{6,0.975} = 2.447$, do not reject H_0 at the 5% level since $0.013 < 2.447$.

Exercise 18.6

- $\hat{X} = 73.2 - 0.719y$
- $\hat{X} = -7.09 + 0.184y$
 - 4.87 inches
- The existing regression line is a y on x regression line, suitable for predicting the distance covered based on their finish time. To predict a finish time based on the distance covered in training, an x on y regression model is needed.
 - $\hat{X} = 7.709 - 0.0367y$
 - 5.14 hours
 - This would be an example of extrapolation, which may not be considered reliable.
- $\hat{X} = 25.46 + 0.0051y$

Exercise 18.7

- The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
 - The residual plot appears to show the points centred on zero ($E(\epsilon) = 0$) in a funnel pattern with decreasing variance as fitted values increase ($V(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
 - The residual plot appears to show the points in the pattern of a curve ($E(\epsilon) \neq 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests that the linear regression model is not suitable.
 - The residual plot shows a random scatter of points not centred on zero ($E(\epsilon) \neq 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is not suitable.
 - The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) \neq \sigma^2$) which suggests the linear regression model is suitable.
 - The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) \neq \sigma^2$) which suggests the linear regression model is suitable.
 - The residual plot appears to show the points in the pattern of a U-shaped curve ($E(\epsilon) \neq 0$) with constant variance ($V(\epsilon) = \sigma^2$) which suggests that the linear regression model is not suitable.
 - The residual plot appears to show the points centred on zero ($E(\epsilon) = 0$) in a funnel pattern with increasing variance as fitted values increase ($V(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
 - The residual plot shows a random scatter of points not centred on zero ($E(\epsilon) \neq 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is not suitable.
 - The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
 - The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
 - The residual plot appears to show the points in the pattern of a curve ($E(\epsilon) \neq 0$), which suggests that the linear regression model is not suitable.
- The residual plot appears to show the points form a curved pattern ($E(\epsilon) = 0$) with increasing variance as fitted values increase ($V(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
- The residual plot shows a random scatter of points centred on zero ($E(\epsilon) = 0$) with constant variance ($V(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
- The residual plot appears to show the points in the pattern of a curve ($E(\epsilon) \neq 0$), which suggests that the linear regression model is not suitable.

Exercise 18.8

- There appears to be a strong negative linear association between 'distance' and ' $\sqrt{\text{power}}$ '.
The coefficient of determination suggests 80.5% of the variation in ' $\sqrt{\text{power}}$ ' is explained by the model.
 - There appears to be a weak positive linear association between 'rating' and ' $\log_{10}$ energy'.
The coefficient of determination suggests 38.2% of the variation in ' $\log_2$ energy' is explained by the model.
 - There appears to be a very strong positive linear association between ' $\log_2$ time' and ' $\log_2$ population'.
The coefficient of determination suggests 97.4% of the variation in ' $\log_2$ population' is explained by the model.
 - There appears to be no linear association between 'quality' and ' $\frac{1}{\text{cost}}$ '.
The coefficient of determination suggests 0.1% of the variation in ' $\frac{1}{\text{cost}}$ ' is explained by the model.
- $\widehat{\frac{1}{Y}} = 2.936 + 2.898x$
 - 0.0574
- $r = -0.984$. This suggests that there is a very strong negative linear correlation between $\log x$ and $\log y$.
- They would have observed the points not showing an equal variance as fitted values increased, such as a funnel shape.
 - They should use model C, as it is the only one whose residual plot appears to show a random scatter of points centred on zero, with a constant variance as fitted values increase. This suggests it is the only model of the three which is appropriate.
- The residual plot appears to show a curved pattern, suggesting a linear model between p and x is not appropriate.
 - The residual plot for this new model appears to show a random scatter of points centred around zero, with a constant variance, so this model appears to be appropriate.
- $\widehat{Y} = 0.0598 + 8.85x$
- 10.3

Exercise 18.9

1. (a) (12.53, 14.52)
(b) (10.14, 16.90)
2. (a) (57.80, 61.17)
(b) (52.95, 66.02)
3. (a) (49.69, 123.0)
(b) (80.52, 92.17)
4. (161.35, 198.62)
5. (a) 81.56 kg
(b) 91.53
(c) Assume that the population of errors are independently and identically distributed as $N(0, \sigma^2)$.
(d) (66.34, 69.24)
(e) This is an example of extrapolation, which may not be reliable.

Chapter 18 Review Exercise

1. (a) 51.46
(b) Since $4.380 > 1.812$ reject H_0 .
2. (a) $R^2 = 0.300$. This suggests that 30.0% of the variation in the tempo of a pop-punk song is explained by the linear model.
(b) Since $-2.620 < -2.921$ do not reject H_0 .
3. (a) 598.9
(b) (57.58, 169.46)
(c) They are interested in a range of values for the number of daily visitors they might receive, rather than the mean number of daily visitors received by websites spending the same amount.
4. (a) The residual plot shows a curved pattern of points, suggesting that the linear model of boiling point on barometric pressure is not appropriate.
(b) A transformation could be taken on one or both of the variables and a new linear model created between those transformed variables.
5. (a) $r = 0.436$, suggesting that there is a moderate positive linear association between the distance a route covers and the ascent.
(b) The regression line given is a y on x regression line, suitable for predicting the ascent of a route given its distance. To predict the distance given the ascent, an x on y regression line would be needed.
(c) $\hat{X} = 2.73 + 0.00539y$
predicted distance = 5.05 miles
(d) This data has only been taken from the Ochil Hills, and any results from this data may not be relevant for routes in the Lomond Hills.